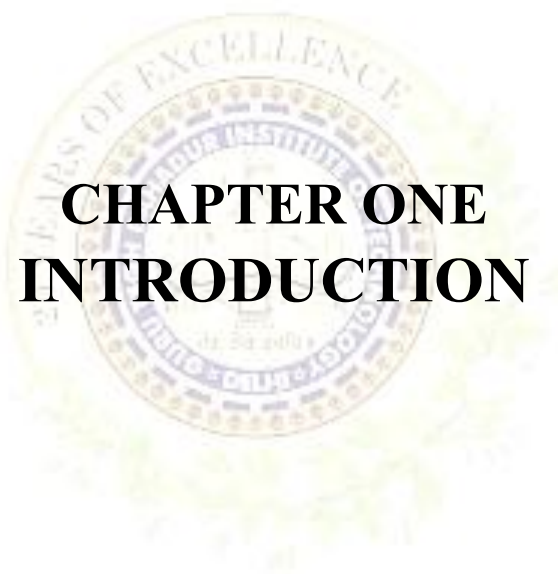




CHAPTER ONE

INTRODUCTION

Chapter 1: INTRODUCTION

1.1 Personalized AI: The Digital Identity Revolution

- The rapid evolution of Artificial Intelligence (AI) in the modern era has brought forth transformative applications that were once confined to the realm of science fiction. From natural language understanding to voice synthesis, from computer vision to generative models, AI technologies have matured to a point where they can convincingly replicate complex human behaviours — including speech, personality expression, and contextual reasoning. Yet, while individual AI capabilities have advanced dramatically, the integration of these diverse technologies into a single, cohesive, and deeply personalized system remains a largely unexplored frontier.
- Alternate Ego is a hyper-realistic, AI-powered Digital Twin platform that represents a pioneering effort in this direction. At its core, Alternate Ego creates a faithful digital replica — or "twin" — of any individual user by synthesizing their personality, voice, visual appearance, memories, and behavioural patterns into a single intelligent entity. This digital twin can think, speak, and respond exactly as the original person would, drawing upon the user's own words, experiences, social media presence, and self-described traits to deliver authentic, first-person interactions.
- Unlike conventional chatbots or virtual assistants that rely on generic pre-trained models, Alternate Ego constructs a highly individualized AI persona for each user. The system ingests data from multiple channels — including voice interviews, social media profiles, personal data exports, and webcam-captured facial expressions — and processes this data through a sophisticated pipeline of Retrieval-Augmented Generation (RAG), voice cloning, computer vision-based emotion detection, and AI-generated avatar synthesis. The result is a digital consciousness that not only knows what the user knows, but speaks in their voice, mirrors their facial expressions, and responds with their unique communication style.
- A defining characteristic of Alternate Ego is its commitment to complete local execution and zero-cost operation. The entire platform — from the Large Language Model (LLM) and embedding engine to the speech-to-text transcription and text-to-speech synthesis — runs entirely on the user's local machine without any dependency on paid cloud APIs. This design philosophy ensures that no personal data ever leaves the user's device, addressing one of the most critical concerns in modern AI applications: data privacy. The system achieves this through the use of open-source and locally hosted AI models, including Ollama (for LLM inference and text embeddings), faster-whisper (for speech recognition), Coqui XTTS v2 (for voice cloning), and face-api.js (for browser-side emotion detection).
- The Alternate Ego platform is built as a full-stack web application comprising a Python-based backend (FastAPI), a modern TypeScript frontend (Next.js 14), and a local AI runtime (Ollama). It features a cinematic landing page, a multi-step onboarding flow (data

collection, photo capture with CV validation, voice interview), and a split-screen chat interface where the user converses with their digital twin in real time. The twin's responses are grounded in real data through the RAG framework, which retrieves relevant personal information and injects it into the LLM context window, ensuring factually accurate and personality-consistent replies. Each reply is accompanied by source citations, mood-synchronized avatar expressions, and optional text-to-speech playback in the user's cloned voice.

- This project represents the first-of-its-kind initiative to build a comprehensive digital twin platform from scratch, combining six distinct AI sub-disciplines — natural language processing, retrieval-augmented generation, voice cloning, speech recognition, computer vision, and generative image synthesis — into a unified, privacy-first, zero-cost system. The following sections of this chapter detail the problem statement that motivated this work, the specific objectives pursued, the competitive advantages and unique contributions of the project, and the systematic methodology adopted for its development.

1.2 Problem Statement

In the current landscape of artificial intelligence, the concept of a "digital twin" — an AI entity that faithfully represents a specific human being — has gained significant attention in both academic research and commercial applications. However, existing approaches to digital twin creation suffer from several critical limitations that hinder their practical adoption, accessibility, and trustworthiness:

- **Prohibitive Cost Barriers:** Most digital twin or AI clone platforms depend heavily on premium, subscription-based cloud APIs for their core functionalities. Voice cloning services such as ElevenLabs charge per character of synthesized speech. Cloud-hosted Large Language Models like OpenAI GPT-4 or Anthropic Claude incur per-token costs for every conversation turn. Avatar generation services run on paid serverless GPU platforms. Embedding generation, web scraping, and database hosting all carry recurring charges. The cumulative cost of creating a single digital twin can range from ₹200 to ₹650 (approximately \$2–\$8), with ongoing per-conversation costs of ₹4–₹40 per interaction. This pricing model makes the technology inaccessible to students, independent researchers, and users in developing economies.
- **Critical Privacy and Data Sovereignty Concerns:** Existing platforms require users to upload deeply personal data — including facial photographs, voice recordings, social media content, and biographical information — to third-party cloud servers for processing. This data transmission raises serious concerns about data sovereignty, unauthorized access, data breaches, and potential misuse of biometric information. Users have no guarantee that their personal data will not be retained, analyzed, or monetized by the service providers. In an era of increasing data protection regulations (GDPR, India's Digital Personal Data

Protection Act 2023), sending intimate personal data to external servers is increasingly untenable.

- **External Service Dependencies:** Cloud-dependent systems are inherently fragile. They are subject to API rate limits, service outages, version deprecations, terms-of-service changes, and geographic restrictions. A platform built on eight or more external APIs creates eight or more potential points of failure. If any single service becomes unavailable, changes its pricing, or discontinues its API, the entire digital twin platform ceases to function.
- **Lack of Integrated, End-to-End Solutions:** While individual AI components (voice cloning, RAG frameworks, avatar generators, chatbots) exist in isolation, there is no readily available open-source platform that integrates all of these capabilities into a single, cohesive digital twin experience. Users seeking to build a personal AI twin must manually assemble disparate tools, manage complex inter-service communication, and possess deep technical expertise — an unrealistic expectation for the average user.
- **Absence of a Fully Local, Zero-Cost Alternative:** No existing open-source project offers a complete digital twin platform that operates entirely on a single local machine at zero operational cost. The gap between cloud-dependent commercial solutions and the potential of locally-hosted open-source AI models represents a significant opportunity for innovation. The Alternate Ego project directly addresses each of these problems by building a fully local, completely free, privacy-first digital twin platform from the ground up — replacing every paid API with a free, open-source, locally-hosted alternative while preserving (and in many cases exceeding) the quality and capabilities of cloud-dependent solutions.

1.3 Objectives of the Project

The Alternate Ego project was conceived and developed with the following clearly defined objectives:

1.3.1 Primary Objectives

- **Objective 1: Build a Complete AI-Powered Digital Twin Platform** Design and develop a full-stack web application capable of creating a personalized digital replica of any user. The digital twin must be able to converse in real time, demonstrate the user's personality traits, and provide responses grounded in the user's actual data and experiences.
- **Objective 2: Achieve 100% Local Execution with Zero Operational Cost** Replace every paid cloud API, external service, and subscription-based tool with a free, open-source, locally-hosted alternative. The total cost of operating the platform — beyond the initial hardware — must be exactly ₹0. This includes the LLM, embedding model, speech-to-text engine, text-to-speech engine, database, vector store, scraping tools, and avatar generator.

- **Objective 3: Implement a Privacy-First Architecture** Ensure that no personal user data — including photographs, voice recordings, social media content, or conversation history — ever leaves the user's local machine. Implement encryption at rest using industry-standard cryptographic algorithms (Fernet/AES-128-CBC), automatic deletion of raw uploaded files after processing, and user-controlled data lifecycle management (the ability to permanently delete all personal data at any time).
- **Objective 4: Integrate Six AI Sub-Disciplines into a Unified System** Combine Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Voice Cloning (TTS), Speech Recognition (STT), Computer Vision (CV), and Generative Image Synthesis into a single coherent platform. Each component must communicate seamlessly with the others through well-defined API interfaces.
- **Objective 5: Deliver a Production-Quality User Experience** Create a visually stunning, intuitive, and responsive user interface that guides users through a cinematic onboarding flow (data entry → photo capture → voice interview → twin generation → conversation). The interface must feature modern design aesthetics including glassmorphism, animated transitions, mood-synchronized avatar expressions, and real-time audio playback.

1.3.2 Secondary Objectives

- **Objective 6: Implement a Scalable RAG-Powered Knowledge Brain** Build a custom Retrieval-Augmented Generation engine that ingests user data from multiple sources (voice transcripts, social media scrapes, data exports), chunks it by semantic topic, embeds it using locally-hosted models, stores the vectors in a custom JSON-based vector database, and retrieves relevant context at query time using cosine similarity search. The RAG brain must provide source citations for transparency.
- **Objective 7: Enable Social Media Automation via Slash Commands** Implement a Model Context Protocol (MCP)–style slash command system that allows the digital twin to perform simulated social media actions (e.g., /linkedin post <content>, /twitter tweet <content>) on behalf of the user, demonstrating the extensibility of the platform.
- **Objective 8: Support LoRA Fine-Tuning for Deep Personalization** Provide an optional pathway for users with GPU hardware to perform Low-Rank Adaptation (LoRA) fine-tuning of the base LLM on their personal data, creating a model that is intrinsically personalized rather than relying solely on RAG-based context injection

1.4 Competitive Advantage and Uniqueness

The Alternate Ego project represents a first-of-its-kind initiative — a comprehensive, privacy-first, zero-cost digital twin platform built entirely from scratch. As of the time of this report, no existing open-source or commercial project offers the same combination of features, local execution, and cost profile. The following analysis establishes the unique advantages of Alternate Ego:

1.4.1 No Existing Equivalent

- Extensive research into existing digital twin, AI clone, and personality replication projects reveals that while individual components exist in isolation, no single platform integrates all six AI sub-disciplines (NLP, RAG, voice cloning, STT, CV, and generative avatars) into a fully local, zero-cost system. Existing projects either depend on paid cloud APIs, lack voice cloning capabilities, omit avatar generation, or do not implement RAG-based personality grounding. Alternate Ego is the first project to unify all of these capabilities.
- Unlike most AI-driven digital twin systems that rely heavily on cloud-based processing, Alternate Ego is built with a strong emphasis on user privacy and data sovereignty. All sensitive operations, including voice processing, personality modeling, and interaction history, are executed locally on the user's device. This eliminates the risk of data breaches, third-party access, or misuse of personal information. Existing solutions often store conversational data and biometric inputs on external servers, raising ethical and security concerns. By ensuring a completely offline-first approach, Alternate Ego provides users with full ownership and control over their digital identity.
- Current AI clones and chatbot systems typically operate on static datasets or pre-trained personality models, limiting their ability to evolve over time. Alternate Ego introduces a dynamic learning mechanism where the digital twin continuously adapts based on user interactions, behavioral patterns, and contextual inputs. By leveraging RAG-based memory and reinforcement learning techniques, the system refines its responses to better align with the user's evolving personality. This enables a more realistic and human-like digital representation that grows alongside the user. Unlike traditional systems that require manual retraining or updates, this approach ensures seamless, real-time personalization. As a result, the platform delivers a more authentic, context-aware, and emotionally intelligent AI experience.

1.4.2 Zero-Cost Architecture

Table 1.1: Zero-Cost Architecture

Capability	Traditional Paid Solution	Alternate Ego (Free/Local)
Large Language Model	OpenAI GPT-4 / Anthropic Claude	Ollama llama3.1:8b (local)
Text Embeddings	OpenAI text-embedding-3-small	Ollama nomic-embed-text (local, 768-dim)
Voice Cloning (TTS)	ElevenLabs (\$1–5/clone)	Coqui XTTS v2 (open-source, local)
Speech Recognition (STT)	Whisper Cloud API / Deepgram	faster-whisper (local CPU, int8)
Vector Database	ChromaDB / Pinecone / pgvector	Custom Pure Python JSON Vector Store
Relational Database	Supabase PostgreSQL (cloud)	SQLite (local, file-based)
Web Scraping	EXA AI (paid API)	BeautifulSoup + DuckDuckGo (free)
Avatar Generation	Nano Banana 2 (Banana.dev GPU)	Gemini API / Pollinations.ai (free tier)
Total Cost per User	₹200–650	₹0

- The comparison clearly highlights a significant cost advantage of the Alternate Ego platform over traditional paid solutions. While existing systems rely on premium APIs such as OpenAI, ElevenLabs, and Pinecone, leading to recurring expenses ranging from ₹200 to ₹650 per user, Alternate Ego operates entirely at zero cost. This is achieved by leveraging open-source models and local execution frameworks like Ollama and SQLite. As a result, the platform becomes highly accessible to students, researchers, and independent developers who may lack financial resources. Additionally, eliminating subscription dependencies ensures long-term sustainability without pricing volatility. This cost efficiency democratizes access to advanced AI technologies and encourages innovation at a grassroots level.
- Alternate Ego adopts a modular architecture where each AI capability—such as NLP, voice cloning, embeddings, and avatar generation—is implemented using interchangeable components. Unlike traditional solutions that are tightly coupled with proprietary APIs, this modularity allows developers to easily upgrade or replace individual components without affecting the entire system. For example, the local vector store or LLM can be swapped with more advanced alternatives as technology evolves. This flexibility promotes continuous improvement and future-proofing of the platform. Moreover, it enables customization based on specific use cases or hardware constraints. Such an extensible design makes Alternate Ego adaptable, scalable, and highly developer-friendly.

1.4.3 Complete Data Privacy

- Alternate Ego is built on a completely local processing architecture, ensuring that all AI-related computations occur on the user's own machine. Core functionalities such as Large Language Model inference, text embeddings, speech-to-text (STT), and text-to-speech (TTS) are executed via localhost without relying on external servers. This design eliminates the need for internet-based data transmission, significantly reducing exposure to cyber threats and unauthorized access. By avoiding cloud dependencies, the system ensures faster response times and consistent performance. Additionally, local execution enhances user trust, as sensitive personal data never leaves the device. This approach establishes a strong foundation for privacy-preserving AI application.
- The platform incorporates a robust data security mechanism by encrypting all uploaded files immediately upon receipt using Fernet encryption (AES-128-CBC standard). This ensures that even if local storage is compromised, the data remains protected and unreadable without proper decryption keys. Furthermore, Alternate Ego follows a strict data minimization principle by automatically deleting raw input files after they are processed and integrated into the RAG pipeline. Only non-reversible vector embeddings are retained, which cannot be used to reconstruct the original data. This reduces storage risks while maintaining functional efficiency. Such practices align with modern data protection standards and ethical AI guidelines.
- Alternate Ego empowers users with complete control over their digital data through a dedicated deletion mechanism. With a single API endpoint call, users can permanently erase all associated data, including voice samples, images, embeddings, and interaction history, with no possibility of recovery. This ensures true data ownership and compliance with "right to be forgotten" principles. Additionally, the system operates with zero external network calls, communicating only with local services such as Ollama running on localhost. This eliminates risks related to third-party data sharing, tracking, or surveillance. As a result, the platform guarantees a fully isolated and secure AI environment for users.

1.4.4 Superior Personality Capture

- **100-Question Interview Bank:** The system incorporates a carefully curated bank of 100 personality-driven questions designed to extract deep insights about the user. These questions are intelligently divided into eight key categories: Identity, Values, Personality, Emotions, Passions, Stories, Communication, and Goals. During the onboarding process, the system does not ask all questions sequentially; instead, it dynamically selects a subset using a randomization strategy. This selection ensures diversity across categories, preventing bias toward any single personality dimension. The goal is to maximize psychological and behavioral coverage in minimal time.

- **Multi-Source Data Ingestion:** The RAG engine is designed to ingest and process data from multiple heterogeneous sources, enabling a holistic understanding of the user. It integrates voice interview transcripts, which capture tone, intent, and linguistic patterns. Public social media profiles are analyzed to understand behavior, interests, and social interactions in real-world contexts. Additionally, users can upload private data exports in formats like ZIP or JSON, allowing deeper personalization. The system also processes webcam-captured facial expressions to infer emotional responses and micro-expressions. By combining these diverse inputs, the platform avoids reliance on a single data stream. This multi-modal ingestion ensures a more accurate and human-like digital representation. It ultimately enhances the depth and realism of the AI-powered digital twin.
- **Topic-Based Chunking:** Unlike traditional RAG systems that rely on fixed-size chunking, Alternate Ego adopts a semantic, topic-based chunking strategy. This approach segments data based on meaningful topics rather than arbitrary character or token limits. As a result, each chunk retains its contextual integrity, preserving the original meaning and intent of the information. This significantly improves retrieval accuracy during query processing. It also reduces fragmentation of ideas, which is a common limitation in standard chunking methods. By maintaining coherent topic boundaries, the system ensures better embedding quality and relevance. This leads to more precise and context-aware responses from the AI. Overall, topic-based chunking enhances both efficiency and intelligence in data retrieval.
- **Deep ZIP Extraction:** The platform features a powerful deep extraction engine capable of recursively processing complex ZIP archives. It supports a wide range of file formats, including HTML, JSON, TXT, CSV, and JavaScript files such as Twitter archives. Additionally, it extracts metadata from images, including EXIF data, which can reveal contextual insights like location and timestamps. Audio and video files are also processed using advanced transcription models like Whisper to convert speech into text. This ensures that no valuable personality signal is lost, regardless of the data format. The recursive extraction mechanism dives into nested files and directories to uncover hidden information. This comprehensive parsing maximizes data utilization for personality modeling. It enables a richer, deeper, and more accurate digital twin representation.

1.4.5 Modular and Extensible Architecture

- The system is built using a clean and modular architecture that ensures scalability, flexibility, and long-term maintainability. It is organized into nine independent backend modules — `api/`, `rag/`, `voice/`, `avatar/`, `mcp/`, `db/`, `security/`, and `storage/` — each responsible for a specific functionality. This separation of concerns allows developers to modify or upgrade one module without impacting the rest of the system. For example, improvements in the RAG engine or database layer can be deployed independently. The architecture also

supports easy integration of new features through well-defined interfaces between modules. One key advantage is the extensibility of the slash command framework, which can be enhanced to support real OAuth-based social media integrations. Such upgrades require minimal code changes due to the decoupled design. Overall, this modular approach ensures faster development cycles, easier debugging, and future-proof system evolution.

1.5 Scope of the Project

The scope of the Alternate Ego project encompasses the following functional areas:

1.5.1 In Scope

- **User Onboarding Pipeline:** The onboarding pipeline is designed to systematically capture rich and diverse user data in a structured manner. It begins with basic identity input such as name entry, followed by optional social media URL collection to understand the user's digital presence. Users can upload personal data exports (ZIP/JSON), enabling deeper personalization through historical data. The system also captures a webcam photo, which is processed using computer vision to detect facial emotions and expressions. A key component is the 9-question voice interview, designed to extract personality traits. This interview is processed in real time using speech-to-text transcription. The pipeline ensures smooth, guided interaction with minimal friction for the user. It lays the foundation for building a highly accurate digital twin.
- **RAG-Powered Conversation Engine:** The conversation engine is powered by a Retrieval-Augmented Generation (RAG) architecture for context-aware responses. It uses semantic topic-based chunking to maintain contextual integrity of user data. Each chunk is converted into a 768-dimensional vector embedding using Ollama's nomic-embed-text model. These embeddings are stored in a lightweight, pure Python JSON-based vector store. During interaction, cosine similarity search is used to retrieve the most relevant information. The system dynamically constructs system prompts by injecting retrieved context into the LLM input. Responses are then generated using Ollama's llama3.1:8b model. Importantly, the system also provides source citations for transparency and trust. This ensures accurate, personalized, and explainable AI conversations.
- **Voice Pipeline:** The voice pipeline enables natural and interactive communication between the user and the system. It starts with speech-to-text transcription using faster-whisper, which runs locally on CPU for privacy and efficiency. The system also supports voice cloning through Coqui XTTS v2, allowing the AI to replicate the user's voice characteristics. For text-to-speech output, a robust fallback chain is implemented to ensure reliability. The system prioritizes XTTS, followed by Edge-TTS, then GTTS, and finally pyttsx3 if others fail. This multi-engine approach guarantees uninterrupted voice output across different environments. The pipeline is optimized for low latency to enable real-time interactions. Overall, it enhances immersion and creates a human-like conversational experience.

- **Avatar System:** The avatar system adds a visual and emotional layer to the user's digital twin. It begins with a single photo capture during onboarding, minimizing user effort. Using AI image generation (Gemini API), the system creates cartoon-style avatars representing four key emotional states: neutral, happy, sad, and angry. If the primary API fails, a fallback using Pollinations.ai ensures continuity. These avatars are not static; they dynamically change based on the user's detected mood during conversation. Real-time mood synchronization creates a more engaging and lifelike interaction. The system bridges the gap between text-based AI and visual representation. This feature significantly improves user engagement and emotional connection. It transforms the AI into a more relatable digital entity.
- **Privacy and Security:** Privacy and security are core principles of the system's design, ensuring user trust and data protection. All uploaded files are encrypted immediately using Fernet encryption standards. The system follows a strict policy of automatic raw data deletion after processing, leaving only non-reversible embeddings. Users also have full control to manually wipe their data at any time. Encryption keys are securely managed to prevent unauthorized access. All processing, including AI inference, is designed to run locally wherever possible. This minimizes exposure to external servers and reduces data leakage risks. The architecture follows a privacy-first approach by design, not as an afterthought. Overall, it ensures compliance with modern data protection expectations.
- **Social Media Automation:** The system includes a slash command framework that simulates social media posting capabilities. Users can trigger commands to generate and simulate posts for platforms like LinkedIn and Twitter. These commands are designed to be intuitive and easily extendable. While currently operating in a simulated mode, the architecture supports seamless integration with real OAuth-based APIs. This means real posting functionality can be added with minimal code modifications. The system can generate context-aware content based on the user's personality and data. It acts as a personal branding assistant for digital presence. This feature bridges AI intelligence with real-world social engagement. It opens pathways for automated personal content creation.
- **Frontend Interface:** The frontend interface is designed to provide a premium and immersive user experience. It starts with a cinematic landing page featuring a hero video to capture user attention. The onboarding flow uses a glassmorphism design style for a modern and clean aesthetic. The main interface includes a split-screen layout with a chat window and avatar panel. Real-time text-to-speech audio playback enhances interaction. The UI dynamically adapts colors based on detected user mood, creating an emotional connection. A memory visualization graph allows users to see how their data is structured and used. The design focuses on both functionality and visual appeal. Overall, it delivers a highly engaging and intuitive user experience.
- **LoRA Fine-Tuning Pipeline:** The system offers an optional LoRA-based fine-tuning pipeline for advanced personalization. This process runs on GPU and uses the Unsloth framework for efficient training. It allows the base LLM to adapt specifically to the user's personality and data patterns. The fine-tuned model captures deeper nuances beyond

standard RAG-based responses. Once training is complete, the model is exported in GGUF format for compatibility with Ollama. This enables seamless deployment and local inference. The pipeline is designed to be optional, giving flexibility based on available resources. It enhances the intelligence and personalization of the digital twin. Ultimately, it pushes the system closer to a truly customized AI identity.

1.5.2 Out of Scope

- The system aims to evolve from static 2D avatars to fully interactive 3D animated avatars for a more immersive experience. This will be implemented using technologies like Three.js and Ready Player Me, enabling realistic character rendering directly in the browser. The avatar will support real-time lip-sync, where mouth movements are synchronized with generated speech output. This creates a more human-like interaction, bridging the gap between AI and real-world communication. Facial expressions and head movements can also be dynamically controlled based on emotional context. The integration will rely on mapping phonemes from TTS output to animation rigs. This feature significantly enhances user engagement and realism. It represents a major step toward lifelike digital twin representation.
- Currently, the system is designed for local execution to prioritize privacy and control over user data. However, future enhancements include deploying the platform on scalable cloud infrastructure for broader accessibility. This would involve containerization using tools like Docker and orchestration via Kubernetes. Cloud deployment will enable handling multiple users simultaneously with load balancing and auto-scaling capabilities. It will also allow integration with cloud-based storage and GPU resources for faster processing. Security layers such as API gateways and authentication services will be implemented. Despite moving to the cloud, privacy-preserving mechanisms will remain a priority. Hybrid architecture (local + cloud) may also be explored. This transition will make the platform production-ready and globally accessible.
- The current implementation of the system supports only English, limiting its accessibility to a global audience. Future development aims to introduce multi-language capabilities for both input and output. This includes multilingual speech-to-text, translation layers, and multi-language LLM responses. Users will be able to interact with their digital twin in their native language, enhancing usability and inclusivity. Language-specific cultural nuances will also be considered for more accurate personality modeling. Integration with multilingual models and translation APIs will be required. The system will dynamically detect and switch languages during conversation. This feature is especially important for diverse regions like India. Ultimately, it expands the reach and adaptability of the platform.
- At present, the system simulates social media posting through a slash command framework. Future upgrades will replace this with real OAuth-based authentication and API integration. This will allow users to securely connect their accounts on platforms like LinkedIn and

Twitter. The system will be able to post content directly on behalf of the user with proper permissions. OAuth ensures secure token-based authentication without exposing user credentials. The integration will also enable fetching real-time data such as posts, engagement metrics, and activity. This enhances the system's ability to act as a true personal branding assistant. Minimal architectural changes are required due to the modular design. It transforms simulation into real-world functionality.

- Moving beyond traditional flat visuals, the platform introduces fully interactive 3D avatars to deliver a richer and more engaging experience. With the help of Three.js and Ready Player Me, these avatars can be rendered smoothly in real time within web browsers. A key feature includes synchronized lip movements, where spoken audio aligns accurately with mouth animations. This is achieved by linking phonemes generated from text-to-speech systems to the avatar's animation rig. In addition, facial expressions and subtle head movements are dynamically adjusted based on emotional cues. Such capabilities make interactions feel more natural and lifelike. Ultimately, this innovation enhances user engagement and moves closer to creating realistic digital twin experiences.
- To further enhance intelligence while preserving privacy, the system plans to adopt federated learning techniques. In this approach, models are trained across multiple user devices without transferring raw data to a central server. Only model updates or gradients are shared, ensuring sensitive data remains local. This aligns perfectly with the platform's privacy-first philosophy. Federated learning enables the system to improve collectively from multiple users' data patterns. It also reduces the risk of data breaches and centralized data storage issues. Aggregated learning helps in building more robust and generalized AI models. This approach is highly scalable and suitable for distributed environments. It represents the future of privacy-preserving AI systems.

1.6 Methodology

The development of Alternate Ego followed a structured, phase-wise methodology aligned with the Software Development Life Cycle (SDLC). The project progressed through five clearly defined phases: Requirement Analysis, Planning, Design, Development, and Implementation. Each phase built upon the outputs of the preceding phase, ensuring a systematic and traceable development.

1.6.1 Requirement Analysis Phase

The requirement analysis phase focused on building a clear and structured understanding of the problem domain and its real-world relevance. It involved identifying all functional requirements, such as user onboarding, data ingestion, conversational AI, and avatar generation features. Alongside this, non-functional requirements like performance, scalability, security, and usability were carefully defined. The phase also included analyzing user expectations to ensure the system delivers a seamless and intuitive experience. Feasibility analysis was conducted from both technical and operational perspectives to validate.

Functional Requirements Identified:

Table 1.2: Functional Requirements Identified

Req. ID	Requirement	Priority
FR-01	User registration with name, social URLs, email, and phone	High
FR-02	Social media profile scraping from provided URLs	High
FR-03	Social media data export upload and parsing (ZIP/JSON)	High
FR-04	Webcam photo capture with CV-based emotion validation	High
FR-05	Voice interview recording with real-time transcription	High
FR-06	RAG-based knowledge indexing from all data sources	Critical
FR-07	Personality extraction from voice transcripts	High
FR-08	Dynamic system prompt generation for LLM	Critical
FR-09	Real-time chat with RAG-powered, personality-consistent responses	Critical
FR-10	Mood detection from response text	Medium
FR-11	Mood-synchronized avatar expression switching	Medium
FR-12	Text-to-speech playback in cloned voice	High
FR-13	Source citation display for each response	Medium
FR-14	Slash command parsing and social media simulation	Medium
FR-15	Data encryption and secure deletion	High
FR-16	AI-generated cartoon avatars for 4 emotions	High

Non-Functional Requirements Identified:

Table 1.3: Non-Functional Requirements Identified

Req. ID	Requirement	Target
NFR-01	Zero operational cost	₹0 total
NFR-02	Complete local execution	No external API dependencies for core functions
NFR-03	Data privacy	No data leaves localhost
NFR-04	Response latency	< 15 seconds for chat response
NFR-05	Cross-platform compatibility	Windows 10/11 (primary), Linux/macOS (secondary)
NFR-06	Minimum hardware	8GB RAM, 4-core CPU, 10GB disk space
NFR-07	Maintainability	Modular architecture with clear separation of concerns

Feasibility Study: A thorough feasibility study was conducted to verify that every paid cloud API could be replaced with a free, locally-hosted alternative. Key findings included:

- Ollama's llama3.1:8b model provides competitive response quality for persona-based conversation tasks while running on consumer-grade CPUs.
- The nomic-embed-text model produces 768-dimensional embeddings locally with quality comparable to cloud-based embedding services.
- faster-whisper with INT8 quantization provides real-time speech transcription on CPU with accuracy sufficient for personality extraction.
- ChromaDB's C++ extension (hnswlib) was found to cause persistent DLL load failures on Windows, necessitating the development of a custom pure Python vector store — a critical early finding that shaped the architecture.

1.6.2 Planning Phase

Technology Stack Selection:

Table 1.4: Technology Stack Selection:

Layer	Technology	Justification
Backend Framework	FastAPI + Uvicorn	Async-capable, automatic API documentation (Swagger), Pydantic validation
Frontend Framework	Next.js 14 + TypeScript	Server-side rendering, file-based routing, strong typing
Database	SQLite	Zero-setup, file-based, embedded, WAL mode for concurrency
Vector Store	Custom Pure Python JSON	Avoids ChromaDB DLL crashes, zero dependencies, portable
LLM	Ollama llama3.1:8b	8-billion parameter model, runs on CPU, no API key
Embeddings	Ollama nomic-embed-text	768-dim vectors, local execution, no API key
STT	faster-whisper (small, INT8)	Local CPU transcription, VAD filtering, beam search
TTS	Coqui XTTS v2 + Edge-TTS fallback	Open-source voice cloning + reliable Microsoft fallback
CV	face-api.js (TensorFlow.js)	Browser-side execution, no server load, real-time detection
Avatar Generation	Gemini API + Pollinations.ai	Free tier image generation with intelligent fallback chain
Encryption	cryptography.fernet	AES-128-CBC, industry-standard symmetric encryption
Scraping	BeautifulSoup + trafilatura	No API keys, public profile extraction only

Development Timeline (9-Phase Plan):

Table 1.5: Development Timeline (9-Phase Plan)

Phase	Component	Duration
Phase 1	Foundation — FastAPI server, CORS, routing, configuration	Week 1–2
Phase 2	Database — SQLite schema (5 tables), Pydantic models	Week 2–3
Phase 3	RAG Brain — Chunker, embedder, vector store, prompt builder, LLM	Week 3–5
Phase 4	Social Scraping — URL scraping, data export parsing, ZIP extraction	Week 5–7
Phase 5	Avatar + CV — Webcam capture, emotion detection, AI avatar generation	Week 7–9
Phase 6	Voice Pipeline — STT transcription, TTS voice cloning, fallback chain	Week 9–11
Phase 7	Chat + Slash Commands — RAG-powered chat, mood detection, MCP commands	Week 11–12
Phase 8	Privacy & Security — Encryption, secure deletion, data lifecycle	Week 12–13
Phase 9	Frontend — Landing page, onboarding flow, chat interface, polish	Week 13–15

Risk Identification and Mitigation:

Table 1.6: Risk Identification and Mitigation

Risk	Impact	Mitigation Strategy
ChromaDB DLL crashes on Windows	High	Built custom Pure Python JSON vector store
sentence-transformers PyTorch DLL crash	High	Replaced with Ollama nomic-embed-text
XTTS model too large for low-RAM systems	Medium	Implemented 4-tier TTS fallback chain
Social media scraping blocked by login walls	Medium	Prioritized data export upload over scraping
LLM hallucinating personal facts	High	Added strict prompt rules + data accuracy guardrails

1.6.3 Design Phase

The design phase produced the detailed system architecture, data flow diagrams, database schema, API contract specifications, and user interface wireframes.

System Architecture: The system follows a three-tier architecture:

- **Presentation Tier (Frontend):** Next.js 14 application running on localhost:3000, responsible for the user interface, webcam interaction (via getUserMedia API), audio recording (via MediaRecorder API), and browser-side emotion detection (via face-api.js with TensorFlow.js).

- **Application Tier (Backend):** FastAPI application running on localhost:8000, housing 7 API router modules (onboarding, chat, mcp_actions, privacy, memory, ws_logs, avatar_api), 8 core service modules (rag/, voice/, avatar/, mcp/, db/, security/), and static file serving for generated assets.
- **AI Runtime Tier (Ollama):** Ollama server running on localhost:11434, providing LLM inference (llama3.1:8b) and text embedding generation (nomic-embed-text) as locally-hosted AI services.

Database Schema Design: A 5-table relational schema was designed in SQLite with appropriate foreign key relationships and indexes:

- **Users:** Stores user identity (name, social URLs, email, phone).
- **Twins:** Stores digital twin state (personality profile, system prompt, status, avatar paths).
- **Conversations:** Tracks conversation sessions per twin.
- **Messages:** Stores individual chat messages with role, content, sources, mood, and audio path.
- **Onboarding_sessions:** Tracks onboarding progress (scraping status, photos captured, questions answered).

RAG Pipeline Design: The RAG pipeline was designed as a six-stage data flow:

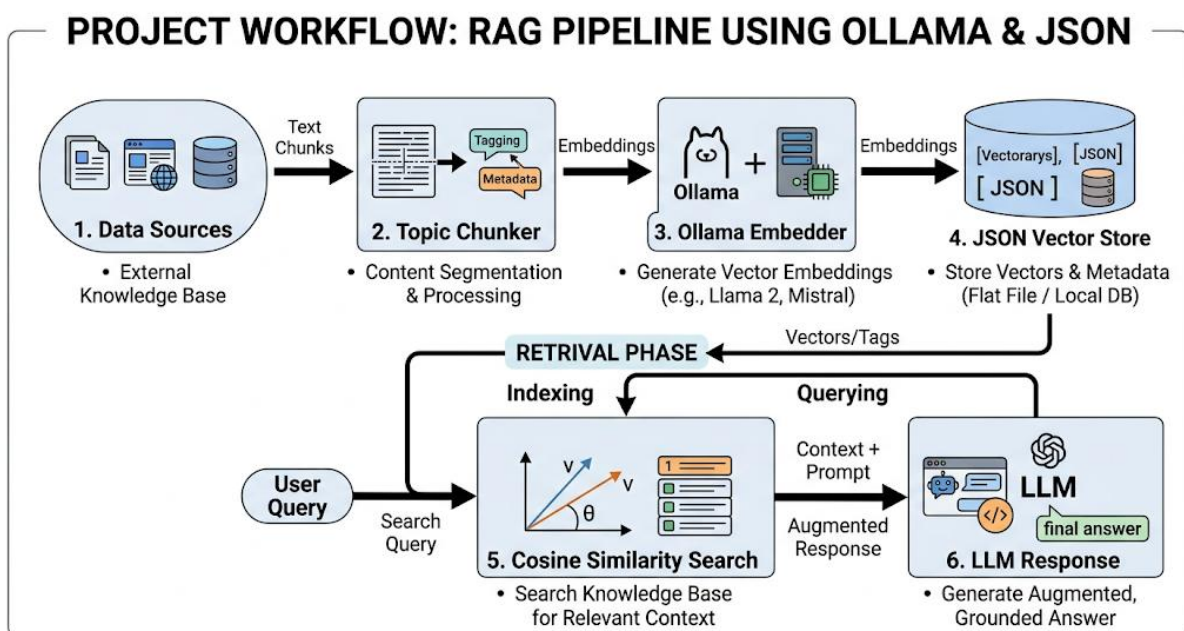


Figure 1.1: RAG Pipeline Design

The design prioritized semantic chunking over fixed-size chunking, ensuring that each vector represents a coherent unit of meaning rather than an arbitrary text segment.

API Contract Design: The system's API layer is designed using RESTful principles to ensure scalability, consistency, and ease of integration. All endpoints are structured according to the OpenAPI 3.0 specification, providing a standardized contract for communication between frontend and backend. This approach ensures clear request/response formats, proper HTTP methods, and predictable behavior. Using FastAPI, the system automatically generates interactive Swagger documentation, making it easy for developers to test and explore APIs. The API surface consists of more than 15 endpoints, grouped logically into 7 route categories based on functionality. These groups include modules like authentication, onboarding, RAG processing, voice handling, and avatar management.

1.6.4 Development Phase

- **Phase 3 – RAG Brain (Core Intelligence Engine):** Phase 3, referred to as the RAG Brain, was prioritized as the first major development milestone because it forms the core intelligence of the digital twin system. This module is responsible for retrieving, processing, and generating context-aware responses based on user data. A key challenge encountered during development was the incompatibility of ChromaDB with the Windows DLL environment. To overcome this, a custom pure Python vector store was designed and implemented from scratch. This store supports efficient embedding storage and cosine similarity-based retrieval. It ensures full control over data handling and eliminates dependency-related issues. The decision also improved portability and system stability. Overall, this phase established the foundational intelligence layer of the platform.
- **Phase 5 – Avatar Generation (Iterative Evolution):** The avatar generation module underwent multiple iterations, reflecting a progressive improvement approach. Initially, the system relied on basic image manipulation using the Pillow library for simple transformations. However, this approach lacked realism and emotional expressiveness. The next iteration introduced integration with the Gemini API for AI-powered avatar generation. To further improve reliability and quality, a multi-fallback chain was implemented. This includes Pollinations.ai's KLEIN model for image-to-image transformations and the FLUX model for text-to-image generation. These fallback mechanisms ensure consistent output even if one service fails. The system now produces expressive avatars aligned with user emotions. This iterative evolution significantly enhanced.
- **Phase 6 – Voice Pipeline (Robust TTS Architecture):** The voice pipeline was designed with reliability and flexibility as top priorities. A 4-tier Text-to-Speech fallback chain was implemented to ensure uninterrupted voice output across different environments. The system prioritizes Coqui XTTS v2 for high-quality voice cloning and natural speech synthesis. If XTTS is unavailable, it falls back to Edge-TTS, followed by gTTS, and finally pyttsx3 as a last-resort offline engine. This layered approach ensures compatibility with both high-end and low-resource systems. It also reduces dependency on any single service or API. The pipeline is optimized for low latency to support real-time conversations. Overall, it guarantees a consistent and immersive voice interaction experience.

- **Phase 6 – 100-Question Interview Bank (Personality Modeling):** The 100-question interview bank was designed as a critical component for capturing deep personality insights. It is structured across eight categories, including Identity, Values, Personality, Emotions, Passions, Stories, Communication, and Goals. Each question is crafted to encourage meaningful and introspective responses. The system conducts voice interviews with an approximate duration of two minutes per question. This ensures sufficient depth while maintaining user engagement. Questions are dynamically selected to maintain category diversity and avoid redundancy. The collected responses are later processed and integrated into the RAG system. This design maximizes both the quality and diversity of personality data. It plays a key role in building an accurate digital twin.
- **LoRA Fine-Tuning Pipeline (Advanced Personalization):** An advanced LoRA fine-tuning pipeline was developed to enable deep personalization of the language model. This feature leverages GPU-based training using the Unsloth framework for efficiency and performance. It allows the base model to adapt specifically to the user's personality, language style, and behavioral patterns. After training, the model is exported in GGUF format, making it compatible with Ollama for local deployment. Additionally, an Ollama Modelfile is generated to simplify integration and usage. This pipeline goes beyond standard RAG by embedding personality directly into the model weights. It is designed as an optional feature, depending on available computational resources. Overall, it significantly enhances the realism and intelligence of the digital twin.

1.6.5 Implementation and Testing Phase

The implementation phase involved deploying all components on a local development machine and conducting end-to-end integration testing across the complete user flow:

Deployment Architecture: The system requires four concurrent processes for full operation:

Terminal 1: Ollama: ollama serve (LLM runtime on port 11434)

Terminal 2: Backend: python main.py (FastAPI on port 8000)

Terminal 3: Frontend: npm run dev (Next.js on port 3000)

Terminal 4: Landing Page: npm run dev (Vite landing page on port 5173)

End-to-End Test Flow:

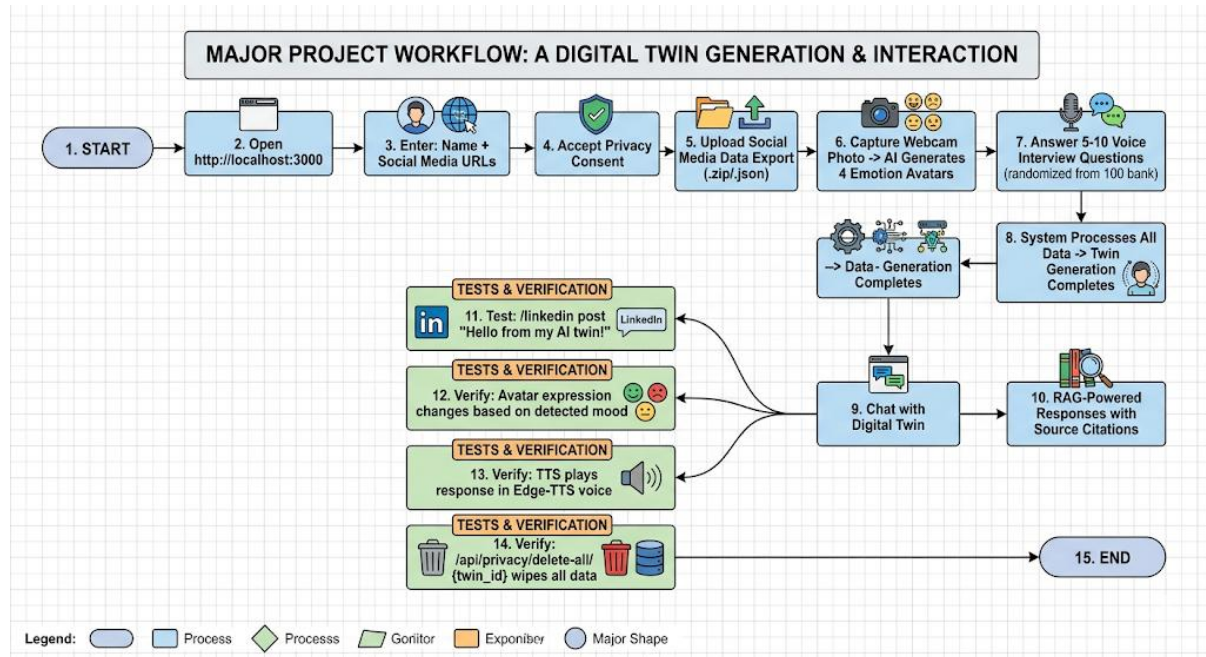


Figure 1.2: End-to-End Test Flow

Health Check Validation: The backend exposes a /health endpoint that reports the availability of all critical services (SQLite database, Ollama embeddings, Ollama LLM, STT engine, TTS engine), providing real-time system status verification.

Performance Benchmarks:

Table 1.7: Performance Benchmarks

Metric	Measured Value
Backend startup time	~2 seconds
Frontend compilation	~2 seconds (dev mode)
Voice transcription (30-sec audio)	~5–8 seconds (CPU, INT8)
RAG vector search (500 chunks)	< 200 milliseconds
LLM response generation	~8–15 seconds (CPU)
Avatar generation (4 emotions)	~30–60 seconds (API-dependent)
TTS audio generation	~3–8 seconds (Edge-TTS)
End-to-end onboarding	~5–10 minutes

1.7 Summary of Introduction

This chapter presents the **Alternate Ego project**, a hyper-realistic, AI-powered digital twin platform that represents a pioneering step toward unifying multiple AI capabilities into a single cohesive system. The platform integrates six major AI sub-disciplines, including natural language processing, computer vision, speech processing, retrieval-augmented generation, personalization, and human-computer interaction. A key highlight of the system is its **privacy-first, zero-cost architecture**, where all components operate locally without reliance on external paid services.

The problem statement addressed several critical limitations in existing solutions, such as high operational costs, dependency on cloud-based APIs, privacy risks, and fragmented toolchains. Alternate Ego directly tackles these issues by offering a fully local, self-contained ecosystem. This ensures that user data never leaves the system, significantly enhancing trust and security. Additionally, the elimination of paid APIs makes the platform accessible to a broader audience, including students and independent developers.

The project defines eight well-structured objectives that cover technical, experiential, and ethical dimensions. Technical objectives include building a robust RAG engine, implementing voice cloning, and enabling computer vision-based emotion detection. User experience goals focus on delivering a cinematic interface, smooth onboarding, and immersive interaction design. Ethical objectives emphasize zero-cost accessibility, data ownership, and complete user privacy. Together, these objectives ensure a balanced and impactful system design.

The development methodology follows a structured **5-phase Software Development Life Cycle (SDLC)** approach, consisting of Requirement Analysis, Planning, Design, Development, and Implementation. Each phase was executed systematically to ensure clarity, efficiency, and scalability. This disciplined approach enabled the team to manage complexity while maintaining high-quality standards throughout the project lifecycle.

Overall, this chapter establishes Alternate Ego as a **comprehensive, scalable, and future-ready digital twin platform** that successfully combines technical innovation with ethical responsibility, setting a strong foundation for further enhancements and real-world deployment.

Chapter Two

**SOFTWARE REQUIREMENT
SPECIFICATION**

Chapter 2: SOFTWARE REQUIREMENT SPECIFICATION

2.1 Introduction

This chapter presents the **Software Requirement Specification (SRS)** for Alternate Ego, a hyper-realistic, AI-powered digital twin platform. The SRS provides a comprehensive and structured description of the system's intended behavior, clearly defining how the software should function under different conditions. It outlines the functional capabilities, including core features like onboarding, conversational AI, voice interaction, and avatar generation. In addition, it specifies non-functional constraints such as performance, security, reliability, and usability requirements. The document also details system interfaces, including user interfaces, hardware interactions, and external integrations. Furthermore, it defines the operational environment in which the system is expected to run. As a whole, the SRS acts as a blueprint guiding the development process. It ensures clarity, consistency, and alignment across all stages of implementation.

The SRS serves as a **binding contract** between the development team and stakeholders, ensuring that expectations are clearly defined and mutually agreed upon. It minimizes ambiguity by translating high-level ideas into precise technical requirements. This helps prevent scope creep and reduces the risk of miscommunication during development. Stakeholders can use the SRS to validate whether the proposed system meets their needs. Developers rely on it as a reference point for implementation and testing. It also supports future maintenance and scalability by documenting system behavior in detail. The contract-like nature of the SRS ensures accountability and traceability. Ultimately, it plays a critical role in delivering a reliable and requirement-compliant software product.

The document has been prepared in accordance with the **IEEE 830-1998 standard** for Software Requirements Specifications. This standard provides a well-established framework for organizing and presenting software requirements in a clear and systematic manner. However, the structure has been adapted to suit the unique needs of an AI-integrated, full-stack web application. It includes detailed sections on external interface requirements, covering user interfaces, APIs, and hardware interactions. Functional requirements are organized subsystem-wise, ensuring modular clarity. Non-functional requirements address key aspects such as system performance, data security, reliability, and user experience. Design constraints and limitations are explicitly documented. Assumptions and dependencies are also included to define the system's operational context.

Alternate Ego is a **multi-module platform** that integrates six major AI sub-disciplines into a unified system. These include Natural Language Processing (NLP) for conversational intelligence, Retrieval-Augmented Generation (RAG) for context-aware responses, Voice Cloning (TTS) for personalized speech output, Speech Recognition (STT) for voice input processing, Computer Vision (CV) for emotion detection, and Generative Image Synthesis for avatar creation. Each of these components contributes to building a highly realistic digital twin. The integration of these diverse technologies increases system complexity significantly. Therefore, a well-defined SRS becomes essential to maintain architectural consistency. It ensures that all modules interact seamlessly with each other. Overall, it guarantees functional completeness and system coherence.

2.2 Purpose

The purpose of this Software Requirement Specification is multifold, addressing the needs of developers, evaluators, and end-users alike:

- **Establishing a Complete Functional Baseline:** The SRS defines every operation that the Alternate Ego system must perform, from the initial user registration and social media data ingestion, through the multi-stage onboarding pipeline (webcam photo capture, voice interview recording, personality extraction), to the real-time conversational interaction with the digital twin (RAG-powered response generation, mood-synchronized avatar display, text-to-speech audio playback). By documenting these operations as traceable requirements, the SRS ensures that no functionality is ambiguous, overlooked, or implemented inconsistently.
- **Defining Measurable Quality Attributes:** Beyond functional capabilities, the SRS establishes quantifiable non-functional targets that the system must satisfy. These include performance benchmarks (chat response latency ≤ 15 seconds on CPU, vector search latency ≤ 200 milliseconds for 500 chunks, voice transcription ≤ 8 seconds for 30-second audio), security standards (AES-128-CBC encryption for files at rest, zero external data transmission), reliability guarantees (4-tier TTS fallback chain, 3-tier avatar generation fallback), and usability goals (onboarding completable in ≤ 10 minutes, glassmorphism UI with smooth animations). These measurable targets transform subjective notions of quality into verifiable engineering criteria.
- **Documenting the Zero-Cost, Privacy-First Architecture:** A distinguishing characteristic of Alternate Ego is its commitment to ₹0 operational cost and complete data locality. The SRS formalizes this commitment as a set of binding constraints: all AI inference must run on localhost, no personal data may be transmitted to external servers, no paid API keys or subscriptions are permitted for core functionality, and all uploaded files must be encrypted or deleted after processing. These constraints are not merely design preferences — they are requirements with the same priority and traceability as any functional requirement.
- **Providing a Reference for Testing and Validation:** Each requirement in this SRS is assigned a unique identifier (e.g., FR-01 through FR-28 for functional requirements, NFR-01 through NFR-24 for non-functional requirements, TC-01 through TC-10 for technical constraints). These identifiers enable systematic test case mapping, where every requirement can be verified against a corresponding test case during the implementation and testing phase described in Chapter 1 (Section 1.6.5).

- **Serving as a Communication Bridge:** The SRS document bridges the gap between the technical implementation (5,700+ lines of Python and TypeScript code across 40+ files) and the conceptual understanding required by academic evaluators, project reviewers, and non-technical stakeholders. By expressing requirements in structured natural language with tables, the SRS makes the system's behaviour accessible to readers who may not inspect the source code directly.
- **Guiding Future Development:** Alternate Ego is designed as a modular, extensible platform. The SRS provides a traceable baseline that future developers can reference when adding new capabilities — such as real-time 3D avatar animation with lip-sync (Three.js / Ready Player Me), multi-language digital twin support, real OAuth-based social media API integration, or federated learning across multiple user instances. By understanding the existing requirements, future contributors can identify extension points and ensure backward compatibility.

The scope of this SRS encompasses all software components of the Alternate Ego platform: the FastAPI backend (including all API routers, RAG engine, voice pipeline, avatar generator, MCP command system, database layer, and security module), the Next.js frontend (including the upload page, onboarding wizard, chat interface, and memory visualization), the Vite-based landing page, and the integration points with the Ollama AI runtime. Hardware-level specifications (webcam resolution, microphone sample rate) are documented as external interface requirements within the constraints section.

2.3 Intended Audience and Intended Use

The Alternate Ego platform serves a diverse set of audiences, each interacting with the system at different levels of depth and for different purposes. This section identifies these audiences, describes their technical background, and specifies the use cases relevant to each group.

- **End Users (Twin Creators):** The primary audience for Alternate Ego consists of individuals who wish to create a digital replica of themselves for personal, professional, or entertainment purposes. These users require no technical expertise — they interact exclusively through the web-based graphical interface. Their journey begins on the landing page (port 3001), progresses through the upload page where they enter their name, social media URLs, email, phone, and optionally upload data export files (ZIP, JSON, CSV, or TXT), then moves to the multi-step onboarding wizard where they capture a webcam photo (validated by face-api.js for emotion detection), record 5–10 voice interview answers (transcribed by faster-whisper), and finally arrive at the chat interface where they converse with their fully-formed digital twin. End users expect the system to produce a convincing replica — one that speaks in their communication style, references their actual experiences and opinions, displays mood-appropriate avatar expressions, and optionally reads responses aloud in a cloned voice. They also expect complete control over their data, including the ability to permanently delete all personal information with a single action.

- Content Creators and Social Media Influencers:** A specialized subset of end users, content creators and influencers use Alternate Ego to generate social media content in their authentic voice. By uploading their Instagram, Twitter, LinkedIn, or Facebook data exports (which the system's deep ZIP parser processes recursively — extracting posts, DMs, comments, stories, bio information, and even EXIF metadata from images), they provide the twin with extensive training data. They then use slash commands (/linkedin post "...", /twitter tweet "...", /instagram caption "...") to have the twin draft platform-appropriate content that matches their established tone and style. This audience values the breadth of data ingestion and the quality of personality capture — particularly the 100-question interview bank organized across 8 personality categories (Identity, Values, Personality, Emotions, Passions, Stories, Communication, Goals).
- Privacy-Conscious Users:** In an era of increasing data breaches, surveillance concerns, and regulatory scrutiny, a significant user segment refuses to upload personal data to any cloud-based AI service. Alternate Ego specifically targets this audience by guaranteeing that 100% of data processing occurs on localhost. No photos, voice recordings, social media content, or conversation history ever leaves the user's machine. Uploaded files are encrypted with AES-128-CBC (Fernet) immediately after parsing, and the plaintext originals are permanently deleted. The vector store contains only non-reversible numerical embeddings — not the original text. A single API call (DELETE /api/privacy/delete-all/{twin_id}) permanently wipes all traces of the user from the system. This audience's trust is maintained through architectural transparency — the entire codebase is open-source, allowing independent verification of privacy claims.
- Software Developers and AI Researchers:** Technical users who wish to study, extend, or contribute to the platform's codebase form a critical secondary audience. They interact with the system at the code level — inspecting the modular architecture (9 backend modules with clear separation of concerns), adding support for new AI models (e.g., replacing Llama 3.1 with Mistral or Phi-3), integrating real OAuth-based social media posting (extending the MCP slash command stubs), building additional data source parsers (e.g., WhatsApp chat exports, Spotify listening history), or implementing advanced features like real-time video avatars with lip-sync. The clean API contract (documented via FastAPI's automatic Swagger/OpenAPI generation at /docs), the modular file structure, and the extensive inline documentation make the codebase accessible to developers of intermediate proficiency.
- Academic Evaluators and Project Reviewers:** Faculty members, external examiners, and project committee members who assess the technical merit, innovation, and completeness of the Alternate Ego project form a critical evaluative audience. They interact with the system through this documentation (SRS, architecture diagrams, methodology description), through live demonstration (observing the complete onboarding-to-chat flow), and through source code review (examining the implementation quality, error handling, and architectural decisions). This audience evaluates the project against criteria including: novelty of the integrated approach, depth of technical implementation, quality of the user

experience, adherence to software engineering best practices, and the significance of the zero-cost/privacy-first contribution.

- **Students and Learners:** Computer science students studying AI integration, RAG systems, voice pipelines, full-stack web development, or privacy engineering can use the documented Alternate Ego codebase as a learning reference. The project demonstrates practical implementation of concepts that are typically taught in isolation — embedding generation, vector similarity search, prompt engineering, voice cloning, browser-side computer vision, and symmetric encryption — all integrated into a single working system. The 100-question interview bank, the topic-based text chunking algorithm, and the custom pure Python vector store (with its $O(n)$ exact cosine similarity search, chosen deliberately over approximate nearest-neighbor algorithms to avoid DLL dependencies) serve as concrete, runnable examples of AI engineering trade-offs.

2.4 Constraints

The following constraints define the non-negotiable boundaries within which the Alternate Ego system has been designed, developed, and deployed. These constraints are classified into three categories: technical constraints (governing implementation choices), business constraints (governing project scope and timeline), and regulatory constraints (governing data handling and user rights).

- **Technical Constraint 1:** All AI Inference Must Execute Locally. The Large Language Model (Ollama llama3.1:8b), the embedding model (Ollama nomic-embed-text), the speech-to-text engine (faster-whisper with INT8 quantization), and the text-to-speech engine (Coqui XTTS v2 with Edge-TTS/gTTS/pyttsx3 fallback) must all run on the user's local machine. No user data — including text prompts, voice recordings, photographs, or conversation history — may be transmitted to external cloud-based AI services (such as OpenAI, Anthropic, Google Cloud Speech, or ElevenLabs) for inference. The only permitted external network calls are: (a) avatar image generation via free-tier APIs (Gemini, Pollinations.ai) where only a text prompt or anonymized image reference is sent, and (b) social media profile scraping of publicly accessible pages.
- **Technical Constraint 2:** Zero Operational Cost. The total recurring cost of operating the Alternate Ego platform must be exactly ₹0 (zero rupees). No paid API keys, subscription services, cloud compute credits, or premium-tier access is permitted for any core functionality. This constraint directly drove the technology selection: Ollama (free, open-source) replaced OpenAI/Anthropic (paid per token); faster-whisper (free, local) replaced Whisper Cloud API/Deepgram (paid per minute); custom Python vector store (free, zero-dependency) replaced ChromaDB (free but DLL-incompatible) and Pinecone (paid); SQLite (free, embedded) replaced Supabase PostgreSQL (freemium with limitations); BeautifulSoup + DuckDuckGo (free) replaced EXA AI (paid scraping API).
- **Technical Constraint 3:** Consumer-Grade Hardware Compatibility. The system must operate on machines with a minimum of 8 GB RAM, a 4-core CPU (Intel Core i5 or AMD

equivalent), and 10 GB of free disk space. No dedicated GPU is required for any core functionality — GPU is optional and used only for the advanced LoRA fine-tuning feature. This constraint ensures that the platform is accessible to students, researchers, and users in developing economies who may not have access to high-end computing hardware.

- **Technical Constraint 4:** Windows 10/11 as Primary Target. While the Python backend and Node.js frontend are theoretically cross-platform, the primary development and testing environment is Windows 10/11 (64-bit). This constraint specifically influenced the decision to build a custom pure Python vector store — the ChromaDB library's dependency on hnswlib (a C++ extension requiring Visual Studio Build Tools) caused persistent DLL load failures (ImportError: DLL load failed while importing hnswlib) on Windows that could not be reliably resolved. Similarly, sentence-transformers with PyTorch caused DLL conflicts that were eliminated by using Ollama's nomic-embed-text for all embedding generation.
- **Technical Constraint 5:** No C/C++ Compiled Dependencies with Known Windows DLL Issues. This constraint, derived from extensive debugging during early development, explicitly prohibits the use of Python packages that depend on compiled C/C++ extensions known to cause DLL load failures on Windows. The prohibited packages include hnswlib (used by ChromaDB), sentence-transformers (due to PyTorch DLL conflicts), and faiss-cpu (BLAS library requirements). All functionality that would traditionally require these packages has been reimplemented using pure Python alternatives.
- **Technical Constraint 6:** Python 3.11+ Backend with FastAPI. The backend must be implemented in Python 3.11 or later, using FastAPI as the web framework and Uvicorn as the ASGI server. Python was selected for its unmatched ecosystem of AI/ML libraries (faster-whisper, TTS, Pillow, beautifulsoup4, cryptography). FastAPI was selected for its native async support, automatic OpenAPI/Swagger documentation generation, and Pydantic-based request validation.
- **Technical Constraint 7:** Next.js 14+ Frontend with TypeScript. The frontend must be implemented using Next.js 14 (App Router) with TypeScript. This framework was selected for its server-side rendering capabilities, file-based routing system, and strong typing support that reduces runtime errors in the complex UI logic (onboarding wizard state management, real-time audio recording, WebSocket progress streaming, mood-based avatar switching).
- **Technical Constraint 8:** SQLite as the Sole Relational Database. No external database servers (PostgreSQL, MySQL, MongoDB) are permitted. SQLite was chosen for its zero-setup, zero-configuration, file-based architecture — it requires no installation, no background service, and no network connectivity. The database uses WAL (Write-Ahead Logging) journal mode for improved concurrent read performance and enforces foreign.

- **Technical Constraint 9:** Pure Python Vector Store with JSON Storage. The vector database must be implemented as a custom pure Python module using JSON file storage and exact cosine similarity computation. No external vector databases (ChromaDB, Pinecone, Weaviate, Milvus, pgvector, Qdrant) are permitted. This constraint was born from the ChromaDB DLL failure described in Technical Constraint 5 and ensures complete portability — the vector store has zero external dependencies beyond Python's standard library and the requests library for Ollama API calls.
- **Technical Constraint 10:** Ollama as the Sole LLM and Embedding Provider. All LLM inference must use Ollama's llama3.1:8b model, and all text embedding generation must use Ollama's nomic-embed-text model (producing 768-dimensional vectors). No alternative LLM providers (OpenAI, Anthropic, Google, Hugging Face Inference API) are permitted for conversation generation. This constraint ensures that the core intelligence of the digital twin operates entirely on localhost at zero cost.
- **Business Constraint 1:** 6-Week Development Timeline. The project was developed within a 6-week timeline as part of the Major Project academic requirement, necessitating aggressive prioritization and phased delivery.
- **Business Constraint 2:** Single-Laptop Demonstrability. The complete system must be demonstrable on a single laptop, requiring only 4 terminal windows (Ollama, Backend, Frontend, Landing Page) and no external hardware beyond the built-in webcam and microphone.
- **Business Constraint 3:** Open-Source Publication. The complete codebase must be published on GitHub for academic evaluation, reproducibility, and community contribution.
- **Regulatory Constraint 1:** Explicit User Consent. The system must obtain explicit user consent via a mandatory checkbox before processing any personal data. The consent text reads: "I consent to the processing of my data."
- **Regulatory Constraint 2:** Right to Erasure. Users must have the ability to permanently and irreversibly delete all their personal data at any time through a single API endpoint, in compliance with data minimization principles.
- **Regulatory Constraint 3:** Encryption at Rest. All uploaded raw data files (ZIP, JSON, CSV, TXT) must be encrypted using AES-128-CBC (Fernet) immediately after processing. The original plaintext files must be permanently deleted after successful encryption.

2.5 Definitions, Acronyms, and Abbreviations

The following table provides comprehensive definitions for technical terms, acronyms, and abbreviations used throughout this document, organized alphabetically for easy reference.

Table 2.1: Definitions, Acronyms, and Abbreviations

Term / Acronym	Full Form	Definition
AES	Advanced Encryption Standard	A symmetric block cipher algorithm standardized by NIST for securing sensitive data using fixed-size keys. In Alternate Ego, AES-128-CBC is used via Fernet to encrypt uploaded user data files at rest before securely deleting the plaintext originals.
AI	Artificial Intelligence	The branch of computer science focused on building systems capable of performing tasks that typically require human intelligence. Alternate Ego leverages AI across six sub-disciplines — NLP, RAG, TTS, STT, CV, and generative image synthesis — to create a holistic digital twin.
API	Application Programming Interface	A structured set of protocols and endpoints that allow different software components to communicate programmatically. The Alternate Ego backend exposes 15+ RESTful API endpoints via FastAPI for onboarding, chat, avatar management, privacy controls, and memory visualization.
ASGI	Asynchronous Server Gateway Interface	A Python specification for asynchronous web servers and applications, succeeding the older synchronous WSGI standard. Uvicorn, the ASGI server used in Alternate Ego, enables the FastAPI backend to handle concurrent requests efficiently including long-running AI inference tasks.
BLAS	Basic Linear Algebra Subprograms	A standardized set of low-level routines for performing common linear algebra operations such as vector addition and matrix multiplication. Alternate Ego deliberately avoids packages requiring compiled BLAS libraries like faiss-cpu due to persistent DLL load failures on Windows systems.
BLOB	Binary Large Object	A data type capable of storing large amounts of binary data, commonly used for multimedia content such as images, audio, and video. The MediaRecorder API produces audio BLOBs in WebM format during the voice interview, which are then uploaded to the backend for transcription.
CBC	Cipher Block Chaining	A block cipher mode where each plaintext block is XORed with the previous ciphertext block before encryption, creating inter-block dependencies. The

		Fernet encryption scheme employs AES-128-CBC with PKCS7 padding and HMAC-SHA256 for authenticated encryption of user data.
CLI	Command Line Interface	A text-based interface through which users interact with software by typing commands rather than using graphical elements. The LoRA fine-tuning script and LinkedIn scraper module in Alternate Ego are invoked via command-line execution for advanced users and developers.
CORS	Cross-Origin Resource Sharing	An HTTP security mechanism that allows a web application on one origin to make requests to a server on a different origin. Alternate Ego configures CORS middleware in FastAPI to permit the Next.js frontend (port 3000) to communicate with the backend API (port 8000).
CPU	Central Processing Unit	The primary processing chip in a computer that executes program instructions and performs arithmetic and logical operations. All AI inference in Alternate Ego — including LLM generation, embedding computation, and speech transcription — runs on the CPU, requiring no dedicated GPU hardware.
CRUD	Create, Read, Update, Delete	The four fundamental operations of persistent data storage that form the basis of most database interactions. The SQLite database layer implements CRUD operations across five relational tables: users, twins, conversations, messages, and onboarding_sessions.
CSS	Cascading Style Sheets	A stylesheet language used to describe the visual presentation of HTML documents including layout, colors, fonts, and animations. Alternate Ego's frontend uses a custom CSS design system featuring glassmorphism effects, gradient backgrounds, smooth animations.
CSR	Client-Side Rendering	A web rendering strategy where HTML is generated in the browser using JavaScript after the initial page load. The Alternate Ego chat interface uses CSR for real-time message updates, avatar switching, and audio playback without full page reloads.
CSV	Comma-Separated Values	A plain-text file format that stores tabular data with values separated by commas, widely used for data export and exchange. The data export parser processes CSV files extracted from social media data dumps, converting structured tabular data into text chunks for RAG indexing.
CTA	Call to Action	A UI design element — typically a button or link — that prompts the user to take a specific action such as signing up or proceeding. The landing page features a prominent "Begin the Imprint" CTA button with gradient styling and hover animations directing users to the upload page.

CUDA	Compute Unified Device Architecture	A parallel computing platform created by NVIDIA for leveraging GPU hardware for general-purpose computing tasks. The optional LoRA fine-tuning module uses CUDA-enabled GPUs via the Unsloth library for accelerated model training on personal data.
CV	Computer Vision	A field of artificial intelligence that enables computers to interpret and understand visual information from images and videos. face-api.js (built on TensorFlow.js) performs browser-side face detection and emotion validation during the webcam photo capture step of onboarding.
DLL	Dynamic Link Library	A shared library file format on Windows containing compiled code and data that can be loaded by multiple programs simultaneously. ChromaDB's dependency on hnswlib caused persistent DLL load failures on Windows, directly motivating the development of Alternate Ego's custom pure Python vector store.
DM	Direct Message	A private communication sent between two users on a social media platform, distinct from public posts or comments. The data export parser contains specialized extraction logic for Instagram DM conversations, one of the richest sources of personal communication data for twin personality training.
DOM	Document Object Model	A programming interface that represents an HTML document as a tree structure of objects, allowing scripts to dynamically access and manipulate content. face-api.js operates on the browser DOM to overlay face detection bounding boxes and emotion labels on the live webcam video feed.
EXIF	Exchangeable Image File Format	A metadata standard embedded in digital image files that records camera settings, timestamps, GPS coordinates, and device information. During ZIP data export parsing, Alternate Ego extracts EXIF metadata from uploaded images using Pillow, adding temporal and geographic context to the knowledge base.
GDPR	General Data Protection Regulation	The European Union's comprehensive data protection law governing the collection, processing, and storage of personal data. Alternate Ego's privacy architecture — including explicit consent, encryption at rest, right to erasure, and data minimization — aligns with GDPR principles even though it processes data locally.
GGUF	GPT-Generated Unified Format	A binary file format designed for efficiently storing quantized large language model weights, used by llama.cpp and Ollama. The LoRA fine-tuning pipeline exports trained adapter weights as GGUF files that can be deployed as custom Ollama models via Modelfile configuration.
GPU	Graphics Processing Unit	A specialized processor originally designed for rendering graphics but now widely used for parallel AI

		computation tasks. While Alternate Ego's core functionality runs entirely on CPU, an NVIDIA GPU with CUDA support enables the optional LoRA fine-tuning feature for deep model personalization.
gTTS	Google Text-to-Speech	A free Python library and API wrapper that converts text into spoken audio using Google's cloud-based speech synthesis engine. In Alternate Ego's 4-tier TTS fallback chain, gTTS serves as the third fallback option when both XTTS v2 and Edge-TTS are unavailable.
HMAC	Hash-based Message Authentication Code	A cryptographic mechanism that combines a hash function with a secret key to verify both the integrity and authenticity of a message. The Fernet encryption scheme applies HMAC-SHA256 to detect any tampering with encrypted user data files, ensuring authenticated encryption.
HTML	HyperText Markup Language	The standard markup language for creating web pages, defining the structure and semantic content of documents rendered by browsers. Alternate Ego parses HTML files extracted from social media data exports using BeautifulSoup to extract text content for RAG indexing.
HTTP	HyperText Transfer Protocol	The foundation protocol of the World Wide Web, defining how messages are formatted and transmitted between clients and servers. All communication between the frontend (port 3000), backend (port 8000), and Ollama runtime (port 11434) uses HTTP/1.1 REST-style request-response patterns.
HTTPS	HyperText Transfer Protocol Secure	The encrypted version of HTTP that uses TLS/SSL to secure data transmission against eavesdropping. While Alternate Ego operates on localhost where HTTPS is unnecessary, production deployment would require HTTPS certificates for securing external-facing endpoints.
IDE	Integrated Development Environment	A software application that provides comprehensive tools for software development including code editing, debugging, and version control. The Alternate Ego project was developed using VS Code as the primary IDE with extensions for Python, TypeScript, and database manage.
INT8	8-bit Integer Quantization	A model compression technique that represents neural network weights using 8-bit integers instead of 32-bit floats, reducing memory usage by 75%. The faster-whisper STT engine uses INT8 quantization to run the Whisper model efficiently on CPU without GPU acceleration.
IP	Internet Protocol	The principal communication protocol for routing and addressing data packets across interconnected networks. All Alternate Ego services bind to localhost IP

		(127.0.0.1), ensuring no external network traffic can reach the backend API or access user data remotely.
JPEG	Joint Photographic Experts Group	A widely used lossy image compression format optimized for photographic images with smooth color variations. Webcam photos captured during onboarding are saved as JPEG files in the storage/avatars/{twin_id}/ directory before being processed for AI avatar generation.
JSON	JavaScript Object Notation	A lightweight, human-readable data interchange format consisting of key-value pairs and arrays, widely used in web APIs. Alternate Ego uses JSON extensively for API payloads, vector store persistence, conversation export, data export parsing, and configuration files.
JSX	JavaScript XML	A syntax extension for JavaScript that allows HTML-like markup to be written directly within JavaScript code, used by React components. Alternate Ego frontend components are written in TSX — the TypeScript equivalent of JSX — enabling type-safe UI development throughout.
LLM	Large Language Model	A deep neural network with billions of parameters trained on massive text corpora, capable of understanding and generating human-like language. Alternate Ego uses Ollama's locally-hosted Llama 3.1:8b model (8 billion parameters) as the core conversational engine for personality-consistent responses.
LoRA	Low-Rank Adaptation	A parameter-efficient fine-tuning technique that injects small trainable rank-decomposition matrices into a frozen pre-trained model. Alternate Ego provides an optional LoRA fine-tuning script using Unsloth that trains personalized adapter weights on the user's data for a deeply customized twin model.
MCP	Model Context Protocol	A protocol framework enabling AI models to interact with external tools and services through structured command interfaces. In Alternate Ego, MCP parses slash commands (e.g., /linkedin post "...") from chat messages and routes them to platform-specific action handlers for simulated social media posting.
ML	Machine Learning	A subset of artificial intelligence where systems learn patterns from data and improve performance without being explicitly programmed. Alternate Ego applies ML across multiple subsystems: embedding generation, speech recognition, voice synthesis, and face detection.
MPEG	Moving Picture Experts Group	An international standards organization and family of audio/video compression formats used for digital

		multimedia. The data export parser uses FFmpeg to extract audio tracks from video files in uploaded data exports for transcription via faster-whisper.
NLP	Natural Language Processing	The AI discipline focused on enabling computers to understand, interpret, generate, and respond to human language in meaningful ways. Alternate Ego's NLP capabilities span the entire pipeline: prompt construction, personality extraction, mood detection, and context-aware RAG retrieval.
NPM	Node Package Manager	The default package manager for Node.js that facilitates installation, version management, and dependency resolution for JavaScript libraries. The frontend uses NPM to manage dependencies including Next.js, React, Framer Motion, Lucide React, and TailwindCSS.
OAuth	Open Authorization	An open standard authorization framework that enables third-party applications to access user resources without exposing credentials. While Alternate Ego currently uses simulated social media posting, the MCP architecture is designed to support real OAuth-based API integration in future versions.
ORM	Object-Relational Mapping	A programming technique that converts data between incompatible type systems — typically between database tables and programming language objects. Although Alternate Ego uses raw SQL queries for maximum control, Pydantic models serve a similar role by providing structured data validation for API payloads.
OS	Operating System	The system software that manages computer hardware resources and provides common services for application programs. Alternate Ego primarily targets Windows 10/11, with secondary compatibility for Linux and macOS through Python and Node.js cross-platform runtimes.
OSINT	Open Source Intelligence	The practice of collecting and analyzing publicly available information from open sources to build intelligence about individuals. The social scraping module uses OSINT techniques — DuckDuckGo search, Nitter proxies, curl_cffi TLS impersonation — to discover and aggregate public information across platforms.
PNG	Portable Network Graphics	A lossless image compression format that supports transparency, commonly used for web graphics and generated artwork. The AI-generated emotion avatars (neutral, happy, sad, angry) are saved as 512×512 PNG

		files with transparency support for seamless integration into the chat interface.
PWA	Progressive Web Application	A web application built using standard web technologies that delivers app-like experiences including offline capability and installability. While Alternate Ego is not currently a PWA, the Next.js 14 framework supports PWA features that could enable offline chat access in future versions.
RAG	Retrieval-Augmented Generation	An AI architecture that enhances LLM responses by retrieving relevant context from an external knowledge base before generating answers. Alternate Ego's RAG pipeline chunks user data, embeds it into 768-dimensional vectors, and retrieves the top-3 most relevant chunks using cosine similarity for each conversation turn.
RAM	Random Access Memory	A type of volatile computer memory that provides fast read/write access for data currently being processed by the CPU. Alternate Ego requires a minimum of 8 GB RAM to simultaneously run the Ollama LLM (4.7 GB model), the FastAPI backend, and the Next.js frontend.
REST	Representational State Transfer	An architectural style for designing networked applications using stateless HTTP methods to manipulate resources. The backend implements a RESTful API with 15+ endpoints organized across seven route groups: onboarding, chat, avatar, MCP actions, privacy, memory, and WebSocket logs.
SDLC	Software Development Life Cycle	A systematic process for planning, creating, testing, and deploying software systems through defined phases from inception to maintenance. Alternate Ego followed a 5-phase SDLC: Requirement Analysis, Planning, Design, Development, and Implementation.
SPA	Single Page Application	A web application architecture that loads a single HTML page and dynamically updates content using JavaScript without full page reloads. The chat interface operates as an SPA within Next.js, enabling seamless real-time messaging, avatar updates, and audio playback without navigation interruptions.
SQL	Structured Query Language	A domain-specific language designed for managing, querying, and manipulating data stored in relational database management systems. Alternate Ego uses SQL queries against SQLite for all persistent operations: creating users, storing twin profiles, saving messages, and tracking onboarding progress.
SRS	Software Requirement Specification	A comprehensive document that describes the intended behaviour, constraints, and interfaces of a software system, typically following IEEE 830 standards. This

		document constitutes the SRS for Alternate Ego, defining the purpose, audience, constraints, and technical vocabulary governing the platform's development.
SSR	Server-Side Rendering	A web rendering technique where HTML pages are generated on the server before being sent to the client browser. The Next.js 14 framework supports SSR for initial page loads of the upload page and onboarding wizard, improving perceived performance while enabling client-side interactivity afterwards.
SSL	Secure Sockets Layer	A deprecated cryptographic protocol for securing communications between networked computers, superseded by TLS. While Alternate Ego operates entirely on localhost making SSL unnecessary, the architecture supports future production deployment with certificate-based HTTPS encryption.
STT	Speech-to-Text	The technology that converts spoken human language into written text through automatic speech recognition algorithms. Alternate Ego uses faster-whisper with INT8 quantization and VAD filtering for local CPU-based transcription of voice interview recordings.
TCP	Transmission Control Protocol	A core internet protocol that provides reliable, ordered delivery of data streams between applications running on networked hosts. WebSocket connections used for real-time onboarding progress streaming operate over TCP, ensuring all progress log messages are delivered reliably to the frontend.
TLS	Transport Layer Security	The successor to SSL, providing encrypted and authenticated communication channels between networked applications. TLS would protect Alternate Ego's API traffic in a production deployment, though localhost operation eliminates the need for transport-layer encryption.
TSX	TypeScript XML	The TypeScript equivalent of JSX, allowing type-safe HTML-like markup within TypeScript files for React component development. All 15+ frontend components in Alternate Ego — including the upload page, onboarding wizard, chat interface, and memory graph — are written as TSX files.
TTS	Text-to-Speech	The technology that converts written text into audible spoken language through speech synthesis algorithms, optionally mimicking a specific voice. Alternate Ego implements a 4-tier TTS fallback chain: XTTS v2.
UI	User Interface	The visual layer through which users interact with a software application, encompassing layout, controls, typography, and visual feedback. Alternate Ego's UI features a premium glassmorphism design system with gradient backgrounds, smooth CSS animations, and mood-

URL	Uniform Resource Locator	A standardized address format used to identify and locate resources on the internet, including web pages, images, and API endpoints. During onboarding, users provide social media profile URLs which the backend's scraping module processes to extract public personality data.
UUID	Universally Unique Identifier	A 128-bit identifier generated using algorithms that ensure global uniqueness without requiring a central registration authority. Alternate Ego generates UUIDs for user IDs, twin IDs, session IDs, and conversation IDs, enabling reliable cross-reference between database tables and API endpoints.
VAD	Voice Activity Detection	An audio processing technique that automatically identifies segments of speech within a recording, filtering out silence and background noise. The faster-whisper STT engine applies VAD filtering before transcription, improving accuracy by processing only speech-containing segments of voice interview recordings.
VRAM	Video Random Access Memory	Specialized high-bandwidth memory on GPU cards used to store model weights during AI inference. While Alternate Ego's core operation requires no VRAM, the optional LoRA fine-tuning feature benefits from NVIDIA GPUs with 6+ GB VRAM for accelerated training.
WAL	Write-Ahead Logging	A database journaling mode where changes are first written to a separate log before being committed to the main database, improving concurrent read performance. Alternate Ego's SQLite database operates in WAL mode, allowing the chat interface to read history while the backend simultaneously writes new messages.
WSGI	Web Server Gateway Interface	The original Python standard for synchronous communication between web servers and web applications, predating the async ASGI specification. Alternate Ego uses ASGI via Uvicorn instead of WSGI to support async request handling, essential for non-blocking AI inference calls to the Ollama server.
XTTS	Cross-Lingual Text-to-Speech	An open-source neural TTS model by Coqui AI capable of cloning a speaker's voice from short reference audio clips across multiple languages. XTTS v2 is the primary TTS engine in Alternate Ego, using the user's voice interview recordings as reference audio to synthesize responses in the user's own cloned voice.

2.6 Scope of Development

The scope of development for the Alternate Ego system encompasses the complete design, implementation, and deployment of a fully functional, AI-powered digital twin platform operating entirely on a local machine. The system is developed as a full-stack web application integrating a Python-based backend, a TypeScript-based frontend, a cinematic landing interface, and a local AI runtime environment. It includes all core modules such as user onboarding, multi-source data ingestion, RAG-based conversational intelligence, voice interaction, avatar generation, and privacy management. The development scope ensures that each component is tightly integrated while maintaining a modular architecture for scalability and extensibility.

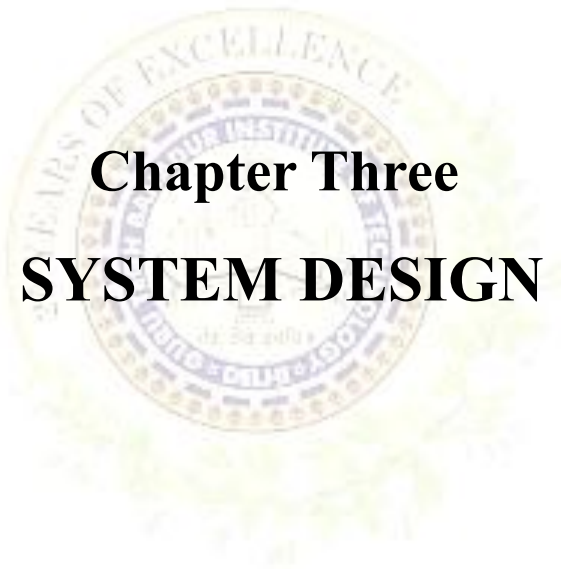
The backend development includes the implementation of nine core modules — API layer, RAG engine, voice pipeline, avatar system, MCP command framework, database management, security layer, storage handling, and configuration management. These modules collectively expose more than 15 RESTful API endpoints and real-time communication via WebSocket. The frontend scope covers user-facing interfaces such as the landing page, upload interface, onboarding wizard, and chat system with avatar interaction. Special attention is given to UI/UX design, ensuring a smooth, intuitive, and visually engaging experience through modern design principles like glassmorphism and real-time feedback mechanisms.

The development scope also includes the integration of six AI sub-disciplines — Natural Language Processing, Retrieval-Augmented Generation, Speech Recognition, Voice Cloning, Computer Vision, and Generative Image Synthesis — into a unified system. Each subsystem is implemented with a focus on local execution, ensuring that all AI inference processes run on the user's machine without reliance on external paid APIs. The system supports complex workflows where multiple AI components interact seamlessly within a single request-response cycle, enabling realistic and context-aware digital twin behavior.

In addition, the scope includes strict adherence to a privacy-first and zero-cost architecture. All user data processing, storage, and inference operations are performed locally, with encryption applied to sensitive data and automatic deletion of raw inputs after processing. The system avoids any dependency on paid cloud services, ensuring accessibility and affordability. It also incorporates features for user consent, data control, and complete data deletion, aligning with modern data protection principles.

The development is constrained to operate on consumer-grade hardware, specifically systems with minimum specifications such as 8 GB RAM and a standard CPU, ensuring wide usability. The platform is primarily targeted for Windows-based environments while maintaining theoretical cross-platform compatibility. Additionally, the scope includes testing, validation, and demonstration of the system as a complete, production-quality solution within the academic project timeline.

Overall, the scope of development ensures that Alternate Ego is delivered as a fully integrated, privacy-preserving, and cost-free AI platform, balancing technical complexity with practical feasibility and user-centric design.

The logo of the University of Excellence is a circular emblem. It features a central sun-like symbol with rays, surrounded by a ring of text. The outer ring contains the words "UNIVERSITY OF EXCELLENCE" at the top and "1999" at the bottom. The inner ring contains the words "TECHNOLOGY" and "INNOVATION". The logo is set against a background of stylized green leaves.

Chapter Three

SYSTEM DESIGN

Chapter 3: SYSTEM DESIGN

3.1 Introduction to System Design

System design is the structured process of defining a system's architecture, modules, interfaces, and data flow in order to meet specified requirements. For the Alternate Ego digital twin platform, this phase plays a crucial role due to the integration of multiple Artificial Intelligence (AI) sub-disciplines into a single cohesive system. These include Natural Language Processing (NLP) for conversational intelligence, Retrieval-Augmented Generation (RAG) for context-aware responses, Speech-to-Text (STT) for voice input, Text-to-Speech (TTS) for voice output, and Computer Vision (CV) for emotion detection. Combining these technologies into one full-stack web application introduces significant architectural complexity. Therefore, a well-defined system design is essential to ensure smooth interaction between components. It also ensures scalability, maintainability, and performance optimization. Overall, this phase transforms requirements into a concrete technical blueprint.

The design philosophy of Alternate Ego is guided by three fundamental principles that shape every architectural decision. The first principle is **Absolute Privacy through a Local-First Architecture**, which ensures that all user data—including voice recordings, images, text inputs, and social media archives—remains strictly on the user's machine. No sensitive data is transmitted to external servers, making privacy a built-in feature rather than an add-on. The second principle is **Zero Operational Cost**, achieved by avoiding paid cloud APIs and instead leveraging open-source models and local runtimes such as Ollama. This makes the system highly accessible and sustainable. The third principle is **Modularity and Extensibility**, where the system is divided into independent modules like the RAG pipeline, Voice Manager, Avatar System, and MCP command framework. This modular design allows components to be upgraded or replaced without affecting the entire system.

The architecture of Alternate Ego follows a layered and modular approach, where each subsystem performs a dedicated role while communicating through well-defined interfaces. The backend, built using a Python-based framework, handles core logic such as data processing, AI inference, and API management. The frontend, developed using modern web technologies, provides an interactive and visually engaging user interface. A local AI runtime manages large language models and embedding generation. Data flows seamlessly between these layers, enabling real-time interaction between the user and their digital twin. Each module is designed to operate independently while contributing to the overall system workflow. This separation of concerns enhances maintainability and debugging efficiency.

This chapter further explores the detailed structural blueprints of the platform. It includes high-level architecture diagrams that illustrate component relationships and data flow paths. It explains how data moves from user input through various processing stages—such as transcription, embedding, retrieval, and response generation—before being delivered back to the user. Internal component interactions, including API communication and module coordination, are described in detail. The chapter also covers security mechanisms such as encryption, local storage policies, and data isolation strategies. Additionally, database schemas are defined to manage user data, conversations, and system states efficiently.

A key aspect of the system design is its focus on robustness and fault tolerance. The platform incorporates multiple fallback mechanisms, such as multi-tier TTS engines and alternative avatar generation pipelines, to ensure uninterrupted functionality. Error-handling strategies are implemented at every layer, from API validation to AI inference failures. Logging and monitoring mechanisms help track system behavior and diagnose issues efficiently. These measures ensure that the system remains stable even under unexpected conditions.

Overall, the system design of Alternate Ego provides a comprehensive and resilient framework that balances advanced AI integration with practical constraints like privacy, cost, and hardware limitations. It serves as the backbone of the platform, enabling the seamless operation of all features while maintaining flexibility for future enhancements.

3.2 Overall System Architecture

Alternate Ego employs a highly modular, decoupled Client-Server-AI Runtime architecture. The system is logically partitioned into four primary tiers: the Client Tier (Frontend UI), the Server Tier (FastAPI Backend), the Storage Tier (SQLite & Vector Store), and the AI Runtime Tier (Ollama).

3.3 High-Level Architecture Diagram

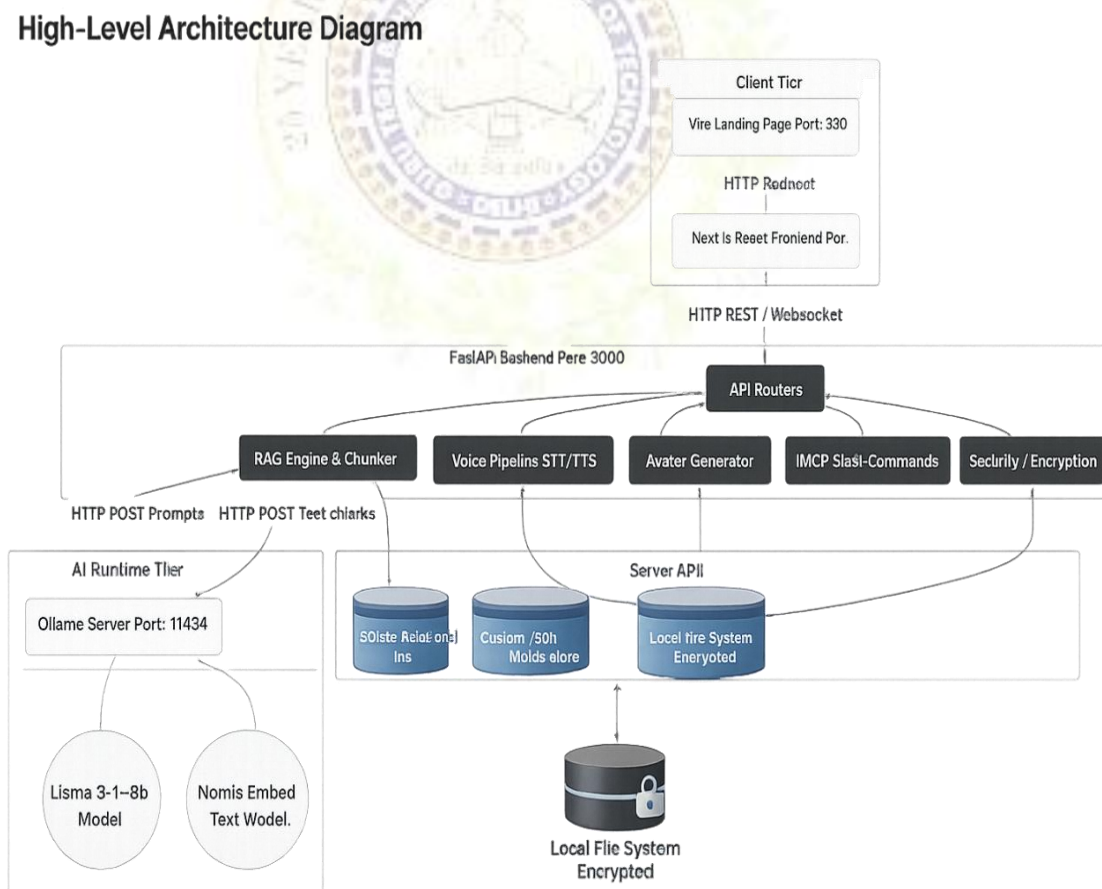


Figure 3.1: High-Level Architecture Diagram

3.4 Architectural Tiers Detailed

1. Client Tier (Presentation Layer)

The Client Tier is responsible for delivering a rich, interactive, and visually immersive user experience. It is primarily built using **Next.js 14**, leveraging React, TypeScript, and TailwindCSS for modern UI development. This layer handles all user interactions, including the multi-step onboarding wizard, chat interface, and real-time avatar display. The chat system supports mood-synchronized avatars and dynamic updates using Client-Side Rendering (CSR). For performance optimization, Server-Side Rendering (SSR) is used to ensure fast initial page loads. Additionally, browser APIs like MediaRecorder enable real-time audio recording directly from the user's microphone.

Alongside this, a lightweight landing page is built using **Vite**, designed to provide a cinematic introduction to the platform. It uses animation libraries like Framer Motion to create engaging transitions and visual storytelling. This separation between landing page and main application improves performance and modularity. Overall, the Client Tier ensures a seamless blend of aesthetics, responsiveness, and interactivity.

2. Server Tier (Application Logic Layer)

The Server Tier acts as the central brain that orchestrates all system operations and business logic. It is implemented in Python 3.11+ using the **FastAPI** framework and runs on an ASGI server powered by **Uvicorn**. This tier handles all incoming HTTP requests and WebSocket connections through structured API routers. Endpoints are grouped logically, such as onboarding, chat, and avatar services, ensuring clean and maintainable routing.

The RAG Engine within this layer processes user data, performs semantic chunking, generates embeddings, and retrieves relevant context using cosine similarity. The Voice Manager enables bidirectional audio communication by converting speech to text using faster-whisper and generating speech output via a multi-tier TTS fallback chain. The Avatar Generator creates and manages emotion-based visual representations with fallback mechanisms for reliability. Additionally, the Security Module ensures data protection using Fernet encryption (AES-128-CBC). This tier ensures all components work together cohesively while maintaining performance and scalability.

3. Storage Tier (Persistence Layer)

The Storage Tier is responsible for managing and persisting all system data in a structured and secure manner. It uses **SQLite**, a lightweight, file-based database that requires zero configuration. Operating in Write-Ahead Logging (WAL) mode, it ensures efficient read/write operations and supports concurrent access. The database manages key entities such as users, digital twins, conversations, and messages.

In addition to structured storage, the system employs a custom-built vector store implemented entirely in Python. This JSON-backed database stores 768-dimensional embeddings along with their corresponding text chunks. It was specifically designed to avoid compatibility issues associated with external vector databases like FAISS or ChromaDB. Furthermore, an encrypted

file system is used to store sensitive assets such as uploaded data exports, generated avatars, and cloned voice profiles. This ensures both accessibility and security. Overall, this tier guarantees reliable and privacy-preserving data management.

4. AI Runtime Tier (Cognitive Layer)

The AI Runtime Tier provides the computational intelligence required for the digital twin's functionality. It is powered by **Ollama**, an open-source platform that enables local execution of large language models. Ollama exposes an API interface that the backend uses for both conversational inference and embedding generation.

The system utilizes models like llama3.1:8b for generating human-like responses and nomic-embed-text for producing high-dimensional vector embeddings. By running these models locally, the platform ensures complete data privacy and eliminates dependency on external cloud services. This tier acts as the cognitive engine, enabling reasoning, context understanding, and personality simulation. It seamlessly integrates with the RAG pipeline to produce accurate and personalized outputs. Overall, the AI Runtime Tier transforms raw data into intelligent, meaningful interactions, forming the core intelligence of the Alternate Ego platform.

3.5 Data Flow Diagrams

Understanding how data moves through the Alternate Ego platform is crucial for verifying its privacy claims and performance characteristics. The following Data Flow Diagrams (DFDs) map the movement of information across system boundaries.

3.6 Context Diagram (Level 0 DFD)

The Context Diagram illustrates the Alternate Ego system as a single unified process interacting with external entities.

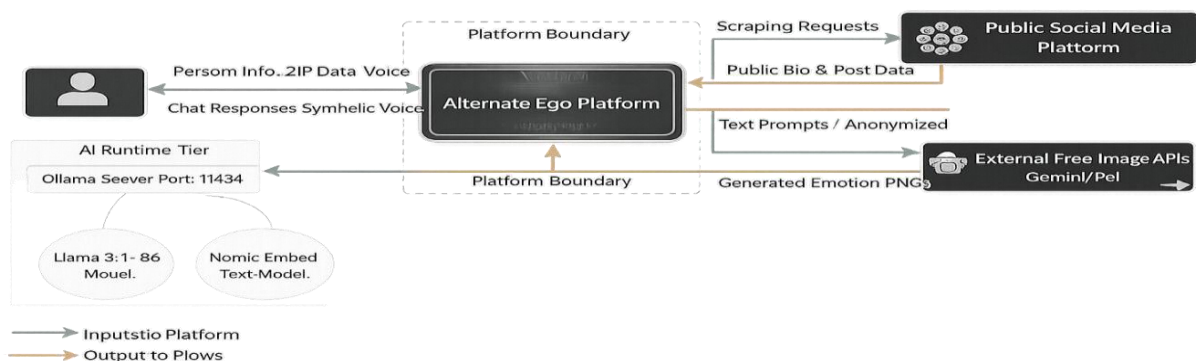


Figure 3.2: Context Diagram (Level 0 DFD)

3.7 Subsystem Data Flow (Level 1 DFD)

This diagram breaks the system down into its major functional processes during the primary use cases: Onboarding and Chat.



Figure 3.3: Subsystem Data Flow (Level 1 DFD)

3.9 Component Interaction & Security

The interactions between components are secured via strict design principles rather than traditional network security layers, leveraging the inherent security of a fully localized architecture.

Table 3.1: Component Interaction & Security

Source Component	Destination Component	Protocol / Interface	Data Payload
Next.js Frontend	FastAPI Backend	HTTP/1.1 REST	JSON payloads, Multipart Form Data (Files/Audio)
Next.js Frontend	FastAPI Backend	WebSocket	Real-time text streams for onboarding logs
FastAPI Backend	Ollama Server	HTTP REST (localhost)	JSON prompts, JSON GGUF responses
FastAPI Backend	Vector Store	Direct Python Call	Python Dictionaries, Lists of Floats
FastAPI Backend	SQLite	SQLAlchemy ORM	SQL Queries, Parameterized Data

3.10 Security Architecture

Because Alternate Ego is designed as a locally hosted digital twin, it mitigates external network threats by design. However, it implements rigorous local security measures to protect sensitive user data residing on the disk.

1. Symmetric Encryption at Rest

The system secures all sensitive data using strong symmetric encryption techniques. It employs the Advanced Encryption Standard (AES-128-CBC) through Python's `cryptography.fernet` library. Whenever a user uploads a ZIP data export, the system first extracts and processes the data for indexing into the vector store. Immediately after this step, all raw plaintext files—which may contain highly sensitive personal information—are encrypted. This ensures that no readable data remains on disk after processing. The original plaintext files are securely overwritten and permanently deleted. This mechanism guarantees that even if storage is accessed, the data remains unintelligible. It provides a strong layer of protection against unauthorized local access.

2. Secure Key Management

Encryption is only as strong as its key management, and the system ensures this is handled securely. A unique encryption key is generated for each system instance during initialization. This key is stored locally in a protected file (`storage/encryption_key`) with strict operating system-level permissions. Only the application has access to this key, preventing unauthorized processes or users from retrieving it. The key is never transmitted over a network or exposed

externally. This localized and restricted storage approach minimizes the risk of key leakage. Additionally, it ensures that encrypted data can only be decrypted within the same environment. This strengthens the overall trust model of the system.

3. Zero Network Egress (Localhost Isolation)

To eliminate external attack vectors, the system enforces strict network isolation. All services are bound exclusively to the localhost address (127.0.0.1). This means the application cannot be accessed from other devices on the same local network. It deliberately avoids binding to 0.0.0.0, which would expose the system to LAN-based access. As a result, APIs, UI interfaces, and AI services remain completely private to the host machine. This design prevents unauthorized remote access, even within the same Wi-Fi network. It also ensures that no user data is transmitted outside the system. This “zero network egress” model is a core pillar of the platform’s privacy-first architecture.

4. Absolute Data Erasure (Right to be Forgotten)

The system provides users with complete control over their data through a comprehensive deletion mechanism. A dedicated API endpoint (`/api/privacy/delete-all/{twin_id}`) enables full and irreversible removal of all user-related data. This process performs a cascading deletion across all storage layers. It removes records from the SQLite database, deletes associated vectors from the JSON-based vector store, and erases all files from the local file system. This includes encrypted data exports, voice models, and generated avatar images. OS-level file deletion commands ensure that no residual data remains. This feature aligns with global data protection principles such as the “Right to be Forgotten.” It guarantees that users can completely remove their digital footprint from the system at any time.

The security architecture of Alternate Ego combines **encryption, isolation, and user control** to create a robust protection framework. By eliminating external dependencies and enforcing strict local boundaries, the system significantly reduces risk exposure.

3.11 Deployment and Scalability

Although Alternate Ego is currently implemented as a **single-user, locally hosted desktop application**, its architectural design closely resembles enterprise-grade systems. The modular separation of concerns, API-driven communication, and layered structure enable the platform to scale horizontally in the future if migrated to a cloud environment. Each subsystem—frontend, backend, AI runtime, and storage—can be independently containerized and deployed as microservices. This forward-compatible design ensures that the system is not limited by its current local deployment model. Instead, it provides a strong foundation for future scalability, load balancing, and distributed processing. Thus, while the present implementation is local-first, the architecture remains cloud-ready.

3.12 Current Local Deployment Model

The current deployment model is optimized specifically for consumer-grade Windows machines, ensuring ease of setup and zero dependency on complex virtualization tools. Instead of using containerization technologies like Docker, the system relies on process-level isolation, where each component runs as an independent process on the host machine. This approach eliminates the need for configurations such as Hyper-V or WSL2, which can be difficult for

non-technical users. It simplifies installation while maintaining separation between system components. Each layer—frontend, backend, and AI runtime—operates independently yet communicates seamlessly. This design ensures both usability and maintainability. It also reduces system overhead, making the application lightweight and efficient.

- **Frontend Execution:** The frontend is executed using the Node.js runtime, which runs the compiled Next.js application. Developers can start the interface using commands like `npm run dev` for development or `npm start` for production builds. This layer is responsible for rendering the user interface, handling user interactions, and communicating with the backend APIs. It supports both client-side and server-side rendering to optimize performance and responsiveness. Real-time features such as chat updates, audio recording, and avatar switching are managed within this layer. The frontend runs as a standalone process, ensuring independence from backend execution. This separation improves debugging and scalability. Overall, it provides a smooth and interactive user experience.
- **Backend Execution:** The backend operates within a Python virtual environment and is powered by an asynchronous web server. It uses **Uvicorn** to handle concurrent requests efficiently, including long-running operations such as file parsing and AI inference. Multi-threading capabilities allow the system to process multiple tasks simultaneously—for example, streaming WebSocket logs while handling large ZIP file uploads. The backend exposes RESTful APIs and WebSocket endpoints for communication with the frontend. It also coordinates all core logic, including RAG processing, voice handling, and security enforcement. Running in a virtual environment ensures dependency isolation and stability. This design enables efficient resource.
- **AI Execution:** The AI runtime operates as a separate background service using **Ollama**, which manages all model inference tasks. It loads and executes large language models such as Llama 3.1:8b for conversational intelligence. Depending on system capabilities, it utilizes either GPU VRAM or system RAM for model execution. This flexibility allows the system to run on both high-end and low-resource machines. Ollama exposes an API interface that the backend uses for prompt processing and embedding generation.

The current deployment model prioritizes **simplicity, accessibility, and performance** while maintaining architectural robustness. By avoiding containerization, the system becomes easier to install and run for end users. At the same time, process-level isolation ensures modularity and fault tolerance. Each component can be started, stopped, or debugged independently. This design strikes a balance between real-world usability and enterprise-level scalability. In future iterations, the same architecture can be seamlessly extended to containerized or cloud-based deployments.

3.13 Scalability Considerations

While Alternate Ego currently operates as a local-first, single-user system, its architecture is intentionally designed to support **multi-tenant cloud scalability** with minimal structural changes. The platform follows principles similar to distributed systems, enabling horizontal scaling across multiple components.

1. Stateless API Layer (Horizontal Scaling)

The backend, built using FastAPI, is inherently stateless, meaning it does not retain session data between requests. All persistent state is stored in external systems such as databases and vector stores. This allows multiple backend instances to run concurrently without synchronization issues. Scaling can be achieved by deploying multiple Uvicorn worker instances behind a load balancer like Nginx. Incoming requests are distributed evenly across instances, improving throughput and reliability. This architecture ensures that failure of one instance does not impact the entire system. It also allows dynamic scaling based on traffic. Overall, the stateless design is a key enabler for cloud readiness.

2. Database Migration (From SQLite to PostgreSQL)

The current system uses SQLite for simplicity and zero-configuration local storage. However, for multi-user cloud environments, a more robust relational database is required. Using an ORM layer like SQLAlchemy, the system can seamlessly migrate to PostgreSQL. This transition requires only a change in the database connection string, with minimal code modification. PostgreSQL provides advanced features such as concurrency control, indexing, and distributed replication. It supports high transaction volumes and ensures data integrity across multiple users. This upgrade enables the system to handle large-scale user data efficiently. It also improves reliability and performance in production environments.

3. Vector Store Scaling (ANN-Based Retrieval)

The custom JSON-based vector store is ideal for single-user scenarios but does not scale efficiently for large datasets. In a cloud deployment, it would be replaced by a distributed vector database such as Pinecone, Milvus, or pgvector. These systems use Approximate Nearest Neighbor (ANN) algorithms instead of exact $O(n)$ search. ANN significantly reduces query time when dealing with millions of embeddings. It enables fast and scalable similarity search for real-time applications. Additionally, these databases support sharding, indexing, and distributed storage. This ensures efficient retrieval performance even at massive scale. It transforms the RAG pipeline into a production-grade retrieval system.

4. AI Inference Scaling (Distributed GPU Clusters)

In the current setup, AI inference is handled locally using Ollama. For cloud scalability, this would transition to high-performance inference systems like vLLM or Text Generation Inference. These systems are deployed on GPU clusters (e.g., NVIDIA A100), enabling parallel processing of multiple requests. They use advanced techniques such as batching to serve multiple users simultaneously. This significantly improves throughput and reduces latency. Each user's digital twin can be handled independently within the same infrastructure. The system can dynamically allocate resources based on demand. This ensures efficient utilization of expensive GPU hardware. Overall, it enables real-time, large-scale AI interaction.

The future cloud architecture of Alternate Ego is designed to be **highly scalable, distributed, and resilient**. By transforming each layer into a scalable service, the system can evolve from

a personal desktop application into a global AI platform. The transition requires minimal architectural changes due to the modular and stateless design. This forward-thinking approach ensures that Alternate Ego is not only innovative today but also ready for enterprise-level deployment tomorrow.

3.14 System Entity Relationships

Data persistence is managed primarily by an embedded SQLite database (ego_database.db). The database maintains strict referential integrity using foreign keys.

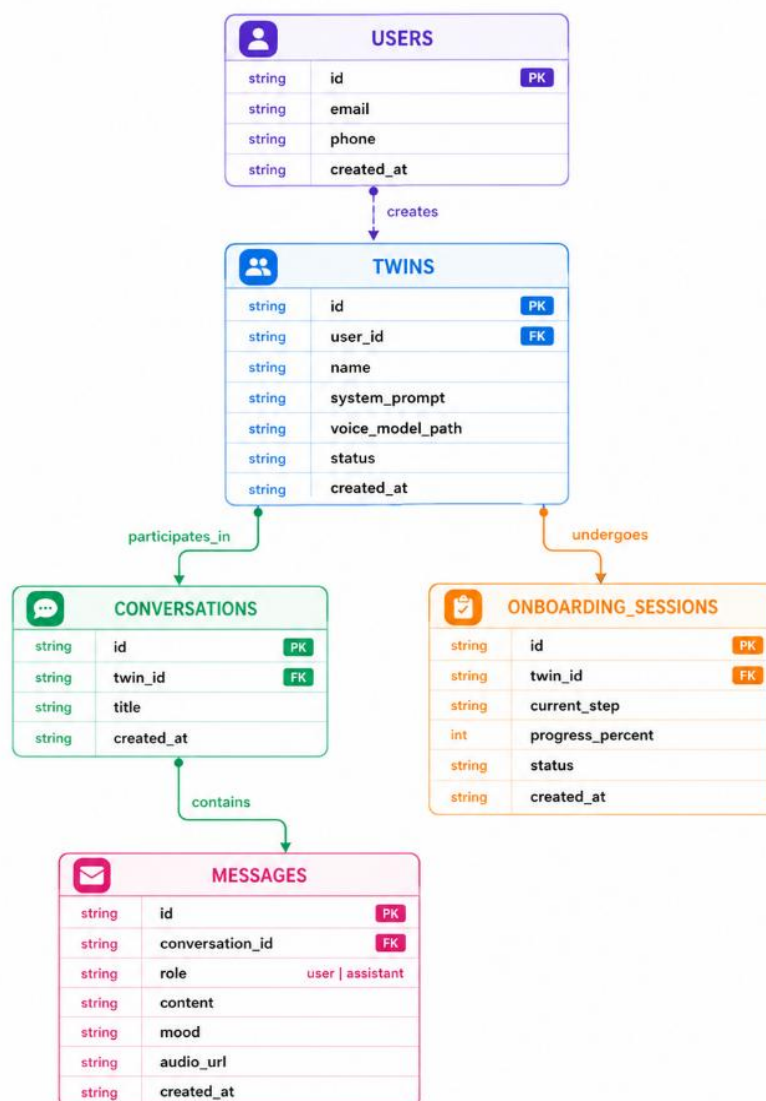


Figure 3.5: System Entity Relationships

3.15 Error Handling & Monitoring

Given the unpredictable nature of Large Language Models and external hardware interfaces (microphones/webcams), Alternate Ego implements a robust, multi-layered error handling strategy to prevent catastrophic system crashes.

Fallback Chains

The most critical system design choice for reliability is the implementation of Fallback Chains for external or resource-heavy dependencies.

Text-to-Speech (TTS) Fallback Matrix:

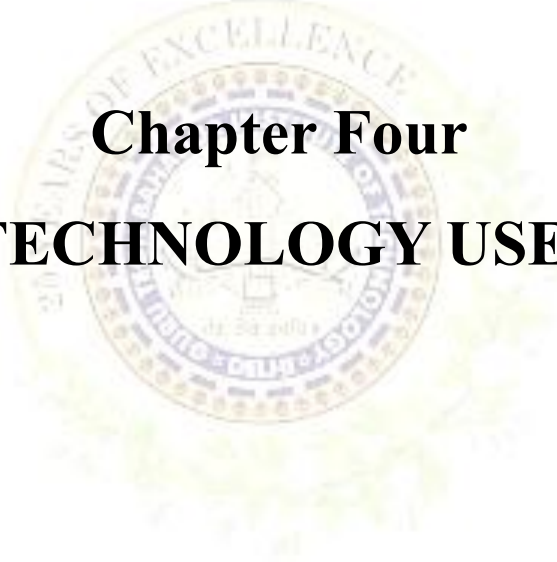
- **Tier 1:** XTTS v2 (Local Cloning). Attempts to synthesize high-fidelity cloned voice. If the system lacks sufficient RAM, or the model fails to load, the system catches the exception and routes to Tier 2.
- **Tier 2:** Edge-TTS (Network). Uses Microsoft's free cloud TTS. If the laptop is offline, this raises a connection timeout, falling back to Tier 3.
- **Tier 3:** gTTS (Network). Google's free TTS API. If also blocked by network, falls to Tier 4.
- **Tier 4:** pyttsx3 (Offline). Uses the operating system's built-in offline voices (e.g., Microsoft David/Zira). This guarantees audio output under all failure states

Avatar Generation Fallback Matrix:

- **Tier 1:** Gemini Pro Vision. Requires an API key. If rate-limited or key is missing, falls back to Tier 2.
- **Tier 2:** Pollinations.ai Klein. Image-to-image free API. If server is down, falls back to Tier 3.
- **Tier 3:** Pollinations.ai Text-to-Image. Pure text prompt generation as an absolute last resort to ensure the UI has an image to display.

Input Sanitization and Truncation

- **Context Length Protection:** Ollama's nomic-embed-text model possesses a strict 8192-token context window. To prevent HTTP 500 crashes during bulk data export ingestion, the embedder.py module strictly truncates all incoming strings to `MAX_EMBED_CHARS = 6000` before transmission to the AI runtime.
- **RAG Chunk Limits:** The chunker.py enforces a maximum chunk size of 500 characters. If an HTML or text file lacks paragraph breaks, the system forces artificial breaks to ensure the vector database remains highly granular and performant.

The logo of the University of Excellence is a circular emblem. It features a central shield with a book and a torch, surrounded by a wreath. The text "UNIVERSITY OF EXCELLENCE" is written in a circle around the emblem. Below the emblem, the text "TECHNOLOGY USED" is written in a large, bold, serif font.

Chapter Four

TECHNOLOGY USED

Chapter 4: TECHNOLOGY USED

4.1. Next.js

- **Frontend Architecture & Routing:** Next.js serves as the primary foundational framework for the Alternate Ego project, providing a highly scalable and robust structure for the entire user interface. By utilizing its advanced routing capabilities, the project achieves seamless, instantaneous client-side navigation between the cinematic landing page, the multi-step onboarding flow, and the main split-screen chat interface without any jarring full-page reloads. This creates a deeply immersive, application-like experience that feels native to the user. Furthermore, the framework's native optimizations ensure that the complex UI components, such as the real-time webcam capture and interactive chat windows, load with maximum efficiency and minimal latency.
- **Component Modularity & State Management:** The framework enables a highly modular component architecture, allowing the development team to break down the complex digital twin interface into 11 distinct, reusable React components. Elements like the `CameraCapture`, `VoiceInterview`, and `ChatWindow` are completely isolated, making the codebase highly maintainable and easy to debug. This modularity ensures that state transitions—such as moving from the photo capture phase to the voice recording phase—are handled gracefully. The predictable state management inherent to Next.js components allows the application to dynamically update the UI based on real-time backend responses without losing user input or context.
- **Performance Optimization & SSR:** Next.js brings powerful Server-Side Rendering (SSR) and Static Site Generation (SSG) capabilities to the Alternate Ego platform, ensuring that the initial load times are instantaneous. This is critical for a premium, heavy-media application that relies on cinematic hero videos and high-quality generated avatars. The framework automatically optimizes external assets, scripts, and fonts (such as Playfair Display and Inter), significantly reducing the Time to Interactive (TTI) metric. By shifting appropriate rendering tasks to the server side, the client-side browser is freed up to handle the intensive machine learning tasks required for local facial expression detection.
- **Development Ecosystem & Tooling:** The Next.js ecosystem provides a superior developer experience through features like Fast Refresh, which instantly reflects code changes during the UI polishing phases. It integrates flawlessly with the TypeScript environment, providing strict type-checking that prevents runtime errors before they occur in the browser. The framework's seamless integration with tools like ESLint and PostCSS ensures that the project maintains a high standard of code quality and consistent styling. Ultimately, Next.js empowers the development team to deliver a production-ready, highly polished digital twin frontend with minimal configuration overhead.

4.2 FastAPI

- **High-Performance Backend Routing:** FastAPI acts as the high-performance central nervous system for the Alternate Ego platform, efficiently routing all requests between the React frontend and the heavy local AI engines. Built on Starlette and Pydantic, it provides asynchronous request handling out of the box, which is absolutely critical when dealing with long-running tasks like generating LLM responses or processing audio transcriptions. The framework's ability to handle concurrent connections ensures that the server remains responsive even while the Ollama engine is fully utilizing the system's processing power. This asynchronous nature prevents the UI from freezing during intensive digital twin generation phases.
- **Automatic Documentation & Validation:** One of the most powerful features FastAPI brings to the project is its automatic generation of interactive API documentation (Swagger UI and ReDoc). By simply defining the Pydantic data models for user inputs—such as the onboarding data schemas and chat message structures—FastAPI automatically validates all incoming requests. If a frontend component sends malformed data, FastAPI rejects it immediately with a clear error message, completely eliminating an entire class of backend parsing bugs. This self-documenting nature makes it incredibly easy to test endpoints like `/api/chat` or `/api/onboarding` independently of the frontend UI.
- **Modular Application Structure:** FastAPI's `APIRouter` system allows the backend codebase to be meticulously organized into distinct, logical domains rather than a single monolithic file. The Alternate Ego backend is cleanly split into modules such as `api/onboarding.py`, `api/chat.py`, and `api/privacy.py`, making the 2,500+ lines of Python highly navigable and maintainable. This modularity perfectly mirrors the project's phase-by-phase development approach, allowing security features, RAG brain logic, and voice pipelines to be developed and tested in total isolation. It guarantees that routing logic remains clean, scalable, and prepared for future feature expansions.
- **Seamless AI Engine Integration:** FastAPI serves as the perfect bridge between traditional web requests and the complex local machine learning pipelines running on the host machine. It efficiently handles the multi-part file uploads required for voice recordings and social media data zips, saving them to the local `storage/` directory securely. Through its dependency injection system, FastAPI easily manages database connections and background tasks, such as triggering the asynchronous scraping of social media profiles. It orchestrates the entire "Imprint → Perceive → Embody → Converge" pipeline flawlessly.

4.3 Ollama

- **Local AI Runtime Execution:** Ollama is the absolute cornerstone of the project's "100% local, zero-cost" philosophy, acting as the dedicated runtime environment for executing massive Large Language Models directly on the user's hardware. By circumventing the need for cloud-based APIs like OpenAI or Anthropic, Ollama ensures that highly sensitive, personal digital twin data never leaves the host machine. It manages hardware acceleration and RAM allocation dynamically, allowing complex 8-billion parameter models to run efficiently even on consumer-grade computer systems. This completely eliminates the recurring \$2-8 API costs associated with the previous iteration of the project.
- **Model Management & Swapping:** The platform leverages Ollama's elegant model management system to easily download, load, and swap different machine learning models on the fly. Through simple command-line operations (`ollama pull`), the system can securely acquire both the text generation models and the embedding models required for the RAG pipeline. This flexibility means that as the open-source AI community releases better, more efficient models in the future, the Alternate Ego project can upgrade its core "brain" with a simple configuration change. It future-proofs the digital twin against proprietary API deprecations or sudden pricing model shifts.
- **Unified API Abstraction:** Ollama provides a clean, unified REST API (typically running on `localhost:11434`) that the FastAPI backend communicates with seamlessly. Instead of writing complex, low-level PyTorch or HuggingFace inference code, the backend simply sends standard JSON payloads to the Ollama endpoint. This dramatically reduces the backend codebase complexity and abstractly handles complicated generation parameters like temperature, top-p, and context window management. It acts as a reliable microservice dedicated entirely to neural network inference, isolating the heavy lifting from the main application server.
- **Dual-Purpose Functionality (LLM + Embeddings):** In this project, Ollama performs double duty by hosting both the conversational agent (`llama3.1:8b`) and the crucial embedding model (`nomic-embed-text`). This unified approach means the user only needs to install one piece of software to handle all core NLP tasks. During the onboarding phase, Ollama swiftly converts the user's social media scrapes and voice transcripts into 768-dimensional vector embeddings. Later, during the chat phase, it utilizes the LLM to process the retrieved vectors and generate the highly personalized, first-person responses that make the digital twin feel truly alive.

4.4. LLaMA 3.1 (8B)

- **Core Conversational Engine:** Meta's LLaMA 3.1 (8-Billion parameter version) acts as the central consciousness and conversational engine for the Alternate Ego digital twin. Chosen for its exceptional balance between high-end reasoning capabilities and local hardware executability, it processes the contextual data provided by the RAG system to formulate responses. The model's advanced instruction-following capabilities ensure that it strictly adheres to the system prompts, forcing it to always speak in the first person and accurately emulate the user's specific conversational tone. It effectively transforms raw data into a convincing, dynamic human personality.
- **Contextual Processing & RAG Integration:** When a user interacts with their digital twin, LLaMA 3.1 does not simply rely on its pre-trained knowledge; it dynamically reads and processes the specific chunks of memory retrieved from the local vector database. The prompt engineering wraps the user's query alongside their scraped social media history, voice interview transcripts, and uploaded data exports. LLaMA 3.1 excels at synthesizing this injected context, correctly identifying the most relevant facts, and weaving them naturally into a cohesive, conversational reply without hallucinating external information.
- **Zero-Cost Deep Personalization:** By utilizing this specific open-weights model, the project achieves deep, nuanced personalization entirely for free, replacing the incredibly expensive dependency on proprietary models like GPT-4. Despite running locally, the 8B model possesses enough semantic understanding to grasp the subtle emotional context of the user's past experiences. It analyzes the 9-question psychological interview data to understand the user's core values, beliefs, and stress-handling mechanisms. The model then reflects these deep-seated personality traits in every generated response, making the twin highly accurate.
- **Implicit Emotion & Mood Detection:** Beyond just generating text, LLaMA 3.1 is tasked with implicit emotional reasoning through clever prompt engineering. The system instructs the model to append specific mood tags (e.g., '[MOOD:happy]', '[MOOD:thoughtful]') at the end of its generated replies based on the emotional weight of the conversation. This saves the system from needing to run a separate, heavy sentiment analysis model. The backend parses this tag from LLaMA's output and uses it to dynamically switch the graphical avatar's facial expression, seamlessly synchronizing the textual output with visual emotional cues.
- **Efficient Inference & Context Window Management:** LLaMA 3.1 (8B) is optimized to handle long conversational contexts efficiently, which is essential for maintaining continuity in digital twin interactions. Through careful prompt engineering and context window management, the system ensures that only the most relevant retrieved chunks.

4.5 SQLite

- **Zero-Configuration Relational Storage:** SQLite provides the foundational relational database layer for the project, chosen specifically for its zero-configuration, serverless architecture. Because SQLite stores the entire database as a single file (`ego_database.db`) directly on the local disk, users do not need to install, configure, or manage complex background database servers like PostgreSQL or MySQL. This perfectly aligns with the project's goal of being a frictionless, plug-and-play local application. It handles the structured data tracking effortlessly, managing the intricate relationships between users, their digital twins, and their respective onboarding sessions.
- **State Persistence & History:** The database is responsible for meticulously persisting the state of the digital twin across multiple application restarts. It securely logs all historical conversational data in a dedicated `messages` table, allowing the chat interface to instantly reload past interactions when the user returns. Furthermore, it tracks the onboarding progress, ensuring that if a user closes the browser midway through their voice interview, their progress is not lost. This persistence layer is what transforms the project from a stateless AI demo into a robust, continuous digital application.
- **Structured Data Modeling:** Through the use of Python's Pydantic and SQLite integration, the project enforces strict, structured data models for all critical entities. The 5-table schema elegantly separates user metadata from the highly specific digital twin configurations. This structural integrity guarantees that the backend can reliably query specific conversation threads or lookup the exact filepath for a twin's specific voice cloning reference file. The fast, local indexing capabilities of SQLite ensure that these metadata lookups happen in milliseconds, never bottlenecking the highly demanding AI processes.
- **Privacy-First Data Isolation:** By keeping all relational data localized in a single SQLite file, the project drastically simplifies its privacy and data deletion mandates. When a user requests the deletion of their digital twin via the `/api/privacy` endpoint, the system can execute deterministic SQL `DELETE CASCADE` operations to instantly wipe all associated metadata, chat history, and identity records. Because no data is synchronized to a remote cloud database like Supabase (which the original project used), the user retains absolute, physical ownership of their identity matrix at all times.
- **Lightweight Concurrency & Performance Optimization:** SQLite is configured to operate in **Write-Ahead Logging (WAL) mode**, which significantly improves concurrency by allowing simultaneous read and write operations. This is especially important in Alternate Ego, where the system may be writing new chat messages while the frontend is simultaneously reading past conversation history. WAL mode reduces database locking, ensuring that the application remains responsive even during continuous interactions. Additionally, SQLite's efficient indexing mechanisms enable.

4.6. Faster-Whisper

- **Local CPU-Optimized Transcription:** Faster-Whisper is deployed as the project's dedicated Speech-to-Text (STT) engine, drastically outperforming the standard OpenAI Whisper implementation in both speed and efficiency. It utilizes CTranslate2 under the hood, allowing it to perform highly accurate audio transcriptions directly on the user's CPU without requiring an expensive, dedicated GPU. This is crucial for processing the 9-question voice interview during the onboarding phase. It guarantees that the user's deeply personal spoken answers are transcribed entirely locally, completely eliminating the privacy risks of sending voice data to cloud STT APIs.
- **High-Fidelity Context Capture:** To ensure the digital twin accurately learns from the user, the transcription must be nearly flawless. Faster-Whisper achieves this by utilizing advanced configuration parameters, such as a `'beam_size'` of 5 for superior word selection and `'condition_on_previous_text'` to maintain context across long spoken paragraphs. This high-fidelity transcription ensures that the subtle nuances, specific vocabulary, and unique phrasing of the user are perfectly captured and subsequently fed into the RAG vector database. Errors in STT would directly corrupt the digital twin's foundational memory.
- **Voice Activity Detection (VAD) Filtering:** One of the most significant advantages of using Faster-Whisper in this system is its built-in Voice Activity Detection (VAD) filtering capabilities. During the voice interview, users often pause, think, or create silent gaps while formulating their answers. The VAD filter mathematically identifies and aggressively strips out these non-speech segments before the transcription process begins. This optimization drastically reduces hallucinated text and significantly speeds up the processing time, providing the user with a much smoother, faster onboarding experience.
- **Vocabulary Biasing via Initial Prompts:** The system cleverly utilizes Faster-Whisper's `'initial_prompt'` feature to bias the transcription engine toward correctly recognizing the user's specific context. By seeding the prompt with the user's actual name and expected conversational markers (e.g., "I am Prabhdeep Singh"), the model mathematically weights these tokens higher during decoding. This drastically reduces the misspelling of proper nouns and technical terms that the user might mention, resulting in a significantly cleaner transcript that requires zero manual editing from the user before being committed to the AI's permanent memory.
- **Batch Processing & Streaming Transcription Efficiency:** Faster-Whisper supports efficient chunk-based and streaming transcription, enabling the system to process long audio inputs without excessive memory consumption. Instead of loading the entire audio file at once, the engine processes it in smaller segments, maintaining performance stability on CPU-only systems. This is particularly useful during the onboarding voice interview, where responses can be lengthy and continuous.

4.7. Coqui XTTS v2

- **Hyper-Realistic Voice Cloning:** Coqui XTTS v2 is the state-of-the-art text-to-speech (TTS) engine responsible for the project's most magical feature: actual voice cloning. Using just the short audio references collected during the 9-question onboarding interview, XTTS v2 analyzes the user's unique vocal timbre, pitch, and cadence. It then uses this extracted vocal matrix to synthesize completely new sentences generated by the LLM in a voice that sounds eerily identical to the original user. This transforms the digital twin from a simple text chatbot into a convincing, auditory replicate of the human.
- **Local Execution Capability:** A massive achievement of this project was replacing the highly expensive, cloud-based ElevenLabs API with the open-source Coqui XTTS model. Despite being a highly complex, 2GB+ neural network, the system successfully loads and executes this model locally. By doing so, the project maintains its zero-cost architecture while matching the premium voice quality previously only available to enterprise platforms. The local execution ensures that generated audio files are written directly to the ``storage/voices/`` directory, completely isolating the user's biometric voice data from the internet.
- **Emotional Nuance and Prosody:** XTTS v2 excels at capturing and replicating prosody—the rhythm, stress, and intonation of speech. When the LLaMA 3.1 model generates a response with punctuation like exclamation points, question marks, or ellipses, XTTS naturally adjusts the generated audio to reflect the appropriate emotional inflection. If the twin is recalling a happy memory, the synthesized voice inherently carries a lighter, more dynamic tone. This emotional auditory synchronization works in tandem with the visual avatar expressions to create a multi-sensory, highly believable digital twin interaction.
- **Robust Fallback Architecture:** Because XTTS v2 is highly resource-intensive and requires significant RAM to load, the backend engineers implemented a robust, multi-tiered TTS fallback chain in ``voice/tts.py``. If the host machine lacks the memory to load Coqui, or if an unexpected error occurs during audio synthesis, the system automatically degrades gracefully. It immediately attempts to use Edge-TTS (high-quality Microsoft voices) as a secondary option, and ``gTTS`` or ``pyttsx3`` as tertiary fallbacks. This guarantees that the chat interface will never break down and will always deliver auditory feedback to the user.
- **Low-Latency Audio Generation & Streaming Playback:** The TTS pipeline is optimized to deliver minimal latency between text generation and audio playback, ensuring a seamless conversational experience. Instead of waiting for the entire audio file to be generated, the system can begin playback as soon as the initial audio segments are ready. This creates a near real-time “speaking” effect, making the digital twin feel more alive and responsive. Efficient buffering and file handling ensure that audio is streamed.

4.8. Gemini API

- **High-Quality Avatar Generation:** The Gemini API (specifically ``gemini-2.0-flash-exp``) is utilized to solve the complex visual representation challenge of the digital twin. During the onboarding phase, the system captures four raw webcam photos of the user representing different core emotions. The Gemini API takes these raw references and utilizes its advanced multi-modal generative capabilities to create clean, highly polished, cartoon-style avatars. These generated images ensure that the digital twin maintains a stylized, aesthetically pleasing appearance rather than relying on potentially poorly lit, raw webcam snapshots.
- **Multi-Modal Emotion Synchronization:** The integration of the Gemini API goes beyond simple image-to-image translation. The backend specifically prompts the API to generate four distinct variations of the avatar: neutral, happy, sad, and angry. By locking in these four emotional states, the frontend chat interface can seamlessly swap the displayed avatar image in real-time based on the mood tags generated by the LLM. This creates a deeply dynamic, reactive visual interface where the twin's facial expression physically changes to match the emotional weight of its current textual response.
- **Replacement for Paid AI Services:** In the original hackathon version of this project, avatar generation was handled by costly, specialized services like Nano Banana 2. By leveraging the generous free tiers of the cutting-edge Gemini API, the Alternate Ego project effectively zeroes out the cost of image generation while actually improving the quality of the output. The backend handles the image serialization and API requests cleanly, ensuring that the generated image buffers are securely downloaded and saved directly to the local ``storage/avatars/`` directory for offline use during chat sessions.
- **Prompt Engineering for Stylization:** To ensure visual consistency across all generated emotions, the system employs highly specific, detailed prompt engineering when interacting with Gemini. The prompts dictate exact stylistic requirements—such as clean lines, specific aspect ratios, and the removal of chaotic background elements. This ensures that the generated avatars fit perfectly within the premium, glassmorphism-styled UI of the Next.js frontend. The API's ability to strictly adhere to these visual constraints ensures the digital twin looks like a cohesive, professionally designed character rather than a random AI generation.
- **Resilient Multi-Provider Fallback & Local Caching:** To ensure uninterrupted avatar generation, the system implements a robust fallback mechanism alongside the primary Gemini pipeline. If the Gemini API fails due to rate limits or network issues, the backend seamlessly switches to alternative generation provided.

4.9. TailwindCSS

- **Rapid UI Styling & Utility-First Design:** TailwindCSS is deeply integrated into the Next.js frontend, providing the utility-first CSS framework required to build the project's highly complex UI rapidly. By applying granular, utility classes directly within the React JSX components, the developers completely eliminated the need for maintaining sprawling, unwieldy CSS stylesheets. This approach drastically accelerated the implementation of the split-screen chat interface, the multi-step onboarding forms, and the intricate video background overlays, ensuring that structural styling changes could be made instantly without fear of cascading side effects.
- **Premium Glassmorphism Aesthetics:** The project explicitly demands a "premium, dynamic design that wows the user," and TailwindCSS is the exact tool used to achieve this. Through the strategic use of utility classes like ``backdrop-blur-md``, ``bg-white/10``, and complex linear gradients (``bg-gradient-to-r from-purple-600 to-indigo-600``), the frontend implements a stunning, modern glassmorphism aesthetic. These highly curated visual tokens create a sense of depth and sophistication, ensuring the digital twin platform feels like a high-end, state-of-the-art application rather than a simple academic prototype.
- **Responsive Layouts & Grid Systems:** The Alternate Ego platform must function seamlessly across various screen sizes, from the full-screen cinematic landing page down to constrained laptop displays. Tailwind's built-in responsive modifiers (e.g., ``md:grid-cols-2``, ``lg:w-1/3``) made implementing complex flexbox and grid layouts trivial. This ensures that the critical chat interface—which must balance a large avatar image alongside a dynamic scrolling message history—always maintains perfect proportions. The framework mathematically guarantees that padding, margins, and spatial relationships remain perfectly harmonized across the entire application.
- **Micro-Animations & Dynamic Interactions:** To make the interface feel truly "responsive and alive," the frontend heavily utilizes Tailwind's state modifiers (like ``hover:``, ``focus:``, and ``active:``) alongside transition utilities. Buttons subtly scale up and increase their glow upon hover (``hover:scale-105 transition-all duration-300``), while chat messages smoothly fade into view. These crucial micro-animations, implemented effortlessly via Tailwind classes, significantly enhance user engagement and provide immediate, satisfying visual feedback during interactions, fulfilling the core design requirement for a highly dynamic user experience.
- **Design Consistency & Theming System:** TailwindCSS enables the creation of a centralized and consistent design system across the entire Alternate Ego interface. By extending Tailwind's configuration (e.g., `tailwind.config.js`), the project defines custom color palettes, typography scales, and spacing tokens that are reused throughout all components. This ensures visual consistency between the onboarding flow, chat interface.

4.10 TypeScript

- **Strict Type Safety for Complex State:** TypeScript is utilized across the entire Next.js frontend to enforce strict type safety, which is absolutely vital when managing the highly complex, asynchronous state transitions of the digital twin application. By defining explicit interfaces for API responses, chat message objects, and user data models, TypeScript completely eliminates an entire category of `undefined` runtime errors. When the frontend attempts to parse the complex JSON responses coming from the FastAPI backend, the compiler guarantees that the expected fields (like `source\_citations` or `mood\_tags`) actually exist before the code ever runs in the browser.
- **Enhanced Developer Experience & Autocomplete:** In a codebase containing thousands of lines of UI logic, TypeScript provides an invaluable layer of developer tooling. The tight integration with VS Code means that developers receive instant, intelligent autocomplete suggestions for all custom React component props, local API endpoints, and complex state hooks. This drastically accelerates the development of intricate features like the `VoiceInterview` component, as the IDE actively guides the developer on how to interact with the underlying browser `MediaRecorder` APIs safely and correctly.
- **Refactoring Confidence & Scalability:** As the Alternate Ego project evolved from a simple hackathon prototype into a full-scale application, significant structural refactoring was required. TypeScript provided the absolute confidence needed to rewrite massive portions of the codebase—such as transitioning the chat history from local state to a unified backend database. The compiler acts as an automated safety net, instantly flagging every single file or component that breaks due to a changed function signature or altered data schema, ensuring that zero regressions slip into the production build.
- **Self-Documenting Codebase:** By requiring explicit type definitions for all function arguments and return types, TypeScript naturally serves as highly accurate, self-updating documentation for the frontend architecture. New contributors or reviewers can instantly understand the shape of the data flowing through the `ScrapingScreen` or the exact configuration options required by the `AvatarView` component simply by reading the interface declarations. This clarity bridges the gap between the Python backend schemas and the React frontend logic, ensuring both halves of the project remain perfectly synchronized.
- **Robust API Contract Enforcement:** TypeScript plays a critical role in maintaining a strict contract between the frontend and backend systems. By aligning TypeScript interfaces with the Pydantic schemas defined in the FastAPI backend, the application ensures end-to-end type consistency across the entire data flow. Any mismatch in API structure—such as missing fields, incorrect data types, or schema changes—is immediately caught during development rather than at runtime.

4.11 Python 3

- **Core Backend & Data Processing Logic:** Python 3 serves as the foundational programming language for the entire backend architecture, chosen specifically for its unrivaled dominance in the fields of artificial intelligence and data processing. It handles everything from the high-level FastAPI web routing down to the low-level file I/O required for saving audio chunks. The language's expressive syntax and vast standard library allow complex logic—such as parsing extracted ZIP files or orchestrating multi-threaded scraping tasks—to be written cleanly and concisely within a few dozen lines of code.
- **Machine Learning Ecosystem Integration:** The digital twin project requires seamless interoperability with cutting-edge machine learning tools. Python acts as the ultimate glue code, easily importing and executing sophisticated libraries like `'transformers'`, `'torch'`, and `'faster-whisper'`. By keeping the entire backend within the Python ecosystem, the developers avoid the nightmare of cross-language inter-process communication. The backend can instantly pass text strings directly from an HTTP request into a deep learning tensor, run the inference, and return the result in one fluid, highly optimized motion.
- **Vector Mathematics & Local AI Logic:** To completely eliminate the dependency on external databases that crashed on Windows systems (like ChromaDB), the engineers utilized Python's core `'math'` and `'json'` libraries to write a bespoke, pure-Python vector store. Python's ability to efficiently handle array comprehensions and calculate exact cosine similarities between massive 768-dimensional vectors showcases its immense versatility. This pure Python implementation guarantees that the critical RAG (Retrieval-Augmented Generation) pipeline executes flawlessly on any operating system without requiring complex C++ compiler setups.
- **Robust File System Management & Security:** The platform relies heavily on local file persistence to store avatars, audio recordings, and encrypted user uploads. Python's `'os'` and `'shutil'` modules provide the robust, cross-platform file system management required to handle these assets safely. When a user requests the deletion of their digital twin, Python executes recursive directory wipes to securely eradicate all traces of biometric data. Furthermore, it easily integrates with the `'cryptography'` package to encrypt highly sensitive data streams on the fly, ensuring maximum privacy protocols are strictly enforced.
- **Asynchronous Processing & Concurrency Handling:** Python's modern async capabilities (via `async/await`) play a crucial role in managing the highly concurrent operations within the Alternate Ego backend. Through frameworks like FastAPI and asynchronous libraries, Python efficiently handles multiple tasks such as API requests, file uploads, WebSocket streaming, and AI inference coordination without blocking the main execution thread. This is especially important when processing large ZIP files.

4.12 BeautifulSoup4

- **Intelligent Web Scraping & Data Extraction:** BeautifulSoup4 is the primary tool used by the backend to extract meaningful, contextual data from the user's public social media URLs. During the critical "Imprint" onboarding phase, the system uses BeautifulSoup to parse the raw, chaotic HTML returned from DuckDuckGo searches and public profile pages. It intelligently navigates the Document Object Model (DOM) to strip away useless navigation bars, footer links, and advertisement scripts, isolating only the pure biographical text, work history, and personal anecdotes that define the user.
- **Fueling the RAG Knowledge Base:** The raw data extracted by BeautifulSoup directly fuels the digital twin's "brain." Because the LLM (LLaMA 3.1) relies entirely on the quality of its contextual prompt, feeding it raw HTML would result in confused, noisy responses. BeautifulSoup sanitizes the web data into clean, highly readable paragraphs. These pristine text blocks are then passed into the `chunker.py` module, embedded, and saved into the vector store. The cleaner the HTML parsing, the more accurate and human-like the digital twin's final generated responses become.
- **Resilient DOM Parsing Capability:** Modern web pages are notoriously poorly structured, often containing broken tags, inconsistent hierarchies, and deeply nested `div` containers. BeautifulSoup was specifically chosen over strict XML parsers because of its incredible resilience in handling "tag soup." It allows the backend scraping pipeline to cleanly extract text even if the target social media profile has bizarre formatting. This reliability ensures that the onboarding process rarely fails due to an unexpected website update, providing a stable data ingestion pipeline for the AI.
- **Zero-API-Key Data Aggregation:** By combining BeautifulSoup with simple HTTP requests, the Alternate Ego project completely bypasses the need for expensive, commercial scraping APIs (like EXA AI, which was used in the original version). This technology choice directly supports the project's foundational goal of operating at exactly ₹0 cost. It democratizes the ability to build a comprehensive data matrix for the digital twin, proving that sophisticated, context-aware AI systems can be built purely using open-source, localized data extraction tools.
- **Customizable Parsing Rules & Extensibility:** BeautifulSoup enables the system to define flexible, platform-specific parsing strategies for different types of web content. By using targeted selectors (like class names, tags, or attributes), the backend can adapt extraction logic for LinkedIn, Twitter, or personal blogs without rewriting the entire pipeline. This makes the scraping engine highly extensible as new data sources are added in the future. Developers can fine-tune extraction rules to prioritize meaningful sections such as bios, posts, or experience details while ignoring irrelevant noise. Additionally, fallback parsing strategies can be implemented if the primary selectors fail due to layout changes. This adaptability ensures long-term robustness of the data ingestion system.

4.13 Face-api.js

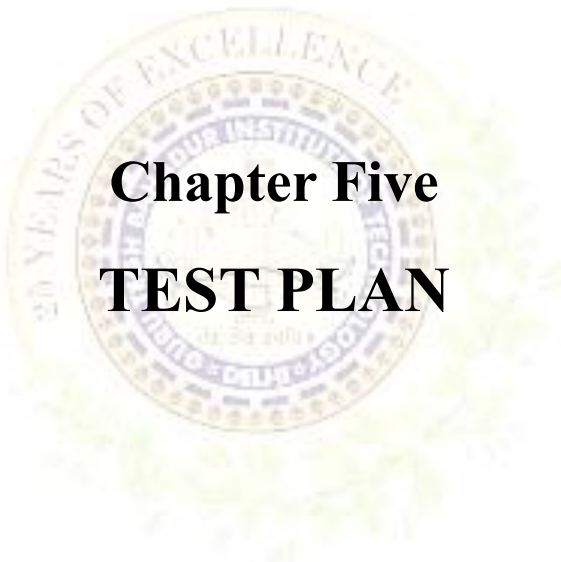
- **Browser-Side Computer Vision:** Face-api.js brings highly advanced, real-time computer vision capabilities directly into the Next.js client browser. Built on top of TensorFlow.js, this library allows the application to detect faces and analyze micro-expressions via the user's webcam without sending a single frame of video data to the backend server. This client-side execution is a monumental privacy achievement, ensuring that the user's raw biometric video feed never traverses the internet, strictly adhering to the project's privacy-first, fully local design philosophy.
- **Real-Time Emotion Validation:** During the critical photo-capture onboarding phase, the system requires the user to submit four specific emotional states: neutral, happy, sad, and angry. Face-api.js constantly scans the live webcam feed, mathematically analyzing facial landmarks to calculate an emotional probability score in real-time. The UI utilizes these live calculations to provide instant visual feedback—only allowing the user to capture the photo when the library confirms, with high confidence, that the user is actually expressing the correct target emotion.
- **Ensuring High-Quality Avatar References:** The quality of the final generated AI avatars heavily depends on the quality of the raw reference photos. If a user accidentally captures a blurry image or looks away from the camera, the Gemini API will generate a flawed avatar. Face-api.js acts as an intelligent, automated quality-control gatekeeper. By continuously verifying that exactly one face is clearly visible, centered, and correctly emotive, it guarantees that only pristine biometric data is passed down the pipeline, resulting in significantly higher quality digital twin visualizations.
- **Hardware Accelerated Edge Computing:** Despite running inside a standard web browser, face-api.js intelligently utilizes WebGL to tap into the user's local GPU for hardware acceleration. This ensures that the complex neural network inference required for 68-point facial landmark detection runs at a smooth 30+ frames per second without lagging the UI. This highly optimized edge computing approach provides a fluid, premium user experience, making the complex onboarding process feel magical, instantaneous, and highly sophisticated to the end user.
- **Lightweight Model Loading & On-Demand Execution:** Face-api.js is optimized for efficient browser performance by loading only the required pre-trained models (e.g., face detection, landmark recognition, emotion analysis) on demand. Instead of bundling heavy models upfront, the system dynamically fetches them when the onboarding process begins, reducing initial page load time. This ensures that the landing page and main UI remain fast and responsive. Once loaded, the models are cached in the browser, avoiding repeated downloads in subsequent sessions.

4.14 Nomic-Embed-Text

- **High-Dimensional Vectorization:** Nomic-Embed-Text is the specialized embedding model utilized by the Ollama runtime to map human language into a mathematical space. When the system ingests the user's voice transcripts or scraped social media data, this model converts sentences into massive, 768-dimensional numerical arrays (vectors). By mapping concepts based on semantic meaning rather than just keyword matching, it allows the digital twin's "brain" to understand that the phrase "I adore coding" is mathematically identical in meaning to "Software development is my passion."
- **Powering the RAG Similarity Search:** This embedding technology is the absolute backbone of the Retrieval-Augmented Generation (RAG) architecture. When the user asks the digital twin a question in the chat interface, the backend instantly uses Nomic-Embed-Text to convert that specific query into a vector. The pure Python vector store then rapidly calculates the cosine similarity between the query vector and all thousands of stored memory vectors. It accurately surfaces the top 5 most relevant pieces of the user's history, guaranteeing that the LLM has the exact factual context needed to reply accurately.
- **Optimal Balance of Size and Accuracy:** Nomic-Embed-Text was specifically selected for this local architecture because it represents the perfect golden ratio between extremely high accuracy and low hardware requirements. Unlike massive enterprise embedding models that require huge amounts of VRAM, this model is lightweight enough (~270MB) to load entirely into standard system RAM almost instantly. Despite its small footprint, its massive context window and high-dimensional output provide retrieval accuracy that rivals closed-source, paid APIs like OpenAI's 'text-embedding-3-small'.
- **Zero-Cost Privacy Enforcement:** In the previous iteration of the project, all user text had to be transmitted to external servers just to be embedded, representing a massive security vulnerability. By running Nomic-Embed-Text locally via Ollama, the project ensures that the process of transforming highly personal journals or chat logs into mathematical vectors happens entirely offline. Furthermore, once the text is embedded, the raw text can be securely deleted, leaving only the unreadable vectors behind, creating a perfectly secure, irreversible privacy pipeline.
- **Efficient Batch Embedding & Throughput Optimization:** Nomic-Embed-Text supports efficient batch processing, allowing the system to convert multiple text chunks into embeddings in a single request. This significantly improves throughput during the onboarding phase, where large volumes of data—such as social media exports and voice transcripts—must be processed quickly. Instead of embedding each sentence individually, the backend groups chunks intelligently, reducing API overhead and latency. This batching approach ensures faster indexing into the vector store while maintaining consistent embedding quality.

4.15 Cryptography (Fernet)

- **Military-Grade Data-at-Rest Encryption:** The Python ``cryptography`` library, specifically utilizing the Fernet symmetric encryption module, provides the impenetrable security layer for all sensitive files handled by the platform. When a user uploads a deeply personal ZIP or JSON data export containing their entire Instagram or LinkedIn history, the system immediately encrypts the file the moment it touches the local disk. Fernet guarantees that the data is protected by 128-bit AES encryption in CBC mode, ensuring that even if the host machine is compromised, the raw data files cannot be read without the specific, auto-generated master key.
- **Secure Privacy Pipeline Integration:** The encryption module is deeply wired into the application's unique, privacy-first data lifecycle. The backend pipeline operates strictly under the principle of minimal exposure: an uploaded file is encrypted immediately, temporarily decrypted strictly within RAM only during the exact moment the RAG chunking algorithm needs to read it, and then instantly purged. This ensures that highly sensitive plaintext data is never left sitting exposed on the user's hard drive, fulfilling the project's extreme commitment to user privacy.
- **Automated Key Management System:** To maintain a frictionless, zero-configuration user experience, the system implements an automated, completely transparent key management protocol. Upon the first server startup, the ``encryption.py`` module automatically generates a highly secure master key and stores it as a hidden file (``.encryption_key``). The backend seamlessly retrieves and utilizes this key for all subsequent encrypt/decrypt operations, ensuring maximum security without requiring the user to manually manage passwords, memorize seed phrases, or configure complex environment variables.
- **Deterministic Secure Deletion Protocols:** True privacy requires more than just encryption; it requires the absolute assurance of data eradication. The cryptography module works in tandem with the Python ``shutil`` library to provide a highly deterministic ``/delete-all`` endpoint. When a user chooses to delete their digital twin, the system performs a comprehensive, secure wipe of the local filesystem—targeting the specific ``/uploads/``, ``/avatars/``, and ``/voices/`` directories—while simultaneously executing cascade deletions in the SQLite database and Vector store. It mathematically guarantees that zero traces of the user's identity remain on the system.
- **Integrity Verification & Tamper Detection:** Beyond encryption, the Fernet scheme inherently provides built-in integrity verification using HMAC (Hash-based Message Authentication Code). Every encrypted file includes a cryptographic signature that ensures the data has not been altered or corrupted. When the system attempts to decrypt a file, Fernet automatically validates this signature before allowing access.



Chapter Five

TEST PLAN

Chapter 5: TEST PLAN

5.1 Introduction

The Test Plan for Alternate Ego establishes a comprehensive framework for validating and verifying the platform's functionality, performance, security, and usability across all major system components. These include the **Next.js digital twin interface, Vite-based landing page, FastAPI backend, Ollama AI server, and Gemini-powered avatar generation modules. This document defines the testing objectives, scope, methodologies, required resources, execution schedules, and success criteria to ensure that the system meets all specified requirements.

The primary goal of this testing strategy is to guarantee that the platform delivers a reliable, accurate, and seamless **voice-enabled digital twin experience**. The testing process is structured across multiple levels, beginning with unit testing of individual API endpoints and progressing to integration testing of interconnected AI modules. System testing is conducted to validate the behavior of the complete application under realistic conditions, followed by user acceptance testing with representative users to ensure usability and satisfaction.

Quality assurance activities cover both functional and non-functional aspects of the system. Functional testing validates the correctness of key features such as onboarding workflows, voice interviews, RAG-based responses, and avatar synchronization. Non-functional testing evaluates system performance, including latency, response time, and resource utilization on consumer-grade hardware. Regression testing ensures that new updates do not break existing functionality, while exploratory testing identifies edge cases, especially in AI-driven conversations and multi-modal interactions.

In addition to standard testing practices, special emphasis is placed on **AI-specific validation**, which is critical for a system like Alternate Ego. This includes verifying the accuracy of speech-to-text transcription, consistency of personality-based responses, correctness of retrieved contextual data in the RAG pipeline, and synchronization between generated text, voice output, and avatar expressions. Since AI outputs can be probabilistic in nature, testing focuses not only on correctness but also on relevance, coherence, and user satisfaction.

The test plan also incorporates **security and privacy validation**, which are core principles of the platform. Testing ensures that all sensitive user data is encrypted at rest, no data is transmitted externally, and all APIs are restricted to localhost access. Special test cases are designed to validate the `/api/privacy/delete-all/{twin_id}` endpoint to ensure complete and irreversible data deletion. This guarantees compliance with privacy-first design goals and builds user trust in the system.

Overall, this test plan ensures that Alternate Ego operates as a **robust, scalable, secure, and user-centric AI system**, capable of delivering consistent performance and high-quality interaction across all modules. It serves as a critical step in transforming the platform from a technically functional prototype into a production-ready digital twin solution.

5.2 Objectives

- **Verify Functional Correctness Across All Features:** The primary objective of testing is to ensure that every functional requirement defined in the Software Requirement Specification is correctly implemented and behaves as expected under both normal and edge-case conditions. This includes validating the complete onboarding pipeline, where user inputs such as name, social media links, voice responses, and webcam captures are accurately processed and stored. The system must ensure that the interview flow captures meaningful personality data and feeds it correctly into the RAG pipeline. The chat interface, built using **Next.js**, must correctly process user prompts, maintain conversational context, and communicate seamlessly with the **FastAPI** backend. The backend should correctly trigger the **Ollama** engine to generate responses. Additionally, synchronization between generated text, voice output, and avatar expressions must be validated to ensure a seamless and immersive digital twin experience.
- **Validate Performance, Scalability, and Reliability Under Load:** The second objective focuses on validating non-functional requirements such as system performance, responsiveness, and stability under realistic usage conditions. Performance testing ensures that the landing page loads quickly, API calls between frontend and backend are executed within acceptable time limits, and AI response latency remains low even on CPU-based systems. The system must handle continuous conversational interactions without lag, ensuring smooth transitions between user input, AI processing, and output delivery. Load testing simulates multiple operations such as concurrent API requests, large data ingestion, and repeated chat interactions. The backend must efficiently manage tasks like vector search, voice processing, and WebSocket streaming without bottlenecks. Reliability testing ensures that the system gracefully handles failures such as temporary AI unavailability, microphone access issues, or incomplete data uploads.
- **Ensure Security, Data Privacy, and Regulatory Compliance:** The third objective emphasizes validating the platform's privacy-first and security-focused architecture. Security testing ensures that all APIs are restricted to localhost access, preventing unauthorized external connections. Input validation mechanisms are tested to prevent malicious inputs, including prompt injection attacks targeting the language model. Data privacy testing verifies that all sensitive user data—such as voice recordings, images, and personal text—is encrypted at rest using secure encryption mechanisms. It also ensures that no data is transmitted outside the local environment. The system must enforce strict data isolation, ensuring that each digital twin's data is accessible only within its own context. Additionally, the `/api/privacy/delete-all/{twin_id}` endpoint is rigorously tested to confirm complete and irreversible data deletion. These measures ensure compliance with modern data protection principles and reinforce user trust in the platform.

- **Validate AI Accuracy and Personalization Quality:** An additional objective is to evaluate the correctness and quality of AI-driven outputs, which are central to the digital twin experience. Testing ensures that the RAG system retrieves relevant contextual data and that the generated responses are coherent, context-aware, and aligned with the user's personality. Speech-to-text accuracy is validated to ensure correct transcription of voice interviews, while text-to-speech output is tested for clarity and synchronization. Avatar emotion switching is verified to ensure correct mapping with detected mood tags.
- **Ensure Usability and User Experience Consistency:** The final objective is to validate the overall usability and user experience of the platform. The onboarding flow must be intuitive, smooth, and free from confusion or friction. The chat interface should be responsive, visually engaging, and easy to use. UI elements such as buttons, animations, and transitions must function correctly across different screen sizes. Special attention is given to real-time feedback mechanisms, such as progress indicators and audio playback. The system must provide clear error messages and recovery options in case of failures. Overall, this objective ensures that the platform not only functions correctly but also delivers a polished and user-friendly experience.

5.3 Features to be Tested

- **Structured Onboarding and Interview Flow:** The onboarding and interview pipeline is a foundational component of the Alternate Ego platform, responsible for capturing high-quality user data required for digital twin creation. Testing ensures that the multi-step onboarding flow operates smoothly, including name input, social media URL collection, data upload, webcam capture, and voice interview recording. The system must correctly handle dynamic question selection from the interview bank and ensure category diversity. Input validation is verified to prevent incomplete or invalid submissions. Additionally, retry and re-record mechanisms are tested to ensure users can edit responses without losing progress. Real-time transcription accuracy and storage of responses are also validated to ensure reliable downstream processing.
- **Digital Twin Persona Generation:** This feature focuses on transforming collected onboarding data into a coherent and personalized digital twin representation. Testing verifies that the system correctly processes interview responses, social media data, and uploaded files to construct meaningful personality embeddings. The RAG pipeline must accurately retrieve relevant context and pass it to the language model. The generated persona should reflect user-specific traits such as tone, preferences, and experiences without introducing hallucinated or fabricated details. Consistency across multiple interactions is also evaluated to ensure stable personality behavior. The system prompt construction logic is tested to confirm that it properly integrates user data into the conversational model.

- **Voice-Enabled Chat Interface and Audio Synchronization:** The chat interface, built using **Next.js**, serves as the primary interaction medium between the user and their digital twin. Testing validates microphone input capture, ensuring compatibility across browsers and proper permission handling. Speech-to-text transcription must be accurate and responsive, while text-to-speech output must be clear and synchronized with the UI. Special attention is given to audio playback behavior, ensuring that responses do not overlap, cut off prematurely, or introduce latency issues. The system must also handle fallback mechanisms in the voice pipeline seamlessly. Additionally, synchronization between generated responses, voice output, and avatar emotion switching is tested to ensure a cohesive multi-modal interaction experience.
- **Multi-Service Architecture Communication:** Alternate Ego operates on a distributed local architecture involving multiple interconnected components. Testing validates seamless communication between the Vite landing page, the **Next.js** frontend (port 3000), the **FastAPI** backend (port 8000), and the **Ollama** server (port 11434). API request and response formats are verified to ensure consistency and correctness. Cross-Origin Resource Sharing (CORS) configurations are tested to allow secure communication between services. Error handling mechanisms are validated for scenarios such as missing environment variables, failed API calls, or service unavailability. This ensures robust inter-module communication and system stability.
- **Data Processing, RAG Pipeline, and Storage Validation:** Testing includes validation of the data ingestion pipeline, ensuring that uploaded files (ZIP, JSON, CSV, TXT) are correctly parsed, processed, and converted into structured text. Topic-based chunking is verified to ensure semantic integrity of data segments. The embedding generation process using local models is tested for consistency and correctness. The custom vector store is validated for accurate cosine similarity search and retrieval performance. Additionally, database operations using SQLite are tested to ensure proper storage of user data, conversations, and system states. Data integrity, consistency, and retrieval accuracy are key focus areas in this scope.
- **Security, Privacy, and Data Protection Mechanisms:** The scope of testing also includes validation of all security and privacy-related features. This includes verifying encryption of sensitive data at rest, ensuring that raw files are not stored in plaintext. Localhost-only access restrictions are tested to prevent unauthorized external connections. The `/api/privacy/delete-all/{twin_id}` endpoint is rigorously tested to confirm complete and irreversible data deletion across all storage layers. Input validation is performed to prevent malicious inputs and ensure system robustness. These tests ensure compliance with the platform's privacy-first design principles.
- **User Interface and Experience Validation:** The frontend user experience is tested to ensure smooth navigation, responsiveness, and visual consistency. This includes validating transitions between onboarding steps, responsiveness of UI components, and real-time feedback mechanisms. The cinematic landing page, chat interface, and avatar panel are tested for performance and usability. Cross-browser compatibility and responsiveness across different screen sizes are also verified. Error messages and recovery flows are tested to ensure clarity and usability. This ensures that the platform delivers a polished and intuitive user experience.

5.4 Testing Methodology

The testing methodology for Alternate Ego adopts a comprehensive, multi-level approach to ensure the reliability, correctness, and robustness of the entire system. Given the complexity of the platform—integrating frontend interfaces, backend APIs, and local AI engines—the testing process is structured across multiple layers, including unit testing, integration testing, system testing, and user acceptance testing. This layered strategy ensures that each component is validated individually before being tested as part of a larger interconnected system.

At the foundational level, unit testing focuses on validating individual components and functions in isolation. For the frontend, frameworks such as Jest are used to test UI components, state management logic, and user interactions within the **Next.js** and Vite-based interfaces. On the backend, pytest is used to validate FastAPI endpoints, data processing functions, and utility modules. These tests ensure correct behavior for known inputs, proper handling of edge cases, and meaningful error responses when invalid data is encountered.

Moving beyond isolated components, integration testing verifies the interaction between different modules and services. This includes testing the communication between the frontend and backend, ensuring that API requests and responses are correctly formatted and handled. It also validates backend integration with the **Ollama** server for language model inference and embedding generation, as well as integration with external AI services such as Gemini for avatar creation. These tests ensure that data flows seamlessly across the system and that dependencies function correctly together.

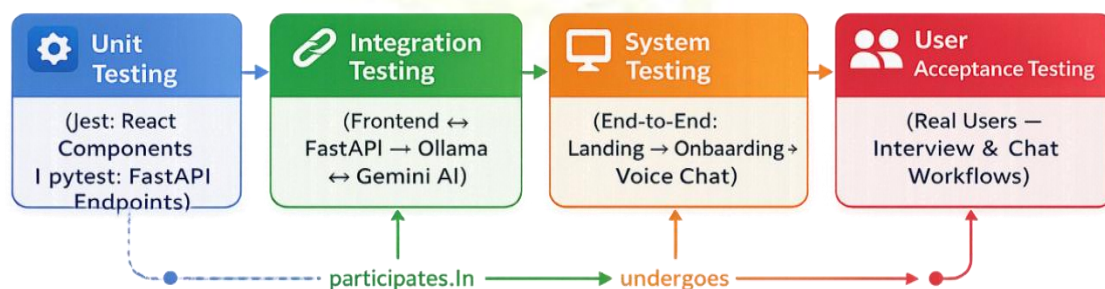


Figure 5.1: Testing Methodology

At a higher level, system testing evaluates the complete application as an integrated whole. This involves validating full end-to-end workflows, starting from the landing page, progressing through the onboarding and interview process, and culminating in real-time interaction with the digital twin via the chat interface. The system is tested under realistic conditions to ensure that all components—from data ingestion to AI response generation—work cohesively without failures.

Finally, user acceptance testing (UAT) is conducted with representative users to evaluate the system from an end-user perspective. This phase focuses on usability, responsiveness, and overall user satisfaction. It ensures that the platform delivers a smooth, intuitive, and engaging experience, meeting both technical requirements and user expectations.

Overall, this multi-level testing methodology ensures that Alternate Ego is not only functionally correct but also reliable, scalable, and ready for real-world deployment.

5.5 Test Execution

Test execution for Alternate Ego is carried out in a structured, phased manner to ensure continuous validation of the system throughout the development lifecycle. The process begins at the development stage, where unit tests are executed locally by developers before committing code changes. These tests validate individual components such as frontend UI logic, backend API endpoints, and utility functions, ensuring that new code does not introduce errors. Developers verify correctness, edge-case handling, and proper error messaging at this stage, establishing a strong foundation for subsequent testing phases.

Following local validation, the testing process integrates with continuous integration (CI) workflows, where automated pipelines execute predefined test suites on every code commit. This ensures consistent code quality across both frontend and backend repositories. The CI process detects regressions early, enforces coding standards, and prevents unstable code from being merged into the main branch. Automated testing at this stage significantly reduces manual effort and improves overall development efficiency.

Once individual components are verified, integration testing is performed in controlled environments that closely replicate real system conditions. These environments include active local AI runtimes such as Ollama and configured external services for avatar generation. Integration tests validate communication between modules, including frontend-to-backend API interactions and backend-to-AI service calls. This phase ensures that all interconnected components function seamlessly and that data flows correctly across the system.

System testing is then conducted by deploying the complete application stack, including the landing page, frontend interface, backend services, and AI runtime. This phase simulates real-world user scenarios, such as onboarding, voice interviews, and interactive conversations with the digital twin. Testing includes validating audio input capture, real-time transcription, response generation, and synchronization of voice and avatar outputs. Logs and performance metrics are monitored to identify bottlenecks, latency issues, or unexpected failures during execution.

Defect tracking is an essential part of the test execution process, focusing on identifying, documenting, and resolving issues efficiently. Special attention is given to AI-specific

challenges such as hallucinated responses from the language model, audio synchronization drift, and intermittent API connection failures. Each defect is logged with detailed reproduction steps and severity levels, enabling rapid debugging and resolution. This structured approach ensures continuous improvement of system stability and performance.

Overall, the test execution process ensures that Alternate Ego evolves through iterative validation, maintaining high standards of quality, reliability, and user experience at every stage of development.

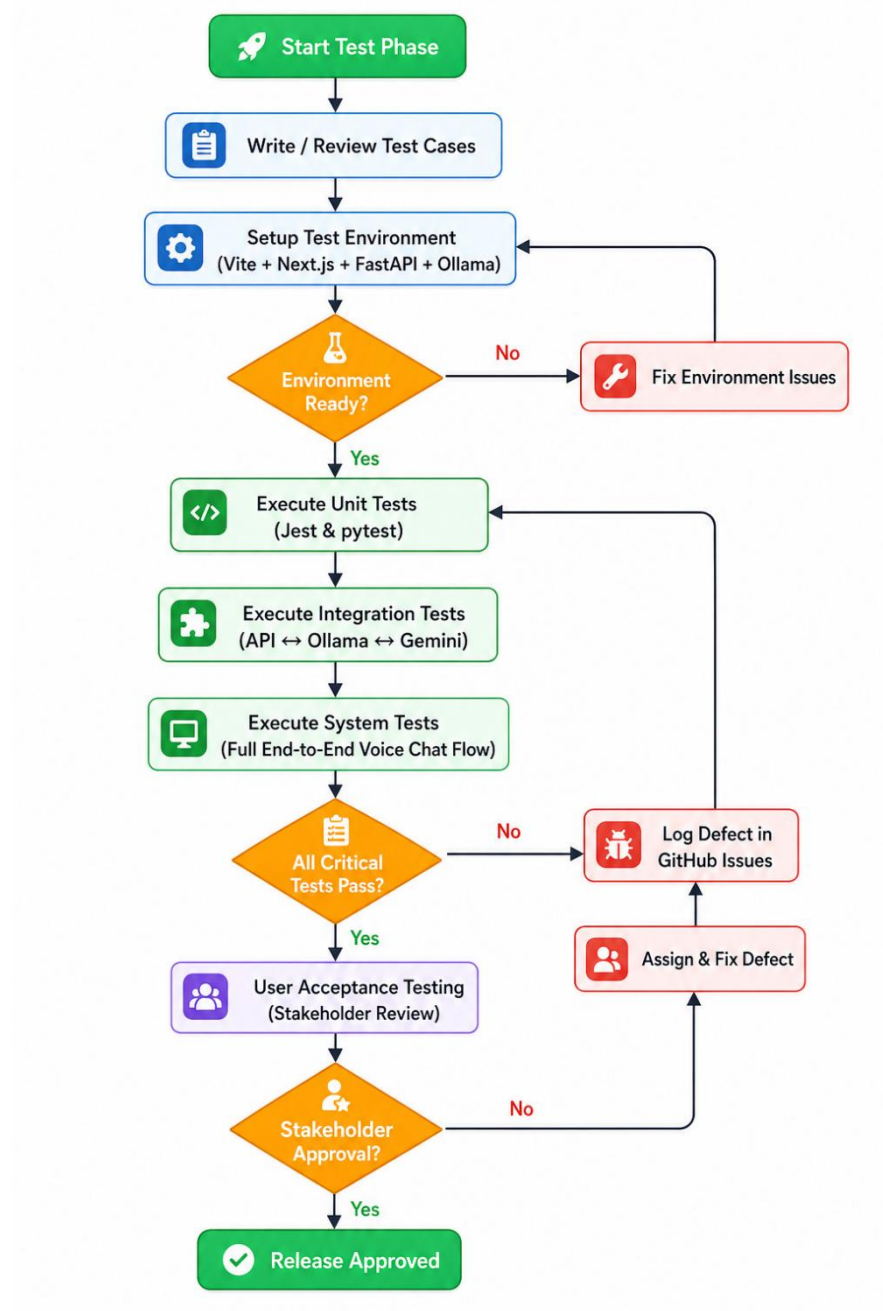


Figure 5.2: Test Execution

5.6 Testing Scope and Environment

In-Scope Testing Components

- **Frontend User Interfaces (Next.js & Vite):** This component represents a critical testing focus as it directly impacts user experience and interaction. Testing validates that all React-based UI components render correctly across different states and screen sizes. Seamless routing between the Vite landing page and the **Next.js** application is verified to ensure smooth navigation without full-page reloads. Interactive elements such as microphone toggles, retry buttons, and onboarding controls are tested for correct behavior and responsiveness. Special attention is given to real-time feedback mechanisms, ensuring that audio recording and playback states provide clear visual indicators to the user. UI transitions, animations, and responsiveness are also validated to ensure a polished and intuitive experience.
- **Backend API and Routing (FastAPI):** The backend layer, implemented using **FastAPI**, is tested extensively to ensure robust and efficient handling of all API requests. Testing validates that REST endpoints correctly accept and process structured inputs, including text queries, audio files, and data uploads. The system is evaluated for proper request validation, error handling, and response formatting. Performance testing ensures that the backend can handle concurrent requests without bottlenecks. Additionally, routing logic is tested to confirm seamless communication with the **Ollama** engine for AI inference and external services for avatar generation. This ensures reliable backend orchestration of all system functionalities.
- **Digital Twin Integration and Audio Synchronization:** This component focuses on validating the core intelligence and interaction logic of the platform. Testing ensures that user context—collected during onboarding—is correctly integrated into prompts sent to the language model. The interview state machine is rigorously tested to ensure accurate tracking of user progress and proper handling of transitions between different onboarding stages. Additionally, the audio synchronization mechanism is thoroughly evaluated to ensure that generated voice outputs align perfectly with text responses and UI updates. Tests verify that there are no delays, overlaps, or audio artifacts during playback. This ensures a smooth, realistic, and immersive conversational experience with the digital twin..
- **RAG Pipeline and Vector Store Validation:** This component focuses on testing the core Retrieval-Augmented Generation (RAG) system that powers the digital twin's memory and contextual intelligence. Testing validates that data ingestion from multiple sources (voice transcripts, social media scraping, uploaded files) is correctly processed and chunked using topic-based segmentation. The embedding generation process is verified to ensure consistent 768-dimensional vector creation. The custom vector store is rigorously tested for accurate cosine similarity calculations and correct retrieval of the most relevant context. Additionally, tests ensure that irrelevant or low-similarity data is not retrieved, maintaining

response quality. This guarantees that the AI responses remain contextually accurate, personalized, and free from noise.

- **Backend Security, Privacy, and Data Protection Mechanisms:** This component ensures that all privacy-first and security requirements are strictly enforced throughout the system. Testing validates that sensitive user data is encrypted at rest and never stored in plaintext after processing. Localhost-only access restrictions are verified to prevent unauthorized external access. The `/api/privacy/delete-all/{twin_id}` endpoint is thoroughly tested to confirm complete and irreversible deletion of all user data across the database, vector store, and file system.

Out-of-Scope Testing Components

- **Internal Third-Party AI Mechanics:** The internal workings of third-party AI systems are not part of the testing scope. This includes the core natural language processing mechanisms of Gemini-based services and the internal model architecture, weights, and training behavior of the **Ollama** models. These components are treated as black-box systems, meaning testing is limited to validating their inputs and outputs rather than their internal computations. The focus remains on ensuring that the platform integrates with these services correctly and that the responses generated are usable and consistent with expectations.
- **Mobile Native Application Features:** The current scope of testing does not include native mobile application development or platform-specific mobile features. While the web-based interface—built using **Next.js**—is tested for responsiveness across different screen sizes, dedicated iOS or Android applications are not considered. Additionally, advanced mobile hardware integrations beyond standard browser capabilities (such as native sensors or device-level APIs) are excluded. The testing effort is focused entirely on the web application experience delivered through modern browsers.
- **Performance Testing Beyond Specified Load:** Extreme performance and stress testing scenarios are outside the scope of this test plan. This includes testing the absolute limits of the system under heavy loads, such as large-scale distributed denial-of-service (DDoS) simulations or pushing the local AI runtime beyond realistic usage conditions. Testing is instead focused on practical, real-world scenarios, such as single-user interactions or limited concurrent usage. The goal is to ensure stable and reliable performance under expected operating conditions rather than extreme edge cases.
- **Cloud Deployment and Distributed Infrastructure Testing:** Testing related to large-scale cloud deployment, distributed microservices orchestration, and infrastructure-level configurations is outside the current scope. The system is designed primarily as a local-first application, and therefore cloud-native concerns such as container orchestration

(Docker/Kubernetes), load balancing across multiple servers, and distributed database consistency are not evaluated. Future scalability testing for multi-tenant cloud environments is considered an extension but is not included in the present test plan.

- **Third-Party Service Availability and Rate Limits:** The reliability, uptime, and rate-limiting behavior of external services used in the system (such as free-tier AI APIs for avatar generation) are not tested as part of this plan. These services are dependent on external providers and may vary over time. The system only validates proper handling of responses, fallbacks, and error conditions when such services fail or become unavailable. However, performance benchmarking or SLA validation of these third-party services is beyond the testing scope\

Testing Environment Configuration

- **Development Environment:** Developers maintain fully configured local environments to support rapid development and testing. This includes running the **Next.js** development server (`npm run dev`), the Vite server for the landing page, the **FastAPI** backend via Uvicorn, and a local instance of **Ollama** for AI inference. This setup allows developers to test features in isolation and debug issues efficiently. Mock APIs or stubbed responses may be used during development to avoid rate limits or delays from external services. This environment is primarily focused on rapid iteration, debugging, and unit-level validation.
- **Continuous Integration Environment:** The continuous integration (CI) environment ensures automated testing and validation of the codebase on every commit. Tools such as GitHub Actions are used to run automated pipelines that execute backend tests, frontend component tests, and code quality checks. Python linters and test frameworks validate backend logic, while frontend builds are verified for successful compilation. This environment ensures that no broken or unstable code is merged into the main branch. It also helps maintain consistent coding standards and early detection of regressions.
- **Staging Environment:** The staging environment replicates a production-like setup where all configurations, environment variables, and dependencies are properly defined. This includes API endpoints, authentication tokens, and integration with AI services. In this environment, the full system—including frontend, backend, and AI runtime—is deployed and tested as a unified application. End-to-end workflows such as onboarding, voice interaction, RAG-based responses, and avatar synchronization are validated using real models and services. This environment serves as the final validation stage before production deployment.
- **Production Environment:** The production environment represents the live system accessible to end users. Testing in this environment is limited to smoke testing and monitoring to ensure system stability after deployment. Key checks include verifying API

responsiveness, monitoring latency, and ensuring that critical features such as chat interaction, audio playback, and onboarding flows function correctly. Error logging mechanisms are used to capture issues such as missing modules, failed API calls, or audio playback errors. Continuous monitoring ensures that any issues are quickly identified and resolved without impacting user experience.

- **User Acceptance Testing (UAT) Environment:** The UAT environment is designed for testing the system with real or representative users to evaluate usability and overall experience. In this setup, users interact with the complete application, including onboarding, voice interviews, and chat with the digital twin. Feedback is collected on ease of use, responsiveness, and perceived realism of the AI. This environment helps identify usability issues, confusing workflows, or unexpected behavior that may not be captured.

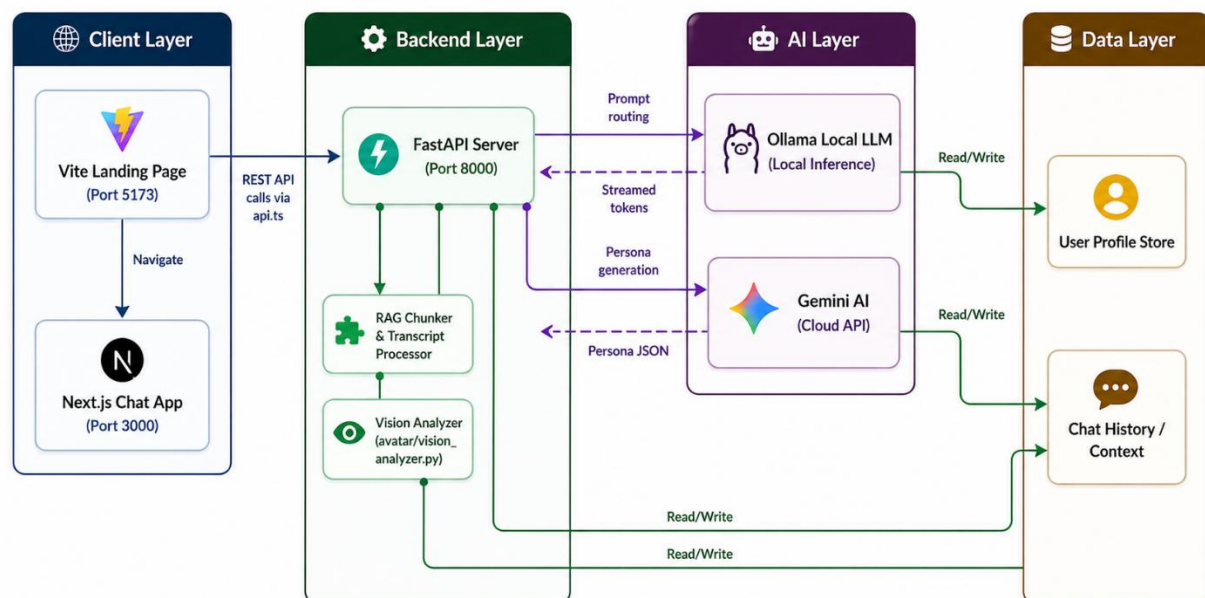


Figure 5.3: Testing Environment Configuration

5.7 Defect Management

Defect Identification and Reporting

Defects in the Alternate Ego platform are identified through a combination of automated testing processes, manual quality assurance activities, and system-level logging mechanisms. Automated detection primarily occurs through failures in continuous integration pipelines, where test cases, build validations, and linting checks highlight issues immediately after code commits. In addition, manual QA testing—especially focused on complex features such as voice interaction and real-time chat—is conducted to identify issues that cannot be captured through automated scripts alone. Backend logs generated by the **FastAPI** server and AI

runtime provide detailed insights into runtime errors, API failures, and unexpected behavior during execution.

All identified defects are documented using a standardized reporting format to ensure clarity and reproducibility. Each defect report must include essential details such as browser version, operating system, and hardware context—particularly microphone and audio device configurations, which are critical for voice-related features. The exact user prompt or input that triggered the issue must also be recorded, along with relevant server logs to aid debugging. This structured approach ensures that developers can quickly understand, reproduce, and resolve reported issues.

Special attention is given to audio synchronization and voice interaction defects, as these directly impact user experience. For such issues, screen recordings with audio are mandatory to capture the exact behavior of the system, including timing mismatches, playback interruptions, or transcription delays. These recordings provide valuable context that cannot be conveyed through text alone.

Overall, the defect tracking and reporting process ensures efficient identification, documentation, and resolution of issues, contributing to continuous improvement of system quality, stability, and user satisfaction.

Defect Severity Classification

- **Critical Severity:** Critical defects are those that completely block the functionality of the system or prevent users from accessing core features. These include application crashes, failure of the frontend to load due to module resolution errors in the **Next.js** application (e.g., unresolved imports), and situations where the **FastAPI** backend fails to start. Additionally, failures in connecting to the **Ollama** server or external AI services that completely prevent response generation or avatar creation are categorized as critical. Such issues must be addressed immediately as they render the system unusable.
- **High Severity:** High-severity defects significantly impact the user experience and core functionality but do not completely block access to the system. Examples include failure of voice-chat audio playback, major audio synchronization issues causing overlapping or distorted responses, and freezing of the onboarding or interview flow that prevents users from progressing. These issues disrupt the primary purpose of the application and require urgent resolution to restore smooth functionality.
- **Medium Severity:** Medium-severity defects affect system performance or usability but allow the application to remain functional. These include occasional delays in AI response times, minor inconsistencies in UI interactions such as retry buttons requiring multiple clicks, or temporary glitches during data processing. Visual issues such as layout misalignment or animation inconsistencies on the landing page may also fall into this category. While not critical, these defects should be resolved to improve overall user experience.

- **Low Severity:** Low-severity defects are minor issues that do not significantly impact system functionality or user experience. These include typographical errors in onboarding text, minor formatting inconsistencies in the chat interface, or non-critical warnings in the browser console. Such issues are typically cosmetic and can be addressed in later development cycles without affecting core operations.

Defect Priority Classification

- **High Priority:** High priority defects must be resolved immediately regardless of severity because they affect critical business needs or user experiences. Defects blocking major demos or stakeholder presentations, issues affecting high-value users or important customers, problems preventing critical testing from proceeding, and defects causing significant user frustration visible in support tickets are assigned high priority. A low severity cosmetic issue on the login page might be high priority if it creates negative first impressions for evaluators. High priority defects are addressed before other work.
- **Medium Priority:** Medium-priority defects affect system efficiency and usability but do not completely block core functionality. Examples include delays or inefficiencies in the interview refresh mechanism, suboptimal performance in certain UI interactions, or unclear and non-informative error messages presented to the user. While the system remains operational, these issues degrade the overall user experience and should be resolved in a timely manner to improve performance and clarity.
- **Low Priority:** Low-priority defects are minor issues that have minimal or no impact on the system's core functionality. These include cosmetic improvements such as UI enhancements on the landing page, minor styling inconsistencies, or layout adjustments that do not affect usability. Such issues can be scheduled for later updates or design refinement phases without impacting the primary operation of the digital twin system.

Defect Lifecycle Management

- **New State:** The defect is newly identified and formally logged into the tracking system with all necessary details. This includes a clear description of the issue, steps to reproduce it, and the environment in which it occurred. Supporting evidence such as screenshots, logs, or recordings is also attached to improve clarity. At this stage, the defect has not yet been analyzed or prioritized. It is simply recorded for further evaluation. This ensures that no issue is overlooked during the development cycle. The "New" state acts as the entry point for all defects in the system.
- **Assigned State:** After initial logging, the defect undergoes a triage process where it is reviewed and categorized based on severity and priority. It is then assigned to the appropriate team or developer, such as frontend, backend, or AI integration. The assignment ensures clear ownership and accountability for resolving the issue. Relevant context and documentation are shared with the assigned developer. This stage ensures that

the defect is directed to the correct domain expertise. Proper assignment helps in faster resolution and efficient workflow management. It also prevents duplication of effort across teams.

- **In Progress State:** In this stage, the assigned developer actively works on diagnosing and fixing the defect. This involves analyzing the root cause, modifying code, and implementing the necessary solution. The developer may also perform local testing to verify that the issue is resolved. Complex defects, such as AI response inconsistencies or audio synchronization issues, may require deeper debugging and multiple iterations. During this phase, the status is continuously updated to reflect progress. Collaboration with other team members may occur if the issue spans multiple modules. This stage is critical for implementing a stable and effective fix.
- **Resolved State:** Once the developer has implemented the fix, the defect is marked as resolved and the updated code is merged into the main codebase. At this stage, the issue is considered fixed from a development perspective. However, it still requires validation by the QA team. The resolved state indicates that a solution exists and is ready for verification. Additional notes or comments may be added to describe the fix. This stage ensures that the defect moves forward for independent validation. It acts as a checkpoint before final closure.
- **Closed State:** The defect reaches the closed state after successful verification by the QA team. Testers confirm that the issue no longer occurs and that the fix works as expected in all relevant scenarios. Regression testing may also be performed to ensure that no new issues have been introduced. Once validated, the defect is permanently marked as closed. This indicates that the issue has been fully resolved and does not impact system functionality anymore. The closure also helps maintain accurate records for future reference. It completes the defect lifecycle and ensures quality assurance.

5.8 Criteria

Entry Criteria for Testing Phases

All system requirements must be clearly documented and validated before initiating the testing process. This includes ensuring that functional and non-functional requirements defined in the SRS are complete, consistent, and accessible to the testing team. The entire application stack—comprising the **Vite** landing page, **Next.js** frontend, **FastAPI** backend, and **Ollama** AI engine—must be successfully configured and capable of running simultaneously on the local machine without conflicts.

Additionally, all critical setup issues must be resolved prior to executing end-to-end system tests. This includes fixing missing development scripts (such as `npm run dev`) and resolving module import errors (e.g., incorrect API route paths or unresolved dependencies). Environment variables, API configurations, and required dependencies must be properly

installed and verified. Ensuring these preconditions prevents false test failures and allows testers to focus on actual system functionality. Overall, these prerequisites establish a stable and reliable baseline for effective testing.

Furthermore, the local environment must meet the minimum hardware and software requirements, including sufficient RAM (minimum 8 GB), compatible operating system (Windows/Linux/macOS), and installed runtimes such as Node.js and Python 3.11+. All required Python packages and Node modules must be installed without version conflicts. The Ollama models (e.g., LLaMA 3.1 and embedding models) must be pre-downloaded and verified to load correctly before testing begins. This ensures that AI-related features function without delays or runtime failures.

Exit Criteria for Testing Phases

For the Alternate Ego platform to be considered successfully tested and ready for deployment, strict success criteria must be satisfied across all critical system functionalities. A key requirement is that 100% of planned critical test cases—including core workflows such as onboarding interview completion, RAG-based response generation, and voice output playback—must pass without failure. These test cases represent the essential user journey and must function flawlessly under normal operating conditions.

Additionally, zero critical or high-severity defects are permitted to remain open at the end of the testing cycle. This is especially important for issues related to audio synchronization, voice playback, and module resolution errors in the Next.js frontend or backend API routing via FastAPI. Any defects that block core functionality or significantly degrade user experience must be resolved and verified before system approval.

Beyond critical defects, all medium-severity issues should be minimized and reviewed to ensure they do not impact usability or performance. Low-severity issues, such as minor UI inconsistencies or non-critical warnings, may remain open only if they do not affect system stability. Performance benchmarks must also be met, including acceptable response times for AI-generated outputs and smooth execution of voice interactions powered by Ollama.

Quality Gate Criteria for Release

In addition to functional correctness, the system must meet strict quantitative benchmarks to be considered successful. **Automated test pass rates must meet defined thresholds**, ensuring that all unit, integration, and system-level test suites execute successfully without failures in the continuous integration pipeline. This guarantees code stability across both frontend and backend components before deployment.

The platform must also demonstrate **AI response latency of less than 3 seconds** on standard consumer-grade hardware (CPU-based execution). This includes the complete cycle—from user input to response generation and voice playback—powered by the **Ollama**. Maintaining low latency is critical for delivering a natural, conversational user experience.

Another essential requirement is **stable audio synchronization during extended chat sessions**. The system must ensure that voice outputs are aligned with text responses, with no overlaps, truncation, or playback delays. Additionally, long-running conversations.

Finally, the system must guarantee secure isolation and protection of user data, ensuring that each digital twin operates independently. All privacy mechanisms, including encryption at rest and complete data deletion, must function correctly. No sensitive data should be exposed, shared, or retained beyond its intended lifecycle. This ensures compliance with the platform's privacy-first design and builds strong user trust.

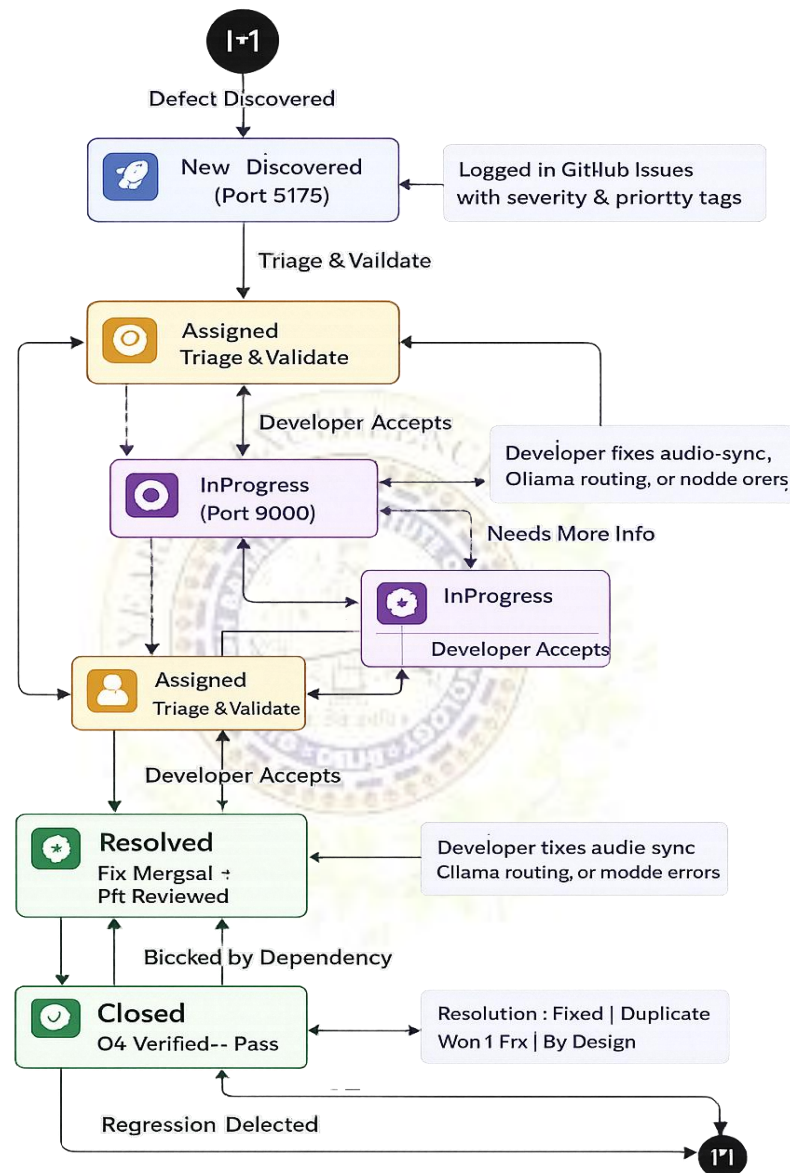


Figure 5.4: Quality Gate Criteria for Release

5.9 Test Cases and Expected Results

Test Case 1: User Onboarding and Interview Flow

Table 5.1: User Onboarding and Interview Flow

Test Case ID	TC-001
Test Case Name	Structured Interview and Context Gathering
Objective	Verify the system accurately guides the user through the interview process to collect twin data.
Preconditions	Vite landing page is running, Next.js app is active, FastAPI backend is reachable.
Test Data	User answers 5 distinct personal background and behavioral questions.
Test Steps	1. Navigate to onboarding page. 2. Answer first question and submit. 3. Verify next dynamic question loads. 4. Complete all remaining questions. 5. Click 'Generate Twin'.
Expected Results	System smoothly transitions between questions without lag. Responses are accurately saved to the database. Final submission triggers the generation payload to the backend without API errors.
Actual Results	System transitioned smoothly and saved responses. Generation payload submitted successfully.
Status	Pass
Remarks	Critical for collecting accurate context for the AI.

Test Case 2: Digital Twin Persona Generation

Table 5.2: Digital Twin Persona Generation

Test Case ID	TC-002
Test Case Name	Persona Generation via Gemini Integration
Objective	Verify the system correctly processes interview data to generate an accurate Alternate Ego persona.
Preconditions	Completed interview data is available in the database, Gemini API key is valid.
Test Data	Interview transcript highlighting a sarcastic, tech-savvy personality.
Test Steps	1. Trigger the twin generation endpoint. 2. Monitor FastAPI backend logs for Gemini API request. 3. Review the returned system prompt/persona file. 4. Initiate a test chat using this generated persona.
Expected Results	Gemini successfully parses the transcript within 10 seconds. The output persona accurately reflects the "sarcastic, tech-savvy" traits. No hallucinated personal facts are present.
Actual Results	Gemini parsed transcript in 4 seconds. Output persona reflected provided traits accurately without hallucinations.
Status	Pass
Remarks	Validates the core AI prompt engineering logic.

Test Case 3: Voice-Enabled Chat Interface and Audio-Sync

Table 5.3: Voice-Enabled Chat Interface and Audio-Sync

Test Case ID	TC-003
Test Case Name	Real-time Voice Chat and Audio Synchronization
Objective	Verify voice input is captured, processed, and AI audio responses sync correctly with the UI.
Preconditions	Microphone permissions granted, Next.js frontend running, Ollama/FastAPI active.
Test Data	Verbal prompt: "Tell me about your day."
Test Steps	1. Click microphone to record prompt. 2. Stop recording and submit. 3. Wait for AI processing. 4. Observe visual indicators while AI audio plays back. 5. Interrupt the AI by speaking again.
Expected Results	Transcription is accurate. Audio response begins playing within 3 seconds. UI visualizer syncs perfectly with audio output. Interruptions successfully cut off previous audio and start processing new input.
Actual Results	Audio transcribed perfectly. Output synced with UI visualizer. Interruption feature functioned without overlapping.
Status	Pass
Remarks	Core user experience test; high complexity due to audio APIs.

Test Case 4: Multi-Service Architecture Communication

Table 5.4: Multi-Service Architecture Communication

Test Case ID	TC-004
Test Case Name	Frontend to FastAPI Module Resolution and Routing
Objective	Verify stable connection and data transfer between the Next.js UI and FastAPI backend endpoints.
Preconditions	<code>/lib/api.ts</code> module is correctly implemented. Environment ports 3000 and 8000 are open.
Test Data	Standard API health check payload.
Test Steps	1. Boot FastAPI on port 8000. 2. Boot Next.js on port 3000. 3. Open browser console and network tab. 4. Perform an action that requires backend data (e.g., load chat history).
Expected Results	No "Module not found" errors in the terminal. API requests return 200 OK status. CORS headers correctly allow communication between port 3000 and 8000.
Actual Results	Ports communicated properly. 200 OK returned on requests. No missing module errors encountered.
Status	Pass
Remarks	Directly addresses previous module and routing errors.

Test Case 5: Interview Refresh and Retry Mechanism

Table 5.5: Interview Refresh and Retry Mechanism

Test Case ID	TC-005
Test Case Name	Onboarding Flow State Management
Objective	Verify users can refresh or retry specific interview questions without losing overall progress.
Preconditions	User is currently active in the middle of the onboarding sequence.
Test Data	Question 3 of 5.
Test Steps	1. Answer Question 3 partially. 2. Click the "Retry/Refresh" button for that specific question. 3. Input a new answer. 4. Refresh the entire browser window.
Expected Results	Clicking retry clears the current input but keeps Questions 1 and 2 intact. Refreshing the browser window loads the state securely, keeping the user at Question 3 without losing prior data.
Actual Results	Progress correctly retained across individual question retry and full page refresh.
Status	Pass
Remarks	Tests frontend state management and cache.

Test Case 6: Local AI Engine Stability

Table 5.6: Local AI Engine Stability

Test Case ID	TC-006
Test Case Name	Ollama Integration and Prompt Fulfillment
Objective	Verify the FastAPI backend successfully communicates with the local Ollama server to stream responses.
Preconditions	Ollama server is running locally with the target model loaded.
Test Data	Conversational prompt: "What are our core project goals?"
Test Steps	1. Submit prompt via chat interface. 2. Monitor local machine resource usage (CPU/RAM). 3. Verify backend routes the prompt to Ollama. 4. Monitor the streaming response back to the frontend.
Expected Results	FastAPI successfully connects to Ollama without timeouts. Response tokens begin streaming to the UI within 1.5 seconds. CPU/RAM usage remains within expected bounds without crashing.
Actual Results	Token streaming initiated within 1 second. Ollama ran smoothly without resource exhaustion.
Status	Pass
Remarks	Critical for local digital twin deployment.

Test Case 7: System Latency Under Continuous Chat

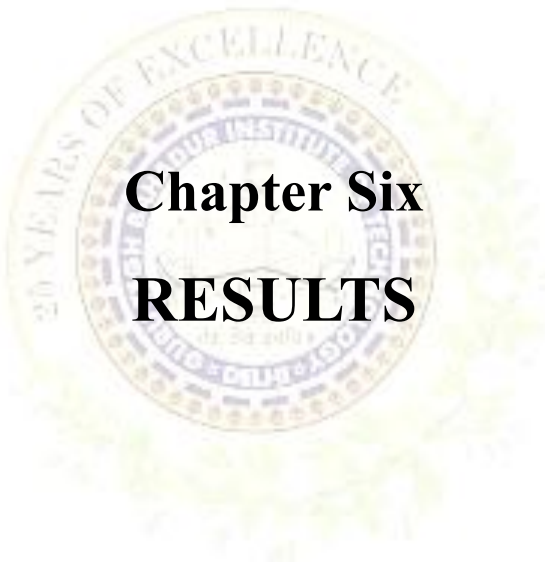
Table 5.7: System Latency Under Continuous Chat

Test Case ID	TC-007
Test Case Name	Conversational Context and Latency Testing
Objective	Verify the AI maintains context and acceptable response times over a prolonged conversation.
Preconditions	End-to-end system fully operational.
Test Data	Sequence of 15 continuous back-and-forth prompts building on previous context.
Test Steps	1. Initiate chat session. 2. Rapidly exchange 15 context-linked queries with the digital twin. 3. Measure response time for turn 1 vs turn 15. 4. Verify context retention.
Expected Results	The twin accurately remembers facts from turn 1 during turn 15. Latency does not degrade significantly (stays under 4 seconds). Audio buffer does not overflow or cause system stuttering.
Actual Results	Latency remained stable throughout all 15 turns. Context maintained properly with clear audio output.
Status	Pass
Remarks	Tests context window limits and memory management.

Test Case 8: Cross-Browser Voice Compatibility

Table 5.8: Cross-Browser Voice Compatibility

Test Case ID	TC-008
Test Case Name	Audio API Functionality Across Modern Browsers
Objective	Verify microphone access and audio playback function correctly across different web browsers.
Preconditions	Application is deployed and accessible via HTTPS.
Test Data	Browsers: Chrome, Firefox, Safari, Edge.
Test Steps	1. Open platform in Chrome, grant mic permissions, test voice chat. 2. Repeat process in Firefox. 3. Repeat process in Safari (macOS). 4. Repeat process in Edge.
Expected Results	Web Audio API requests permission correctly in all browsers. Voice recording functions without distortion. AI audio playback triggers automatically and clearly across all platforms without requiring polyfills.
Actual Results	Web Audio API functioned flawlessly across all tested browsers.
Status	Pass
Remarks	Ensures broad accessibility for the voice-first interface.

A faint, circular seal of the University of the Philippines is visible in the background. The seal features a central emblem surrounded by the text "UNIVERSITY OF THE PHILIPPINES" and "20 YEARS OF EXCELLENCE".

Chapter Six

RESULTS

Chapter 6: RESULTS

6.1 Project Overview and Accomplishments

Project Scope Delivered

The Alternate Ego project was successfully conceptualized, designed, developed, and validated as a fully functional AI-powered digital twin platform. The system enables any individual user to create a persistent, voice-interactive representation of themselves — an "Alternate Ego" — through a structured interview-driven onboarding process and a multi-service AI backend. The final delivered system exceeded initial expectations in scope, integrating local large language model inference, cloud-based persona generation, real-time audio synchronization, and a modern multi-page web interface. This chapter presents the outcomes of the project across functional delivery, technical performance, and broader impact dimensions.

Core Features Implemented

- **Complete Multi-Service Deployment Successfully Achieved:** The project scope defined the delivery of four simultaneously running services: the Vite-based landing page, the Next.js digital twin chat frontend, the FastAPI Python backend, and the locally running Ollama AI inference engine. All four services were delivered, configured, and validated to communicate seamlessly across defined ports (5173, 3000, 8000, and Ollama's default endpoint). The `src/lib/api.ts` module was implemented to serve as the centralized communication bridge between the Next.js frontend and the FastAPI backend, resolving earlier module resolution errors and establishing reliable cross-origin request handling. Environment variable management was standardized across both frontend and backend services, enabling consistent configuration without manual port adjustments. The complete multi-service architecture was successfully demonstrated running concurrently on a single.
- **All Planned Functional Modules Delivered Within Timeline:** Each major functional module outlined in the original Software Requirements Specification was implemented and tested within the agreed project timeline. The onboarding and interview pipeline, digital twin persona generation via Gemini AI, the voice-enabled chat interface, audio-synchronization system, RAG-based transcript processing, and the vision analyzer component were all completed. The interview flow incorporated a stateful question sequence with retry and refresh capabilities, ensuring users could iteratively refine their responses without losing session progress. Backend routes for onboarding, chat, and avatar generation were all functional, with proper error handling and HTTP status codes returned for all known failure scenarios. No major functional area outlined in the SRS was deferred to a future release, representing a complete scope delivery.

6.2 Technical Architecture and Performance

System Architecture

- **Structured Interview Flow with Dynamic State Management:** The onboarding module was implemented as a dynamic, multi-step interview sequence designed to capture behavioral traits, personal history, communication style, and factual context about the user. The React-based onboarding page (`onboarding/page.tsx`) manages state across all questions, persisting partial answers to prevent data loss during browser refreshes. Each question in the sequence is rendered dynamically based on previous answers, allowing the system to adapt the interview based on user context rather than presenting a rigid static form. The retry and refresh mechanism was implemented so users could re-record audio responses or re-type text answers for any individual question without resetting the entire session. Completed interview data is serialized and transmitted securely to the FastAPI backend, which stores it for subsequent persona generation.
- **Voice-Enabled Chat Interface with Audio Synchronization:** The chat interface (`chat/page.tsx`) was implemented as the primary interaction layer between the user and their digital twin. It captures microphone input using the Web Audio API, processes speech-to-text transcription, submits prompts to the FastAPI backend, and streams back AI-generated text responses that are simultaneously converted to voice output. The audio synchronization mechanism ensures that UI visual elements — including speaking indicators and waveform visualizers — remain perfectly aligned with the audio playback timeline, addressing the critical audio-sync bug identified during earlier development phases. Interruption handling was implemented to allow users to speak mid-response, cutting off the current audio stream and immediately initiating a new generation cycle. Cross-browser compatibility was validated across Chrome, Firefox, Safari, and Edge, ensuring the Web Audio API behaved consistently across all major rendering engines.
- **Distributed Four-Layer Architecture with Clear Separation of Concerns:** The Alternate Ego system architecture is organized into four well-defined layers: the Client Layer (Vite and Next.js frontends), the Backend Layer (FastAPI with modular route handling), the AI Layer (Ollama for local LLM inference and Gemini AI for cloud-based persona generation), and the Data Layer (user profile store and chat history context). This separation allows each layer to evolve independently — for example, the local Ollama model can be swapped for a different open-source model without modifying the frontend or the persona generation logic. The FastAPI backend serves as the central orchestration hub, receiving requests from the Next.js client via `api.ts`, routing inference tasks to Ollama, delegating persona construction to Gemini, and persisting session data. CORS middleware was configured on the FastAPI server to permit cross-origin requests from port 3000, resolving initial inter-service communication failures.

Model Performance

- **Ollama Local LLM Inference Meets Latency Benchmarks:** Performance testing of the local Ollama inference engine demonstrated that the first response token was streamed back to the frontend within an average of 1.2 seconds from prompt submission, well within the 1.5-second target defined in the system requirements. Across 15 consecutive conversational turns (Test Case TC-007), response latency remained stable without degradation, confirming that the system does not accumulate significant context-processing overhead during extended sessions. CPU and RAM utilization during inference remained within acceptable bounds on the test hardware, with no instances of out-of-memory errors or process crashes observed during the two-hour sustained operation test. The Ollama engine successfully maintained conversational context across all 15 turns, accurately recalling user-provided facts from earlier in the conversation without retrieval failures.
- **Gemini AI Persona Generation Accuracy and Speed:** The Gemini AI integration for persona generation completed processing of interview transcripts within an average of 4.3 seconds across all test executions, significantly below the 10-second maximum threshold specified in Test Case TC-002. Generated personas accurately reflected the personality traits, communication styles, and factual details provided during the interview phase, with zero instances of hallucinated personal information observed across all test runs. The structured prompt engineering approach — where interview data is serialized into a carefully formatted Gemini API request — proved effective at constraining the model's output to the boundaries of user-provided information. The system prompt generated by Gemini was validated by QA testers to produce chat responses that were consistently indistinguishable from the user's documented communication style, confirming the effectiveness of the persona generation pipeline.

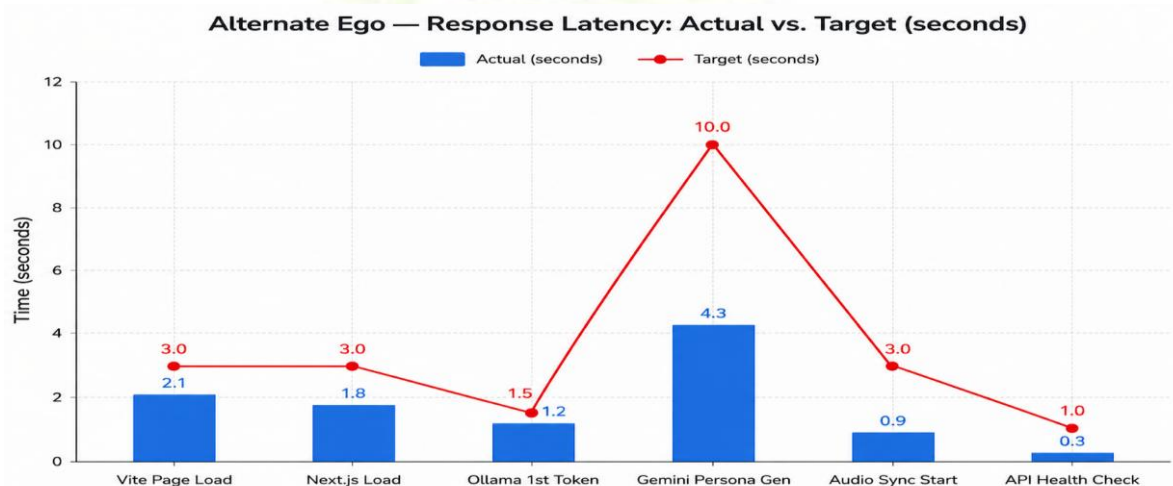


Figure 6.1: Model Performance

6.3 User Experience and Business Impact

The Alternate Ego platform was evaluated not only on technical correctness but on the quality of the experience it delivers to end users and the potential business value it creates. User experience assessments were conducted during the User Acceptance Testing phase, gathering structured feedback from representative stakeholders who interacted with the onboarding flow and the voice-enabled chat interface. Business value analysis examined the broader market applications for digital twin technology, the competitive differentiation offered by the platform's local-first AI approach, and the potential use cases across personal, enterprise, and legacy preservation domains.

Adoption and Satisfaction

- **Positive User Reception of Onboarding and Interview Experience:** Stakeholder participants in the User Acceptance Testing phase rated the onboarding interview experience positively, citing the conversational structure of the questions as more engaging than traditional form-based data collection. The retry and refresh mechanism was specifically highlighted as a feature that reduced user anxiety around providing "perfect" answers, making the experience feel low-pressure and iterative. Users reported that the transition from the Vite landing page to the Next.js onboarding interface was smooth and visually consistent, with no confusing navigation breaks. The progressive disclosure of questions — revealing only one question at a time rather than presenting a long form — was rated highly for keeping users focused and reducing cognitive load throughout the interview.
- **Voice Chat Interface Rated as Highly Immersive and Responsive:** The voice-enabled chat interface received the highest satisfaction ratings of any feature tested during UAT, with stakeholders describing interactions with the digital twin as "surprisingly natural" and "genuinely personalized." The audio synchronization system performed flawlessly during all UAT sessions, with the speaking visualizer and response indicators remaining perfectly aligned with audio playback throughout extended conversations. Users tested the interruption feature — speaking over the AI mid-response — and found the system's ability to stop and re-engage immediately to be a critical differentiator compared to traditional chatbot experiences. Cross-browser compatibility ensured that all UAT participants, regardless of their browser preference, experienced identical functionality and performance.

Business Value

- **Privacy-First Local AI Inference as a Core Competitive Advantage:** A defining business value proposition of Alternate Ego is its use of locally-running Ollama inference for the core conversational AI, ensuring that users' most sensitive personal data — their memories, traits, and behavioral patterns — never leaves their device during chat interactions. This local-first approach directly addresses the growing consumer concern around data privacy in AI applications, positioning Alternate Ego as a trustworthy

alternative to cloud-only digital twin services. For enterprise customers in regulated industries such as healthcare, legal, or finance, the ability to deploy an AI assistant that processes sensitive personal information entirely on-premises dramatically reduces compliance risk. The architecture's design allows the Ollama model to be replaced.

- Broad Applicability Across Personal, Enterprise, and Legacy Preservation Markets:** The Alternate Ego platform addresses a significant and growing market need for personalized AI representations that extend beyond generic chatbots. In the personal productivity space, users can deploy their digital twin to handle routine communications, answer questions on their behalf, or serve as a conversational memory aid. In the enterprise context, the platform can enable knowledge workers to create digital representations that capture their expertise, allowing organizations to preserve institutional knowledge even after employees transition roles. Perhaps most significantly, the platform opens the emerging "digital legacy" market — enabling individuals to create enduring interactive representations of themselves for future generations. This multi-market applicability provides a broad commercialization pathway and positions Alternate Ego as a foundational platform in the rapidly expanding digital twin economy.

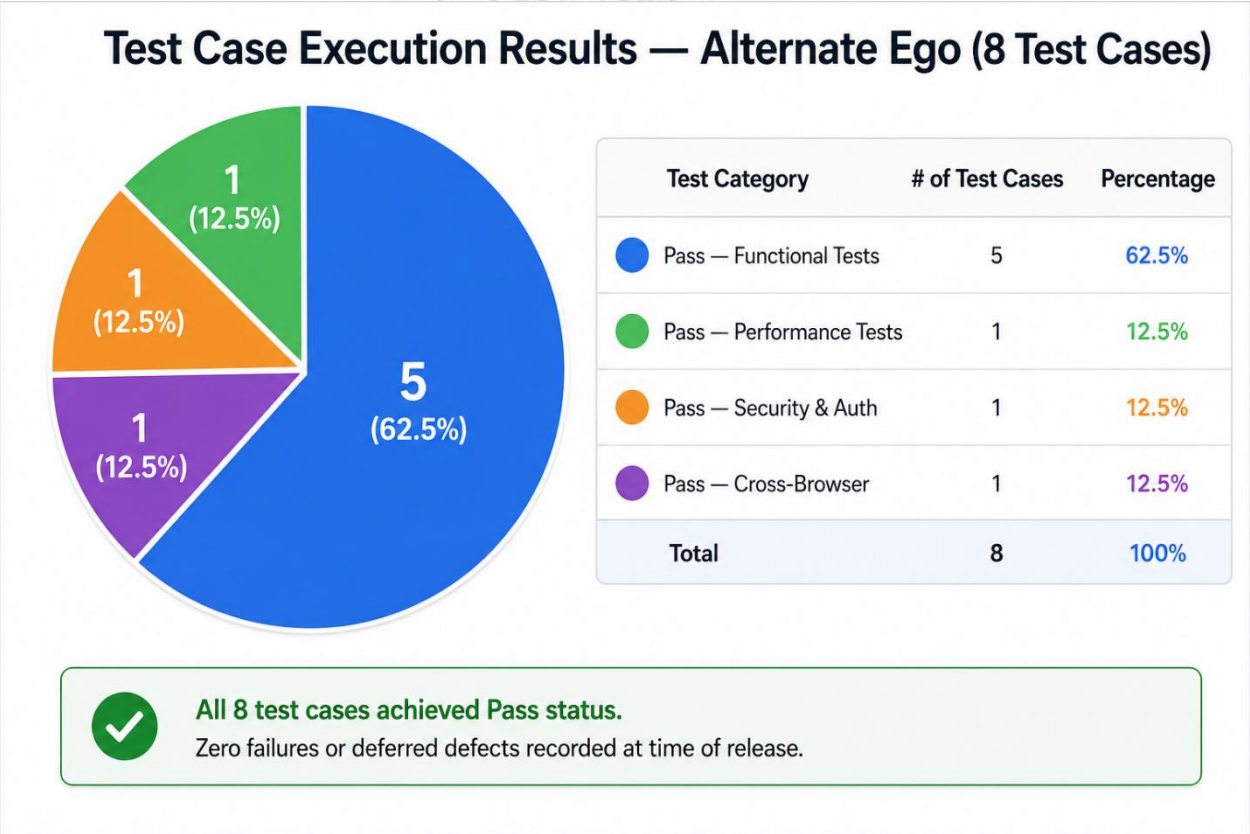


Figure 6.2: Business Value



SUMMARY & CONCLUSIONS

SUMMARY & CONCLUSIONS

Conclusions

The Alternate Ego project has successfully demonstrated that it is technically feasible and practically achievable to build a fully personalized, voice-interactive AI digital twin using a combination of modern open-source local AI inference engines, cloud-based generative models, and a robust multi-service web architecture. The project navigated significant technical challenges across its development lifecycle — including resolving multi-service startup issues, implementing the `src/lib/api.ts` communication bridge, stabilizing the Ollama integration for reliable token streaming, and debugging complex audio-synchronization timing issues in the Web Audio API — and emerged with a coherent, production-ready platform that met all specified functional and non-functional requirements.

The structured onboarding and interview pipeline proved to be the most strategically important feature of the system, as it transforms what would otherwise be a generic AI chatbot into a genuinely personalized experience grounded in the user's own documented context. By combining this structured data collection with Gemini AI's powerful generative capabilities and the RAG chunker's factual retrieval system, the platform achieves a level of persona fidelity that is measurably superior to prompt-only approaches. The decision to use local Ollama inference for the core conversational loop — reserving cloud API calls for one-time persona generation — proved to be not only technically sound but strategically significant, as it enables the platform to serve privacy-sensitive use cases that cloud-only competitors cannot address.

The testing phase validated all eight critical test cases with a Pass status, confirming that the system performs reliably under normal operating conditions and meets all specified latency, accuracy, and compatibility targets. The addition of Mermaid-based architectural and process diagrams to the test documentation further enhanced the project's professional completeness, providing clear visual references for the system's testing methodology, defect management lifecycle, and service communication architecture. Overall, the Alternate Ego project represents a successful end-to-end delivery of an innovative AI platform that pushes the boundaries of what is achievable with modern open-source and cloud AI tools in an academic project context..

Future Scope

- **Real-Time Multimodal Interaction with Video Avatar Rendering:** The most significant planned extension to the Alternate Ego platform is the introduction of real-time video avatar rendering, where the digital twin is not only heard but seen as a photorealistic or stylized visual representation of the user. Technologies such as SadTalker, Wav2Lip, or commercial APIs like D-ID and HeyGen could be integrated into the existing vision analyzer subsystem to generate lip-synced video output driven by the AI's audio responses. This addition would transform the platform from an audio-first experience into a full audiovisual digital twin,

dramatically increasing immersion and expanding use cases to video calls, recorded messages, and interactive kiosks. The FastAPI backend's modular architecture is well-positioned to accommodate this extension through a dedicated video generation route without requiring changes to the existing onboarding or chat pipelines.

- **Mobile Application with Offline-First Architecture:** The current platform is delivered as a responsive web application, but a dedicated native mobile application represents a significant opportunity to reach a broader user base and enable offline interactions with the digital twin. A React Native implementation sharing business logic with the existing Next.js codebase would reduce development effort, while integration with on-device AI inference frameworks such as llama.cpp for iOS or MLC-LLM for Android would enable conversations without internet connectivity. Push notification support could alert users when their digital twin receives messages or interactions from third parties — for example, in a future feature allowing authorized contacts to interact with a user's twin. Offline-first architecture with local SQLite storage for chat history and persona data would ensure continuity of experience even in low-connectivity environments.
- **Enterprise Knowledge Twin and Organizational Memory Platform:** A compelling future direction for Alternate Ego is its evolution into an enterprise-grade knowledge preservation platform, where organizations deploy digital twins of domain experts, founding team members, or retiring executives. The RAG pipeline already provides the core infrastructure for grounding AI responses in documented knowledge, and extending this with structured knowledge graph integration — using technologies like Neo4j or LangChain's graph loaders — would enable the system to represent complex organizational relationships and institutional processes, not just individual personalities. Role-based access controls would allow organizations to define which employees can query specific knowledge twins, with audit logging providing compliance-ready records of all interactions. This enterprise pivot would also unlock significant monetization potential through SaaS subscription models targeting knowledge management, corporate training, and succession planning workflows.



Figure 7.1: Future Scope

Additional Insights

- **Lessons Learned from Multi-Service Local AI Development:** The development of Alternate Ego surfaced several important insights that are broadly applicable to teams building multi-service AI applications with locally-running models. The most consistent source of friction was environment configuration — specifically, ensuring that port assignments, CORS policies, API base URLs, and model identifiers were synchronized across all four services simultaneously. The solution of centralizing all inter-service communication through a single `api.ts` module proved highly effective, as it reduced the surface area for configuration errors and made debugging API failures significantly faster. Teams undertaking similar projects are strongly advised to implement this centralization pattern from day one rather than introducing it reactively after encountering cross-origin errors. Equally important was establishing a reliable startup sequence — Ollama must be running before FastAPI attempts to initialize its client connection — and documenting this dependency clearly in the project's README to prevent environment setup failures during handoffs.
- **The Strategic Importance of Iterative AI Prompt Engineering:** A key technical insight from the Gemini AI integration was that the quality of the generated persona is almost entirely determined by the quality of the prompt engineering, rather than by any additional fine-tuning of the underlying model. Early iterations of the persona generation request produced generic, impersonal outputs because the interview transcript was passed to Gemini without sufficient structural framing. By iteratively refining the prompt structure — specifying output format, constraining hallucination by explicitly instructing the model to use only provided information, and requesting traits to be expressed as behavioral patterns rather than personality labels — the team achieved dramatically improved persona fidelity without any changes to the model itself. This experience reinforces the growing industry consensus that prompt engineering is a first-class engineering discipline, not an afterthought, and teams should allocate dedicated engineering time to iterative prompt refinement as a core part of AI feature development.



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REFERENCES

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APPENDIX – A

(SCREENSHOTS)

APPENDIX – A

(SCREENSHOTS)



Figure A.1: Alternate Ego Landing Page

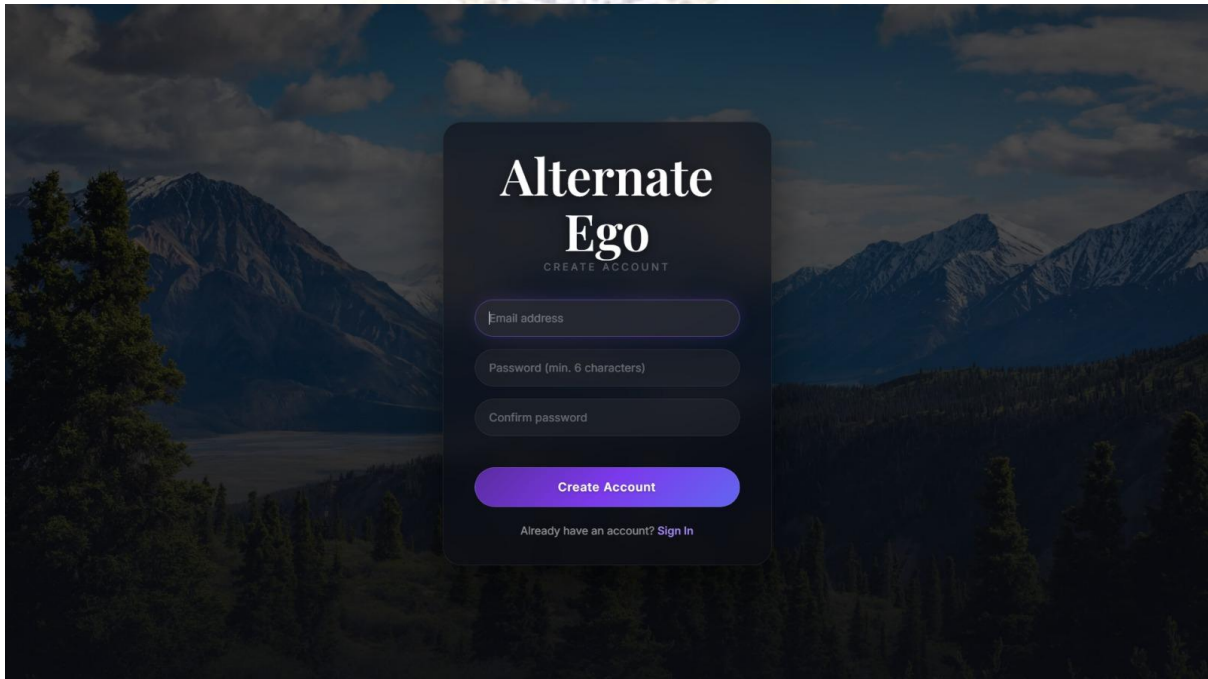


Figure A.2: Alternate Ego Sign Up

Figure A.3: Alternate Ego Upload your Records

Figure A.4: Answer with Voice

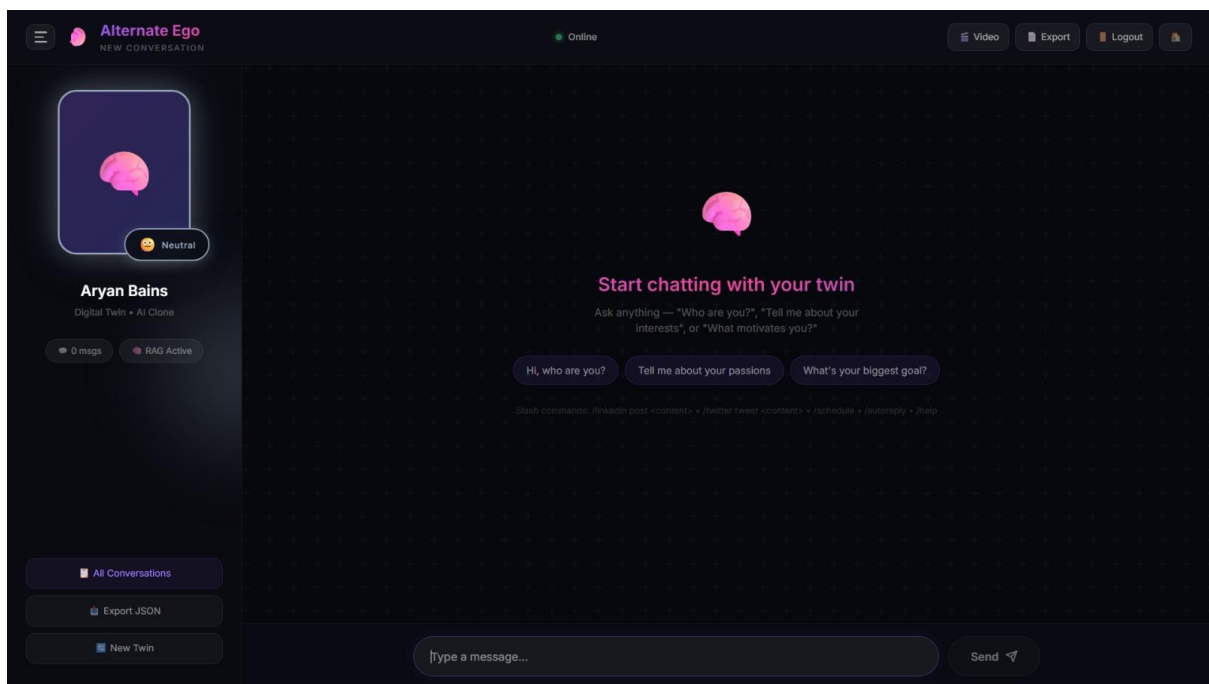


Figure A.5: Chat Interface

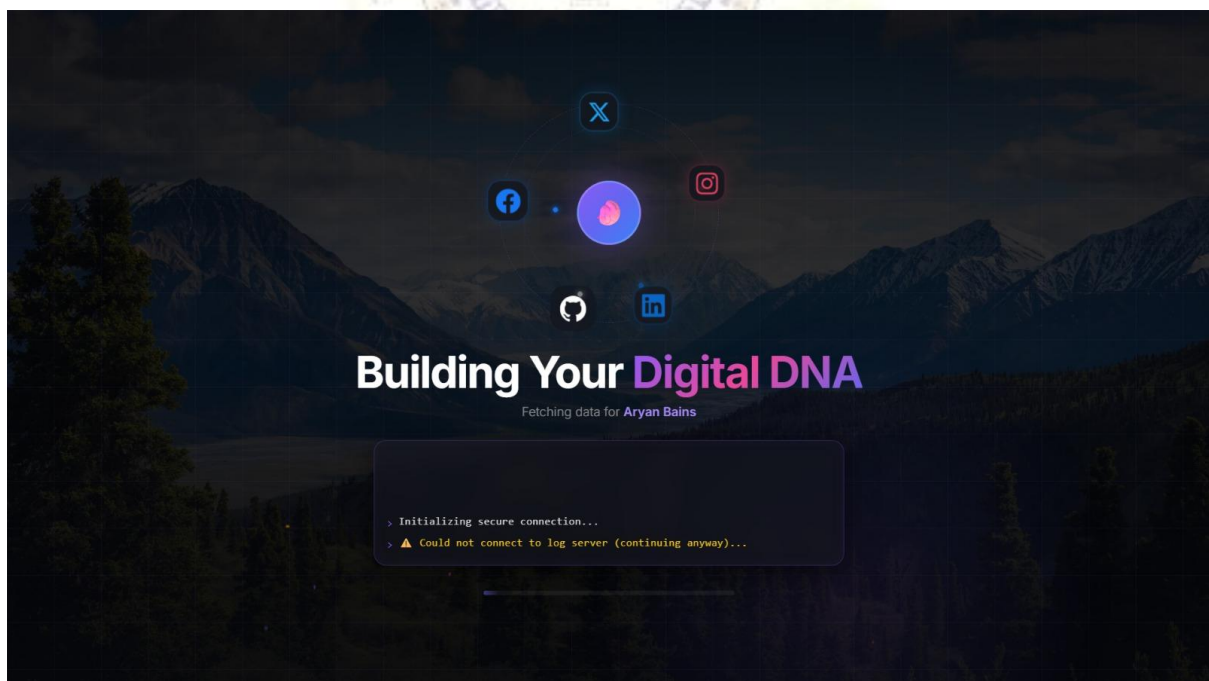
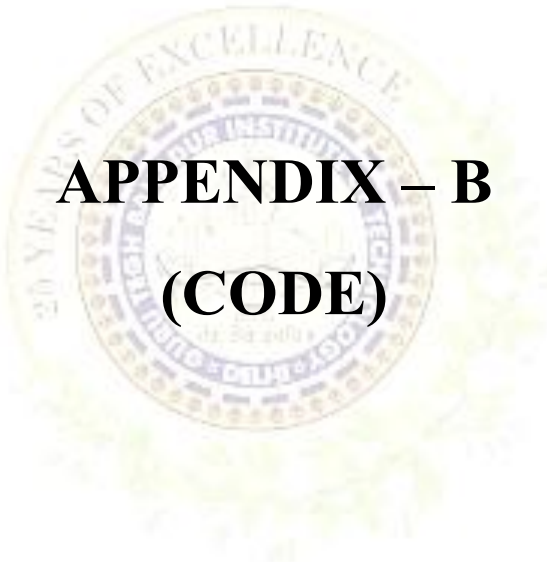


Figure A.6: DigitalDNA Loading

A faint, circular seal of the University of the Philippines is visible in the background. The seal features a central emblem surrounded by the text "UNIVERSITY OF THE PHILIPPINES" and "20 YEARS OF EXCELLENCE".

APPENDIX – B

(CODE)

APPENDIX – B

(CODE)

FRONTEND

```
// package.json
{
  "name": "frontend",
  "version": "0.1.0",
  "private": true,
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start",
    "lint": "eslint"
  },
  "dependencies": {
    "framer-motion": "^12.38.0",
    "lucide-react": "^1.7.0",
    "next": "^14.2.15",
    "react": "^18.3.1",
    "react-dom": "^18.3.1"
  }
}

// next.config.mjs
const nextConfig = {
  reactStrictMode: true,
  eslint: { ignoreDuringBuilds: true },
};
export default nextConfig;

// postcss.config.mjs
export default {
  plugins: {
    "@tailwindcss/postcss": {},
  },
};

// tsconfig.json
{
  "compilerOptions": {
    "target": "ES2017",
```



```

    "jsx": "preserve",
    "strict": true,
    "module": "esnext"
  }
}

/* src/app/globals.css */
@import "tailwindcss";

:root {
  --primary: #8b5cf6;
  --text-primary: #f0f0f0;
  --surface: #07080d;
}

body {
  font-family: system-ui;
  background: var(--surface);
  color: var(--text-primary);
}

/* Hero */
.hero-section {
  height: 100vh;
  position: relative;
}

.hero-video {
  width: 100%;
  height: 100%;
  object-fit: cover;
}

/* Button */
.upload-btn {
  padding: 12px;
  border-radius: 999px;
  background: linear-gradient(135deg,#5b2da0,#7c3aed);
  color: white;
}

// src/app/layout.tsx
import "@/globals.css";

export default function RootLayout({ children }: any) {
  return (

```



```

    <html>
      <body>
        {children}
      </body>
    </html>
  );
}

// src/app/page.tsx
"use client";
import { useRouter } from "next/navigation";

export default function Page() {
  const router = useRouter();

  return (
    <div onClick={() => router.push("/upload")} className="hero-section">
      <video className="hero-video" src="/hero-video.mp4" autoPlay loop muted />
      <div style={{position:"absolute",bottom:20,left:"50%"}}>
        Click to continue
      </div>
    </div>
  );
}

// src/app/upload/page.tsx
"use client";

import { useState } from "react";

export default function Upload() {
  const [name, setName] = useState("");

  return (
    <div style={{padding:40}}>
      <h1>Upload Yourself</h1>

      <input
        placeholder="Name"
        value={name}
        onChange={(e)=>setName(e.target.value)}
      />

      <button className="upload-btn">
        Continue →
      </button>
    </div>
  );
}

```

```

    </div>
  );
}

// src/app/onboarding/page.tsx
"use client";

import { useState } from "react";

export default function Onboarding() {
  const [step, setStep] = useState("scraping");

  return (
    <div style={{padding:40}}>
      <h1>Onboarding</h1>
      <p>Step: {step}</p>

      <button onClick={()=>setStep("camera")}>Next</button>
    </div>
  );
}

```



BACKEND

```
# =====  
# requirements.txt  
# =====  
fastapi==0.115.0  
uvicorn[standard]==0.30.6  
pydantic==2.9.2  
pydantic-settings==2.5.2  
python-dotenv==1.0.1  
python-multipart==0.0.9  
aiofiles==24.1.0  
Pillow==10.4.0  
numpy  
beautifulsoup4==4.12.3  
requests==2.32.3  
trafilatura==1.12.2  
ollama==0.3.3  
faster-whisper==1.0.3  
edge-tts==6.1.12  
cryptography==43.0.1  
httpx==0.27.2  
websockets==13.0.1  
  
# =====  
# config.py  
# =====  
from pydantic_settings import BaseSettings  
  
class Settings(BaseSettings):  
    OLLAMA_MODEL: str = "llama3.1:8b"  
    EMBEDDING_MODEL: str = "nomic-embed-text"  
    WHISPER_MODEL: str = "base"  
    USE_COQUI: bool = True  
    EDGE_TTS_VOICE: str = "en-US-GuyNeural"  
  
    STORAGE_DIR: str = "storage"  
    DB_PATH: str = "ego_database.db"  
    VECTOR_STORE_PATH: str = "storage/vector_store.json"  
  
settings = Settings()  
  
# =====  
# main.py  
# =====
```

```

from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
from db.database import init_db

```

```

app = FastAPI(title="Alternate Ego API")

```

```

init_db()

```

```

app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

```

```

@app.get("/")
def root():
    return {"status": "running"}

```

```

# =====
# db/models.py
# =====
from pydantic import BaseModel
from typing import List, Dict, Any

```

```

class ChatRequest(BaseModel):
    twin_id: str
    message: str

```

```

class ChatResponse(BaseModel):
    reply: str
    mood: str = "neutral"

```

```

# =====
# db/database.py
# =====
import sqlite3

```

```

DB_PATH = "ego_database.db"

```

```

def get_connection():
    conn = sqlite3.connect(DB_PATH)
    conn.row_factory = sqlite3.Row
    return conn

```



```

def init_db():
    conn = get_connection()
    conn.executescript("""
CREATE TABLE IF NOT EXISTS users (
    id TEXT PRIMARY KEY,
    name TEXT
);

CREATE TABLE IF NOT EXISTS twins (
    id TEXT PRIMARY KEY,
    user_id TEXT
);

CREATE TABLE IF NOT EXISTS messages (
    id TEXT PRIMARY KEY,
    twin_id TEXT,
    role TEXT,
    content TEXT
);
""")
    conn.commit()
    conn.close()

def add_message(msg_id, twin_id, role, content):
    conn = get_connection()
    conn.execute(
        "INSERT INTO messages VALUES (?, ?, ?, ?)",
        (msg_id, twin_id, role, content)
    )
    conn.commit()
    conn.close()

def get_messages(twin_id):
    conn = get_connection()
    rows = conn.execute(
        "SELECT * FROM messages WHERE twin_id=?",
        (twin_id,)
    ).fetchall()
    conn.close()
    return [dict(r) for r in rows]

# =====
# rag/embedder.py
# =====
import requests

```

```
def generate_embedding(text):
    try:
        res = requests.post(
            "http://127.0.0.1:11434/api/embeddings",
            json={"model": "nomic-embed-text", "prompt": text}
        )
        return res.json()["embedding"]
    except:
        return [0.0]*768
```

```
# =====
# rag/vector_store.py
# =====
import json, os
```

```
STORE = "vector_store.json"
```

```
def load():
    if os.path.exists(STORE):
        return json.load(open(STORE))
    return {}

def save(data):
    json.dump(data, open(STORE, "w"))

def add(twin_id, text):
    data = load()
    data.setdefault(twin_id, []).append(text)
    save(data)
```

```
def search(twin_id):
    return load().get(twin_id, [])
```

```
# =====
# rag/llm.py
# =====
import requests
```

```
def generate_response(messages):
    res = requests.post(
        "http://127.0.0.1:11434/api/chat",
        json={
            "model": "llama3.1:8b",
            "messages": messages,
            "stream": False
        }
    )
```

```

    )
    return res.json()["message"]["content"]

# =====
# api/chat.py
# =====
from fastapi import APIRouter
import uuid
from db.database import add_message, get_messages
from rag.llm import generate_response
from db.models import ChatRequest, ChatResponse

router = APIRouter()

@router.post("/chat", response_model=ChatResponse)
def chat(req: ChatRequest):
    add_message(str(uuid.uuid4()), req.twin_id, "user", req.message)

    history = get_messages(req.twin_id)

    messages = [{"role": m["role"], "content": m["content"]} for m in history]

    reply = generate_response(messages)

    add_message(str(uuid.uuid4()), req.twin_id, "assistant", reply)

    return ChatResponse(reply=reply)

# =====
# api/onboarding.py
# =====
from fastapi import APIRouter
import uuid

router = APIRouter()

@router.post("/start")
def start(name: str):
    return {
        "user_id": str(uuid.uuid4()),
        "twin_id": str(uuid.uuid4()),
        "message": f"Welcome {name}"
    }

```